Model Analysis

Table of contents

1 Data analysis /Dataframe import and filters

1.1 DataFrame import and filters

The data has been imported with the following(s) function(s):

```python
def build_df():
    if flag_IR:
        df, features_engineered = import_data_IR()
        if flag_IR_maskP:
            df = mask_period_IR(df)
        if flag_IR_balR:
            df = mask_balanceHC_random_IR(df, n_samples=None)
    if flag_ER:
        df, features_engineered = import_data_ER()

    cols_to_check = target_col + features_X
    mask = np.isfinite(df[cols_to_check]).all(axis=1)
    df = df[mask]
```

```
    cols = list(dict.fromkeys(target_col + features_X  +
                              feats_to_balance + feats_grouping +
                              feats_reference+ feats_postproc))
    df = df[cols]
    #df['campaign'] = "ALL"

    return df
```

```
def import_data_IR():
    if target_col[0] == "i_NH3":
        #file_path = "csv/innova/switched/df_signal_decomposition_NH3_current_db2.csv"
        file_path = "csv/innova/notswitched/df_signal_decomposition_current.csv"
    elif target_col[0] == "i_CO2":
        file_path = "csv/innova/switched/df_signal_decomposition_CO2_current.csv"
    else:
        raise ValueError("Target is not supported")

    df = pd.read_csv(file_path, sep=";")
    df["datetime"] = pd.to_datetime(df["datetime"])
    df.sort_values("datetime").reset_index(drop=True)

    if target_col[0] == "i_NH3":
        df["campaign"] = df["datetime"].apply(assign_campaign)
        df = df[df["campaign"] != "other"].copy()
        df["innova_point"] = df.apply(assign_innova_point, axis=1)
        df = df[df["innova_point"] != "no"].copy()
        with open("csv/innova/notswitched/features_decomposition_list.json", "r") as f:
            dict_features = json.load(f)
    elif target_col[0] == "i_CO2":
        df["campaign"] = df["datetime"].apply(assign_campaign_iCO2)
        df = df[df["campaign"] != "other"].copy()
        df["innova_point"] = df.apply(assign_innova_point_iCO2, axis=1)
        df = df[df["innova_point"] != "no"].copy()
        with open("csv/innova/switched/features_decomposition_CO2_list.json", "r") as f:
            dict_features = json.load(f)
    else:
        raise ValueError("The target col do not match any existing function")


    df["hot"] = df.apply(assign_hot_season,axis=1)
    df["umidity_range"] = df.apply(assign_humidity, axis = 1)
    #Removing of rabbit 4
```

```
    df["column"] = df.apply(assign_colonna, axis = 1)
    df['height'] = df.apply(assign_altezza, axis = 1)



    return df,dict_features
```

```
def assign_innova_point(row):
    camp = row["campaign"]
    rid = row["id_rabbit"]

    # if camp == "2025_Jun":
    #     mapping = {2: "sl11", 4: "sl5", 6: "sl4", 1: "sl3", 5: "no"}
    # elif camp == "2025_Sep":
    #     mapping = {2: "sl11", 4: "sl5", 3: "sl4", 6: "sl3", 5: "no"}
    if camp == "2025_Apr":
        #mapping = {1: "no", 2: "no", 4: "no", 6: "sl5"}
        #mapping = {1: "no", 2: "sl5", 4: "no", 6: "sl5"}
        mapping = {1: "no", 2: "no", 4: "no", 6: "no"}
    elif camp == "2025_Jun":
        mapping = {2: "sl11", 4: "sl5", 1: "sl3", 6: "sl4", 5: "no"}
        #mapping = {2: "sl11", 4: "no", 1: "sl4", 6: "sl3", 5: "no"}
    elif camp == "2025_Sep":
        mapping = {2: "sl11", 4: "sl5", 6: "sl3", 3: "sl4", 5: "no"}
        #mapping = {2: "sl11", 4: "no", 6: "sl4", 3: "sl3", 5: "no"}
    elif camp == "2026_Gen":
        #mapping = {1: "no", 4: "no", 5: "sl11", 6: "no"}
        #mapping = {1: "sl11", 4: "no", 5: "sl11", 6: "sl11"}
        mapping = {1: "no", 4: "no", 5: "no", 6: "no"}
    else:
        mapping = {}

    return mapping.get(rid, "no")
```

Function print :

1.2 Metadata

Nombre de lignes df	8168
Nombre de lignes df clean	8168

Nombre de features	5
% de NaN	0.0
Taille dataset entraînement	5717
Taille dataset test	2450
Métrique utilisée	r2

Column target : i_NH3

Features_X : r_NH3, r_temperature, r_umidity, r_CO2, r_THI

feats_to_balance :

feats_grouping : id_channel

feats_postproc : datetime, id_channel, id_rabbit, campaign

feats_ref :

id_cols : campaign

Grouping : id_channel

2 Model(s) analysis

2.1 Explanation legend tables

Columns

- rank: ranking regarding cross validation test
- train_cv: mean score train of the CV
- test_cv: mean score test of the CV
- test: score on the testing dataset
- diff: test-test_cv
- r_test: test/test_cv
- r_train: train_cv/test_cv

Colors:

[Orange] r_test < 0.95 (under performance of the test)

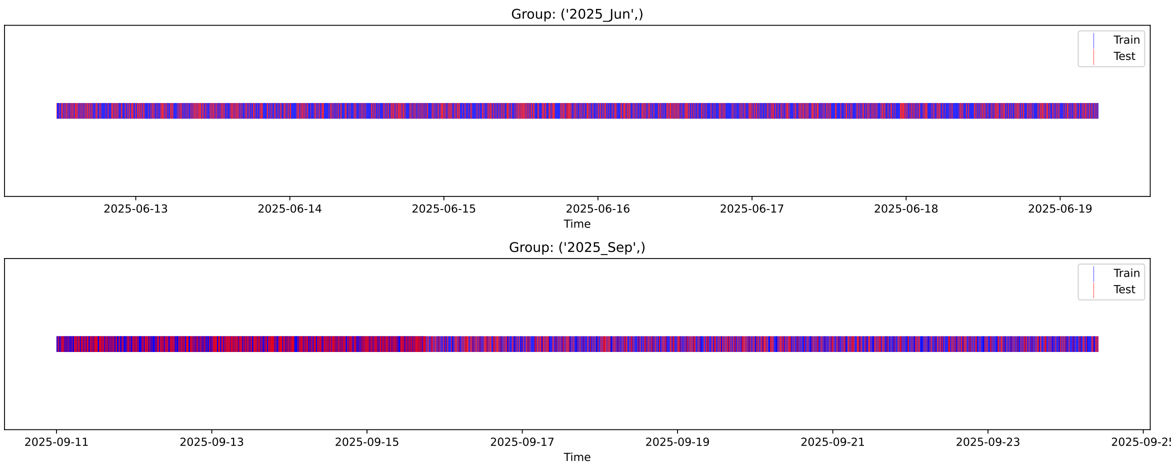
[Green] r_test > 1.05 (over performance of the test)

[Red] r_train not in [0.95, 1.05] (gap between train/test)

2.2 Repartition of train and test

2.2.1 External Split: X_train vs X_test

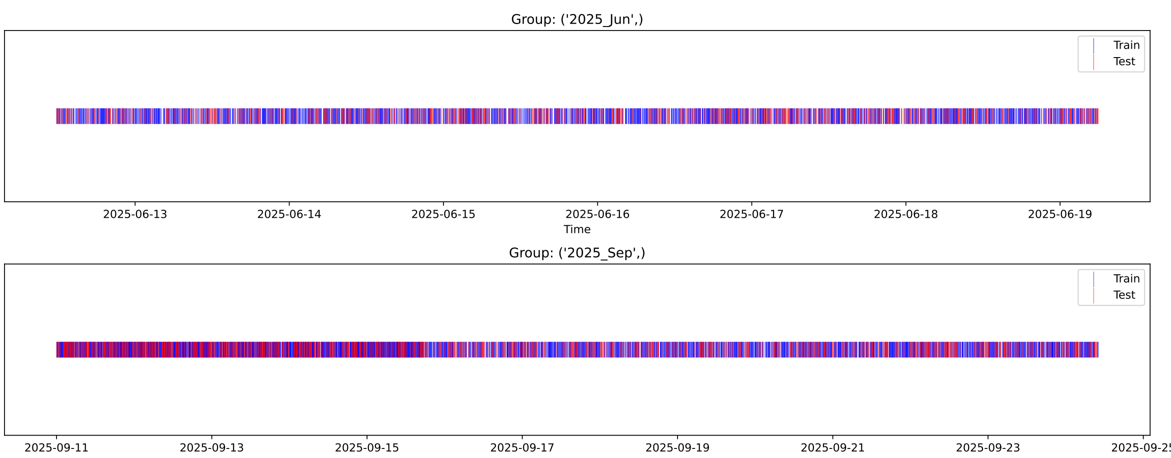
External Split: Train: 5717 samples Test: 2451 samples



2.2.2 Internal CV Folds

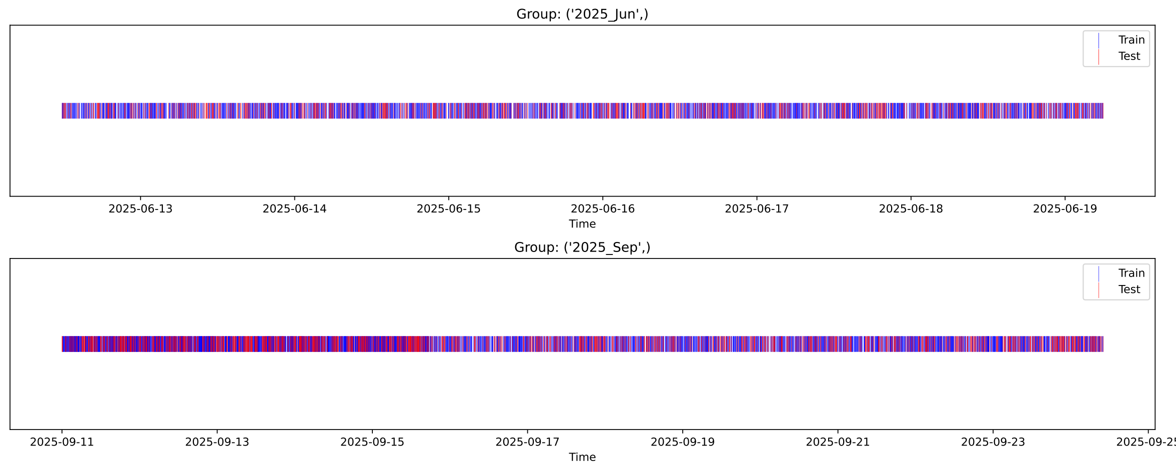
Fold 1/5

Train: 4001 samples Test: 1716 samples Excluded: 0 samples Window exclusion: 0h



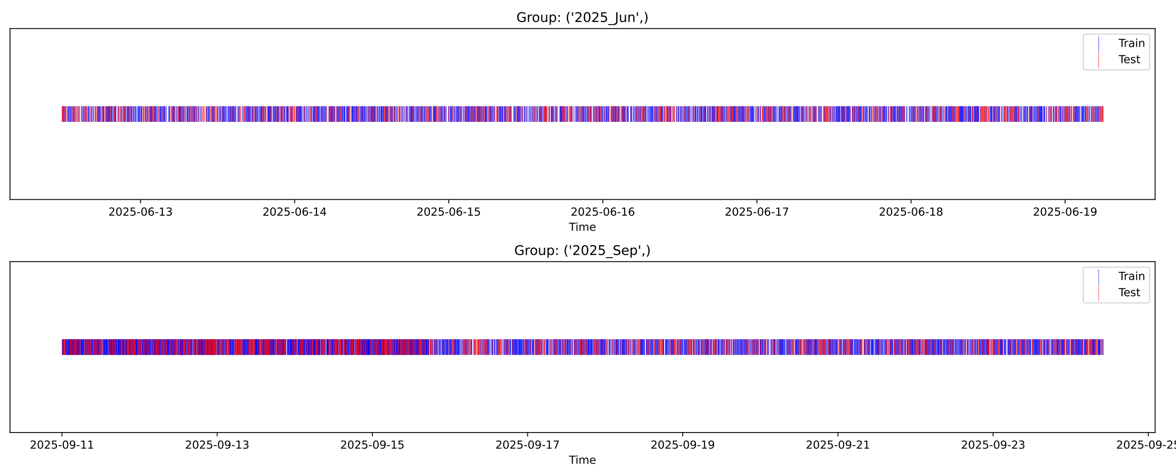
Fold 2/5

Train: 4001 samples Test: 1716 samples Excluded: 0 samples Window exclusion: 0h



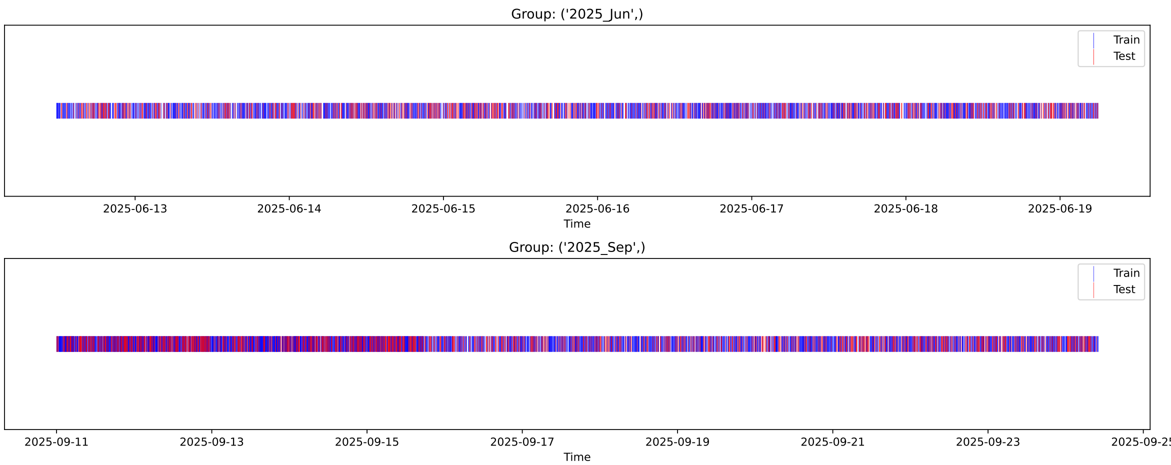
Fold 3/5

Train: 4001 samples Test: 1716 samples Excluded: 0 samples Window exclusion: 0h



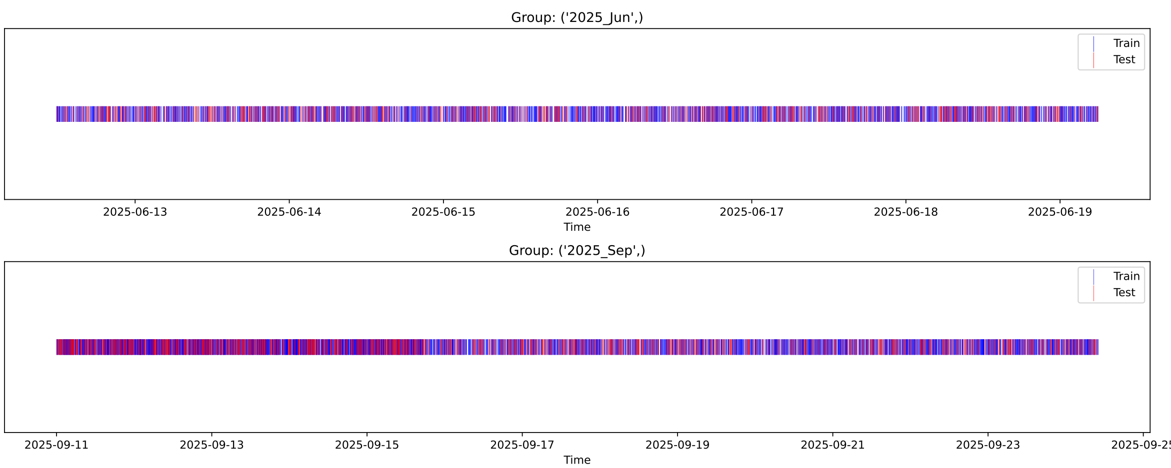
Fold 4/5

Train: 4001 samples Test: 1716 samples Excluded: 0 samples Window exclusion: 0h



Fold 5/5

Train: 4001 samples Test: 1716 samples Excluded: 0 samples Window exclusion: 0h



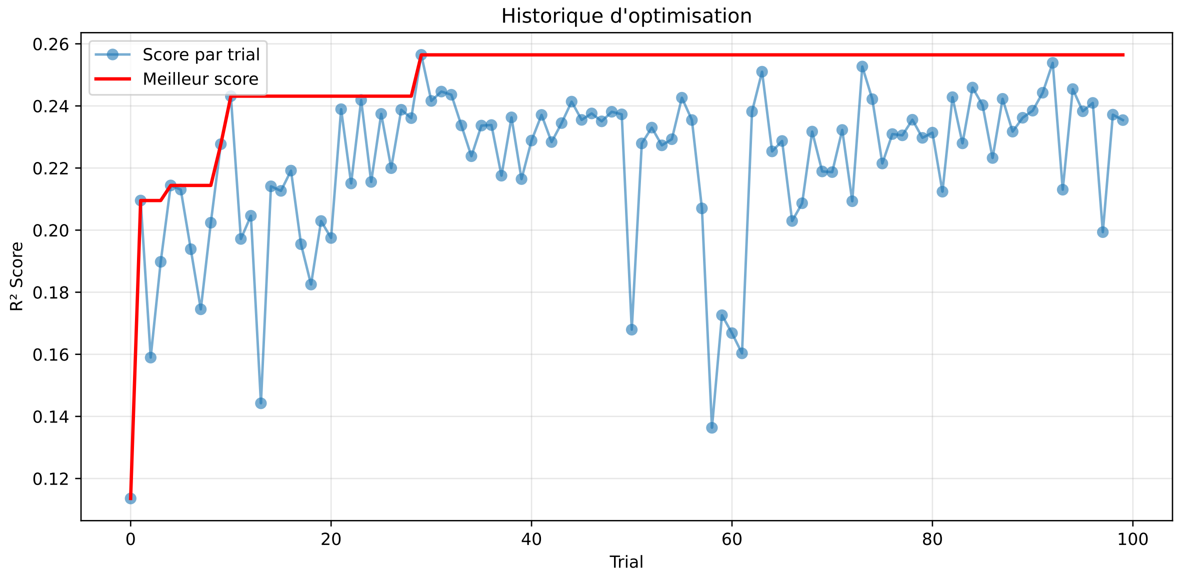
2.3 Model : LGBM

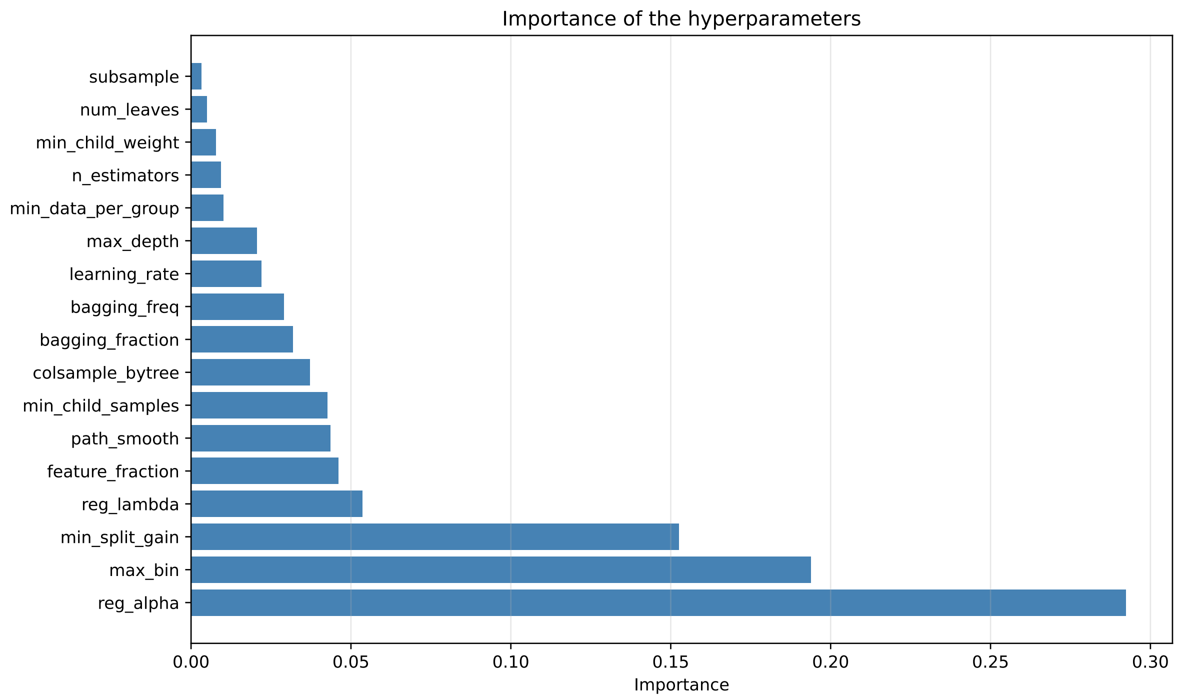
2.3.1 Gridsearch — Group: 3

Len(group) : 1356, **Len(training)** : 950, **Len(test)** : 406

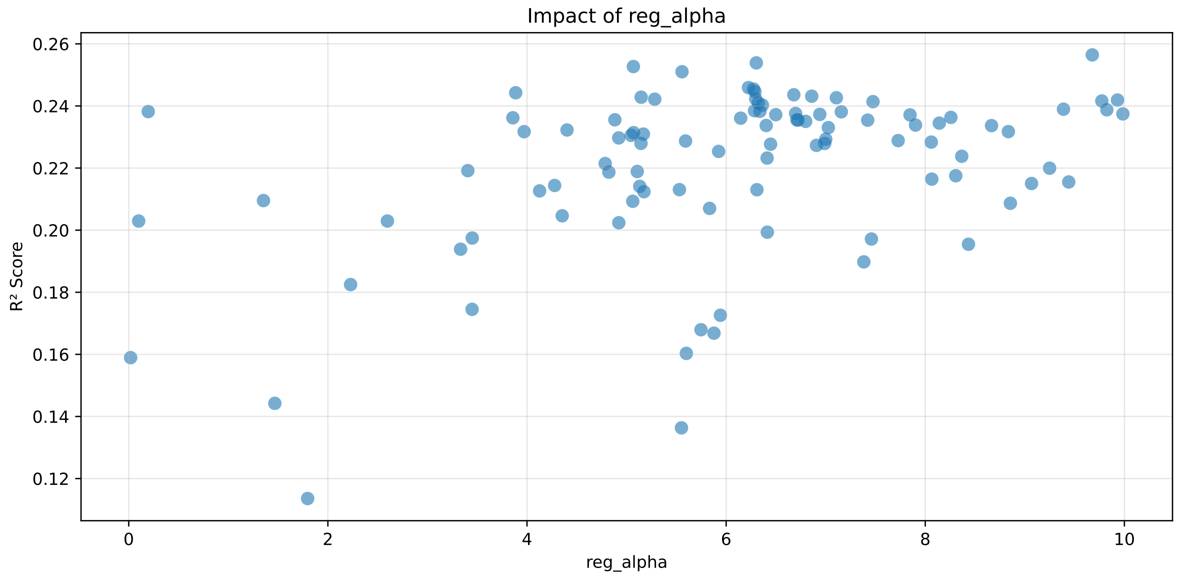
Distribution y_train vs y_test: y_train: mean=2.317, std=0.708, min=1.041, max=6.086 y_test: mean=2.318, std=0.740, min=1.068, max=5.447

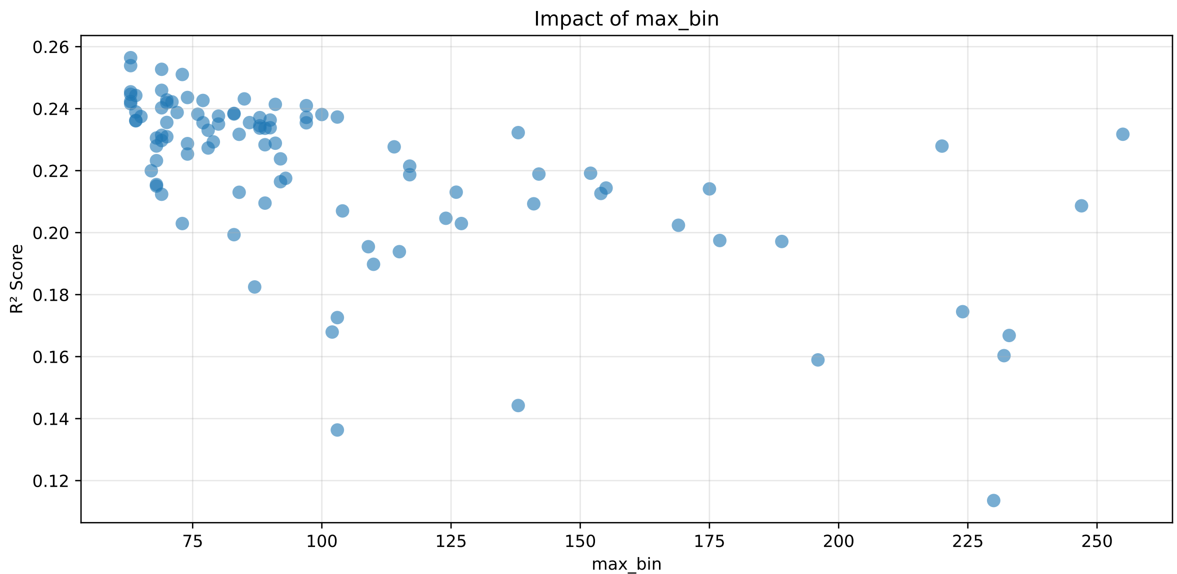
==================== TOP Importance FEATURES ==================== feature importance r_umidity 84 r_NH3 25 r_temperature 24 r_THI 6 r_CO2 2

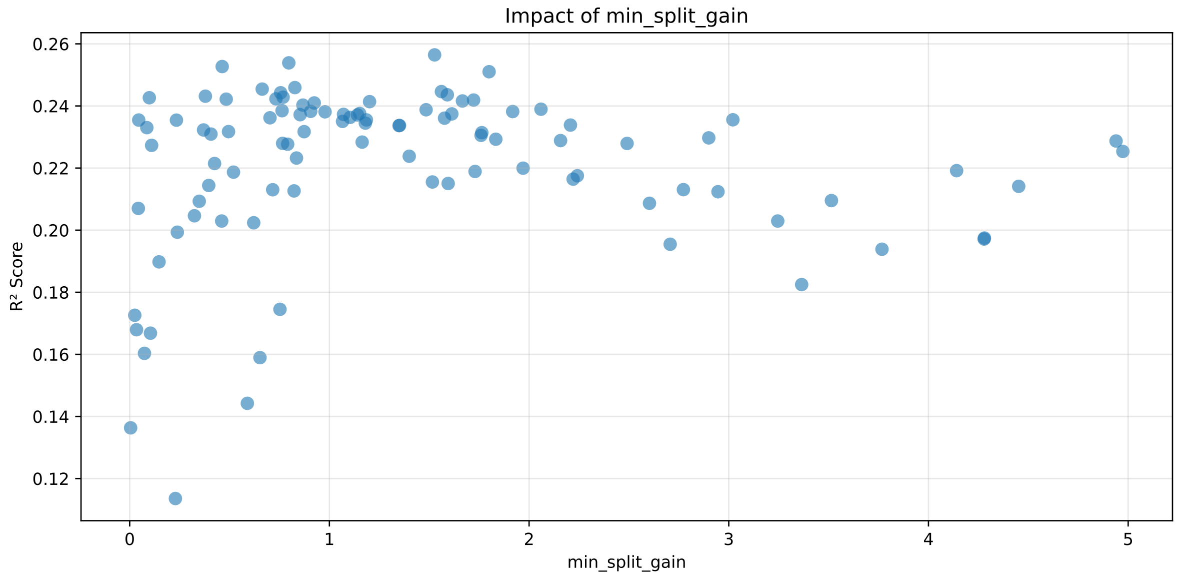


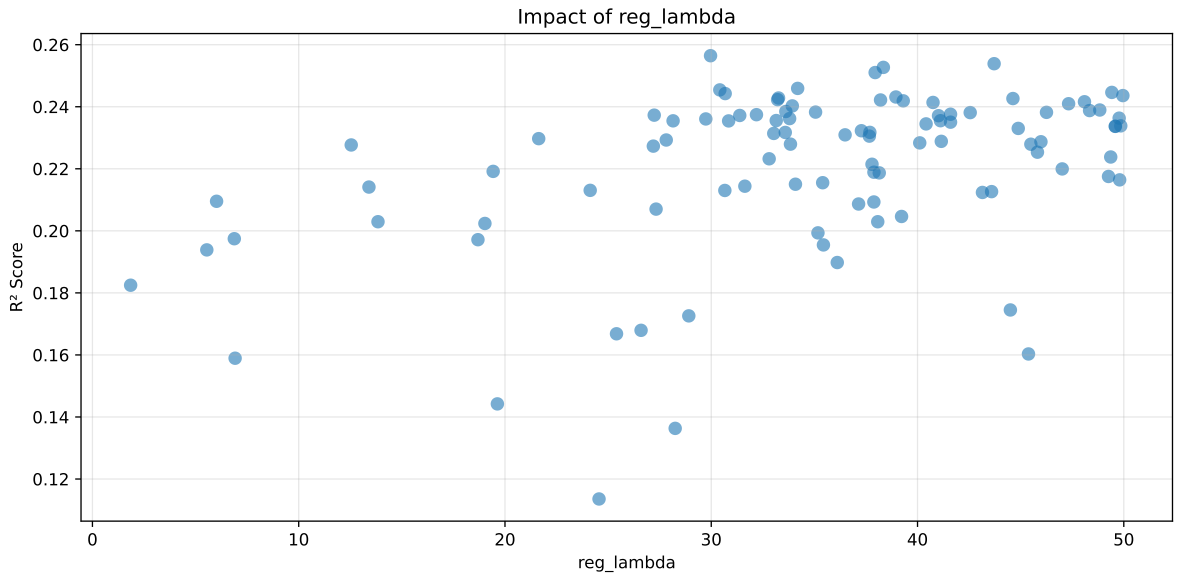


Paramètres avec >5% d'importance : ['reg_alpha', 'max_bin', 'min_split_gain', 'reg_lambda']


Impact of reg_alpha
R² Score
0.26
0.24
0.22
0.20
0.18
0.16
0.14
0.12
0
2
4
6
8
10
reg_alpha


Impact of max_bin
R² Score
0.26
0.24
0.22
0.20
0.18
0.16
0.14
0.12
75
100
125
150
175
200
225
250
max_bin



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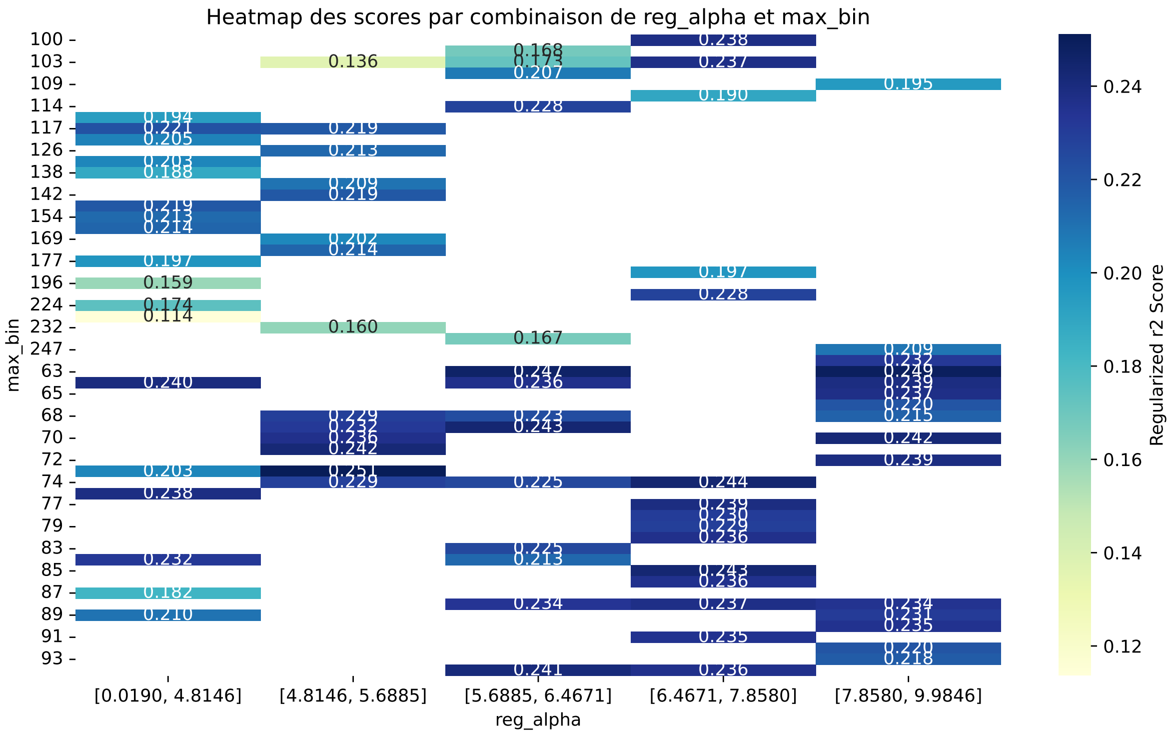
C:\Users\titou\Desktop\UNICT\1- Gas sensor calibration\Python\machine_learning\workflow_main

The default value of observed=False is deprecated and will change to observed=True in a futu
```



rank	mean_train_cv	mean_test_cv	test_score_cv	std_test_cv	penalized_score	test	diff	r_test	r_train
1	0.574	0.5211	0.5953	0.0454	0.2564	0.6396	0.1185	1.2274	1.1016
2	0.6029	0.5447	0.631	0.0366	0.2539	0.6422	0.0975	1.1789	1.1068
3	0.62	0.5587	0.6471	0.0466	0.2527	0.6659	0.1071	1.1917	1.1096
4	0.5892	0.5329	0.6139	0.0442	0.251	0.632	0.0991	1.1861	1.1058
5	0.6155	0.5539	0.6399	0.0403	0.2459	0.6586	0.1047	1.189	1.1112
6	0.6095	0.5488	0.6343	0.0376	0.2454	0.6543	0.1055	1.1922	1.1106
7	0.5799	0.524	0.6007	0.0425	0.2446	0.6391	0.1151	1.2197	1.1066
8	0.6263	0.5626	0.65	0.0411	0.2442	0.6773	0.1147	1.2039	1.1132
9	0.5796	0.5236	0.6019	0.0431	0.2436	0.6397	0.1161	1.2217	1.107
10	0.6265	0.5626	0.6541	0.0461	0.2431	0.6642	0.1016	1.1806	1.1136
---	---	---	---	---	---	---	---	---	---
100	0.7032	0.605	0.6993	0.0377	0.1136	0.7141	0.1092	1.1805	1.1625

Hyperparameters:

Rank 1: {'reg___n_estimators': 600, 'reg___max_depth': 5, 'reg___learning_rate': 0.04258868103893547, 'reg___num_leaves': 54, 'reg___subsample': 0.8871342271179443, 'reg___colsample_bytree': 0.5814155170740922, 'reg___min_child_samples': 31, 'reg___reg_alpha': 9.677548842500514, 'reg___reg_lambda': 29.965696321177724, 'reg___min_split_gain': 1.5269209242583384, 'reg___path_smooth': 2.0950309301273897, 'reg___min_child_weight':

0.004578447315706952, ’reg__feature_fraction’: 0.8175662181293745, ’reg__bagging_fraction’: 0.5814808662746196, ’reg__bagging_freq’: 6, ’reg__max_bin’: 63, ’reg__min_data_per_group’: 157}

Rank 2: {’reg__n_estimators’: 400, ’reg__max_depth’: 5, ’reg__learning_rate’: 0.09041931957355383, ’reg__num_leaves’: 51, ’reg__subsample’: 0.7275970617392259, ’reg__colsample_bytree’: 0.6657573866470484, ’reg__min_child_samples’: 17, ’reg__reg_alpha’: 6.303160122719217, ’reg__reg_lambda’: 43.71363927324215, ’reg__min_split_gain’: 0.7967109191710162, ’reg__path_smooth’: 2.5483653583351047, ’reg__min_child_weight’: 0.10075626655743541, ’reg__feature_fraction’: 0.7736457381040517, ’reg__bagging_fraction’: 0.5592352514614396, ’reg__bagging_freq’: 7, ’reg__max_bin’: 63, ’reg__min_data_per_group’: 106}

Rank 3: {’reg__n_estimators’: 500, ’reg__max_depth’: 4, ’reg__learning_rate’: 0.09942355721995612, ’reg__num_leaves’: 48, ’reg__subsample’: 0.7620423674024694, ’reg__colsample_bytree’: 0.6886897131317807, ’reg__min_child_samples’: 29, ’reg__reg_alpha’: 5.068377891337464, ’reg__reg_lambda’: 38.34411625504232, ’reg__min_split_gain’: 0.4631987954785505, ’reg__path_smooth’: 4.6184484918272375, ’reg__min_child_weight’: 0.03510340091792142, ’reg__feature_fraction’: 0.7294732325661595, ’reg__bagging_fraction’: 0.523853533231399, ’reg__bagging_freq’: 4, ’reg__max_bin’: 69, ’reg__min_data_per_group’: 122}

Rank 4: {’reg__n_estimators’: 500, ’reg__max_depth’: 4, ’reg__learning_rate’: 0.05002667995839095, ’reg__num_leaves’: 24, ’reg__subsample’: 0.8073531375502105, ’reg__colsample_bytree’: 0.6478292756909891, ’reg__min_child_samples’: 25, ’reg__reg_alpha’: 5.55685577551958, ’reg__reg_lambda’: 37.94591387945961, ’reg__min_split_gain’: 1.7997612075110776, ’reg__path_smooth’: 1.0516399284494937, ’reg__min_child_weight’: 0.03884095685261583, ’reg__feature_fraction’: 0.8669252556074326, ’reg__bagging_fraction’: 0.5948053197094818, ’reg__bagging_freq’: 3, ’reg__max_bin’: 73, ’reg__min_data_per_group’: 132}

Rank 5: {’reg__n_estimators’: 600, ’reg__max_depth’: 3, ’reg__learning_rate’: 0.08792437423075875, ’reg__num_leaves’: 47, ’reg__subsample’: 0.7904721083538908, ’reg__colsample_bytree’: 0.691877232074925, ’reg__min_child_samples’: 29, ’reg__reg_alpha’: 6.224215775184975, ’reg__reg_lambda’: 34.18931712538756, ’reg__min_split_gain’: 0.8272652025167805, ’reg__path_smooth’: 2.3953756789286507, ’reg__min_child_weight’: 0.005996667716599095, ’reg__feature_fraction’: 0.8984435634911545, ’reg__bagging_fraction’: 0.7100729197206613, ’reg__bagging_freq’: 4, ’reg__max_bin’: 69, ’reg__min_data_per_group’: 114}

Rank 6: {’reg__n_estimators’: 400, ’reg__max_depth’: 5, ’reg__learning_rate’: 0.08960611668999059, ’reg__num_leaves’: 52, ’reg__subsample’: 0.7378452998083597, ’reg__colsample_bytree’: 0.6356377399591387, ’reg__min_child_samples’: 23, ’reg__reg_alpha’: 6.275506995482527, ’reg__reg_lambda’: 30.414685456592938, ’reg__min_split_gain’: 0.6635216162386934, ’reg__path_smooth’: 2.4217889491670075, ’reg__min_child_weight’:

0.09792228798868058, 'reg__feature_fraction': 0.7193605846397025, 'reg__bagging_fraction': 0.5617506886561858, 'reg__bagging_freq': 4, 'reg__max_bin': 63, 'reg__min_data_per_group': 107}

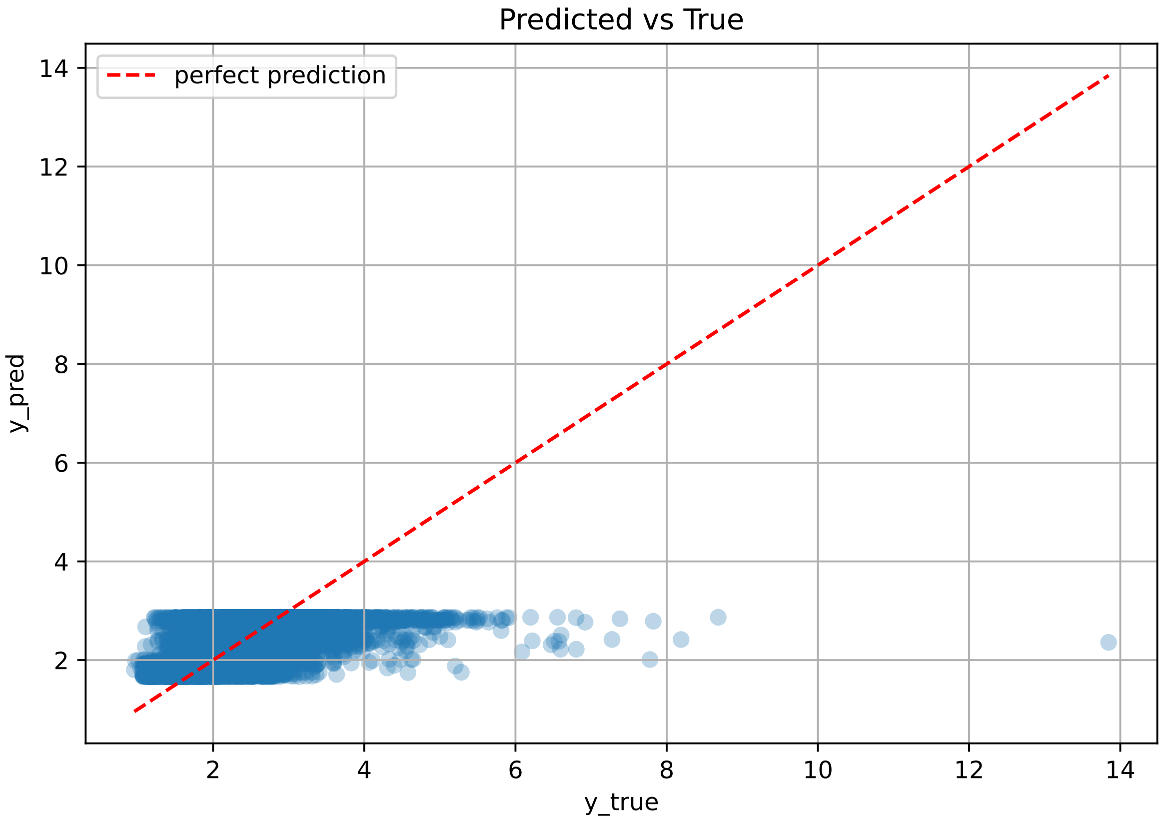
Rank 7: {'reg__n_estimators': 800, 'reg__max_depth': 5, 'reg__learning_rate': 0.024412014430987547, 'reg__num_leaves': 25, 'reg__subsample': 0.8864052212697241, 'reg__colsample_bytree': 0.570016586589859, 'reg__min_child_samples': 31, 'reg__reg_alpha': 6.289038210562215, 'reg__reg_lambda': 49.42043758723043, 'reg__min_split_gain': 1.5610288840543438, 'reg__path_smooth': 0.8727725181287087, 'reg__min_child_weight': 0.005944830394661272, 'reg__feature_fraction': 0.8965629655638414, 'reg__bagging_fraction': 0.5810861221570418, 'reg__bagging_freq': 7, 'reg__max_bin': 63, 'reg__min_data_per_group': 152}

Rank 8: {'reg__n_estimators': 400, 'reg__max_depth': 5, 'reg__learning_rate': 0.09135082862260185, 'reg__num_leaves': 52, 'reg__subsample': 0.7233313344041113, 'reg__colsample_bytree': 0.6658231338610645, 'reg__min_child_samples': 18, 'reg__reg_alpha': 3.8862766833230817, 'reg__reg_lambda': 30.675082365406517, 'reg__min_split_gain': 0.7560298028140426, 'reg__path_smooth': 2.5810768223199205, 'reg__min_child_weight': 0.09248666235485226, 'reg__feature_fraction': 0.76761860252955, 'reg__bagging_fraction': 0.5577007547945038, 'reg__bagging_freq': 2, 'reg__max_bin': 64, 'reg__min_data_per_group': 105}

Rank 9: {'reg__n_estimators': 800, 'reg__max_depth': 5, 'reg__learning_rate': 0.021923015230884378, 'reg__num_leaves': 63, 'reg__subsample': 0.8821211984341676, 'reg__colsample_bytree': 0.8972586340634752, 'reg__min_child_samples': 28, 'reg__reg_alpha': 6.679724928487374, 'reg__reg_lambda': 49.955485934039785, 'reg__min_split_gain': 1.590734790443947, 'reg__path_smooth': 4.191930850462242, 'reg__min_child_weight': 0.0011192297521096608, 'reg__feature_fraction': 0.8883352279998399, 'reg__bagging_fraction': 0.5842246337699092, 'reg__bagging_freq': 7, 'reg__max_bin': 74, 'reg__min_data_per_group': 155}

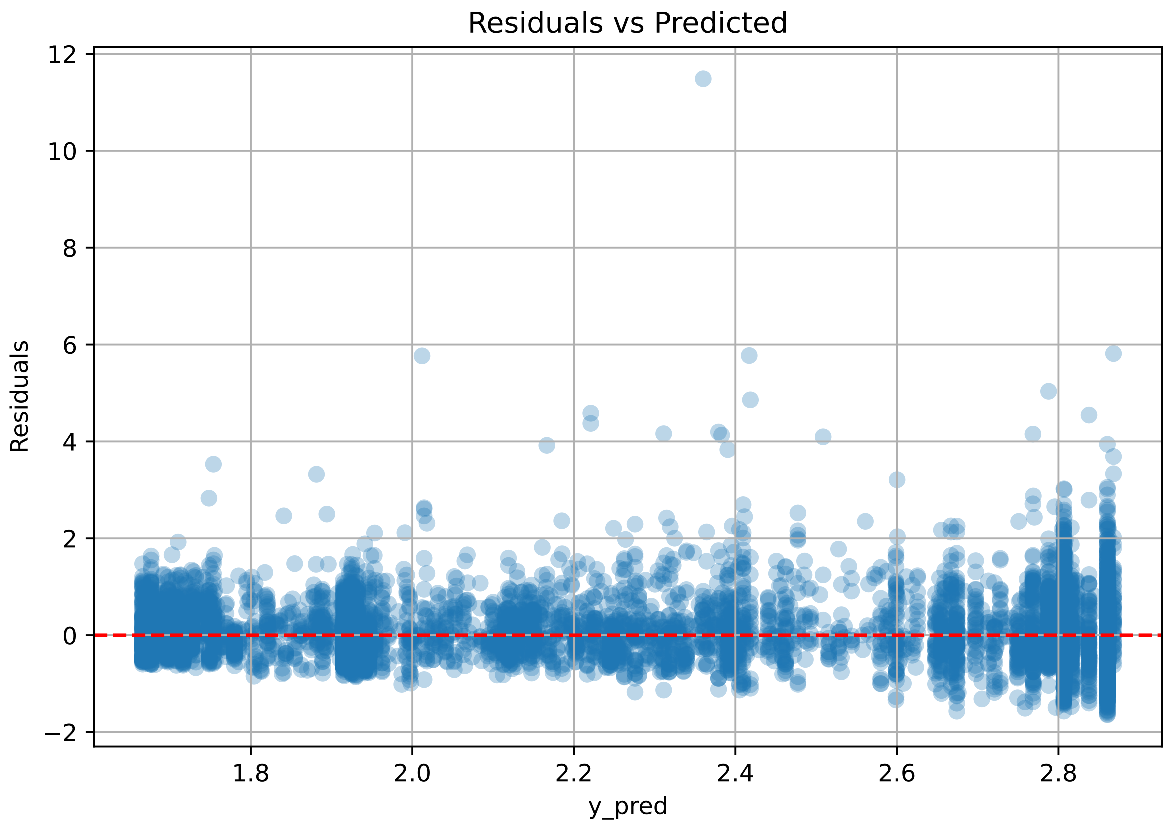
Rank 10: {'reg__n_estimators': 800, 'reg__max_depth': 4, 'reg__learning_rate': 0.09597646754470718, 'reg__num_leaves': 48, 'reg__subsample': 0.7194988417900289, 'reg__colsample_bytree': 0.8542926523875047, 'reg__min_child_samples': 19, 'reg__reg_alpha': 6.859618862388173, 'reg__reg_lambda': 38.95139064886134, 'reg__min_split_gain': 0.37914463755726546, 'reg__path_smooth': 4.277138174755311, 'reg__min_child_weight': 0.0075447984764240245, 'reg__feature_fraction': 0.8928462622283437, 'reg__bagging_fraction': 0.553911650953262, 'reg__bagging_freq': 4, 'reg__max_bin': 85, 'reg__min_data_per_group': 153}

2.3.2 Scatter



Predicted vs True
perfect prediction
y_pred
y_true
14
12
10
8
6
4
2
2
4
6
8
10
12
14

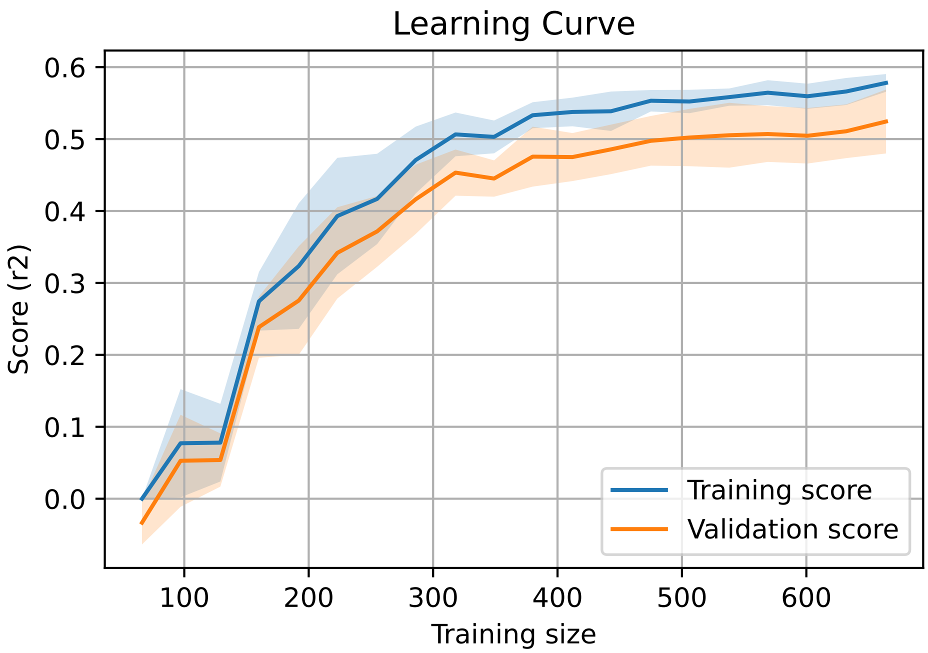
2.3.3 Scatter residuals



2.3.4 Learning curve — Group: 3

Len(group) : 1356, **Len(training)** : 950,

Len(training_real) : 665, **Len(test_of_training)** : 285, **Len(test)** : 406

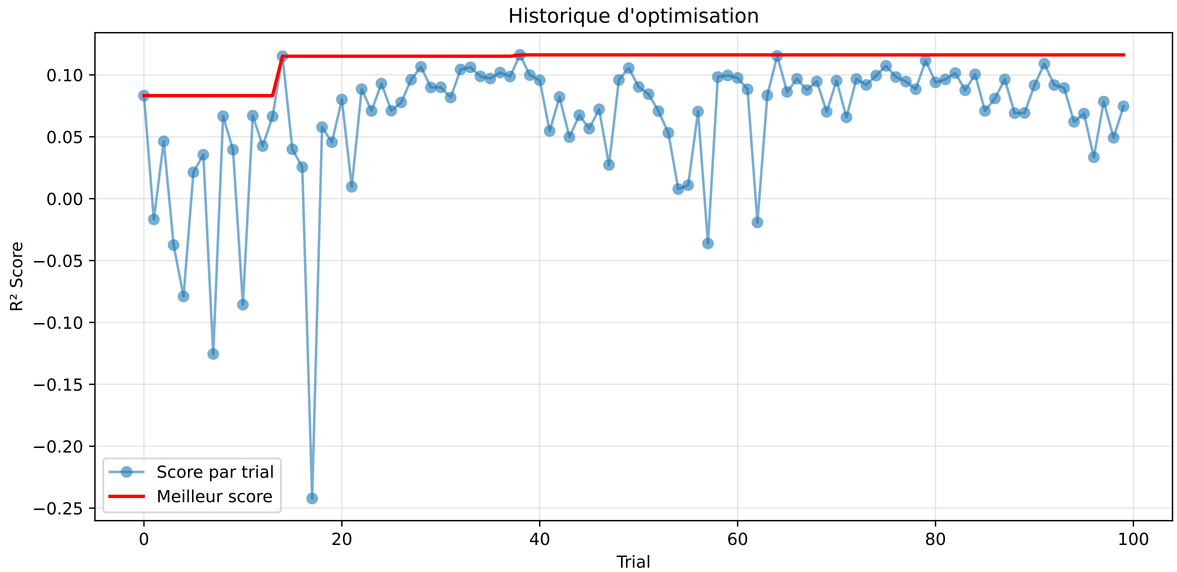


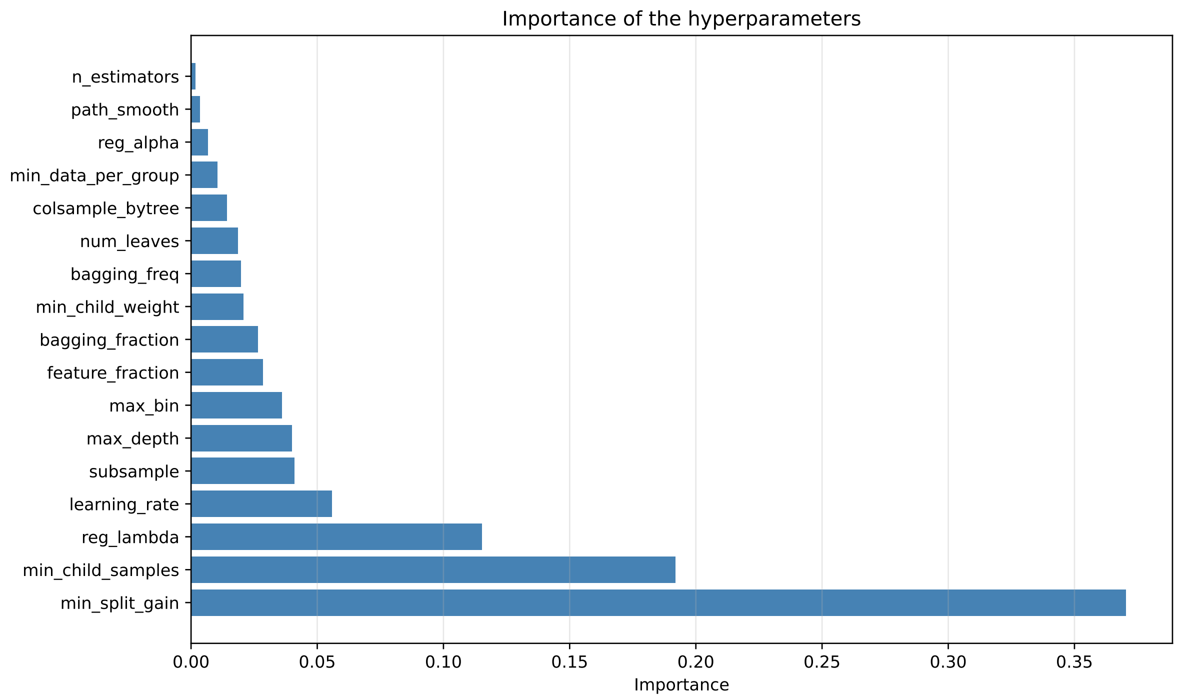
2.3.5 Gridsearch — Group: 4

Len(group) : 1360, **Len(training)** : 952, **Len(test)** : 408

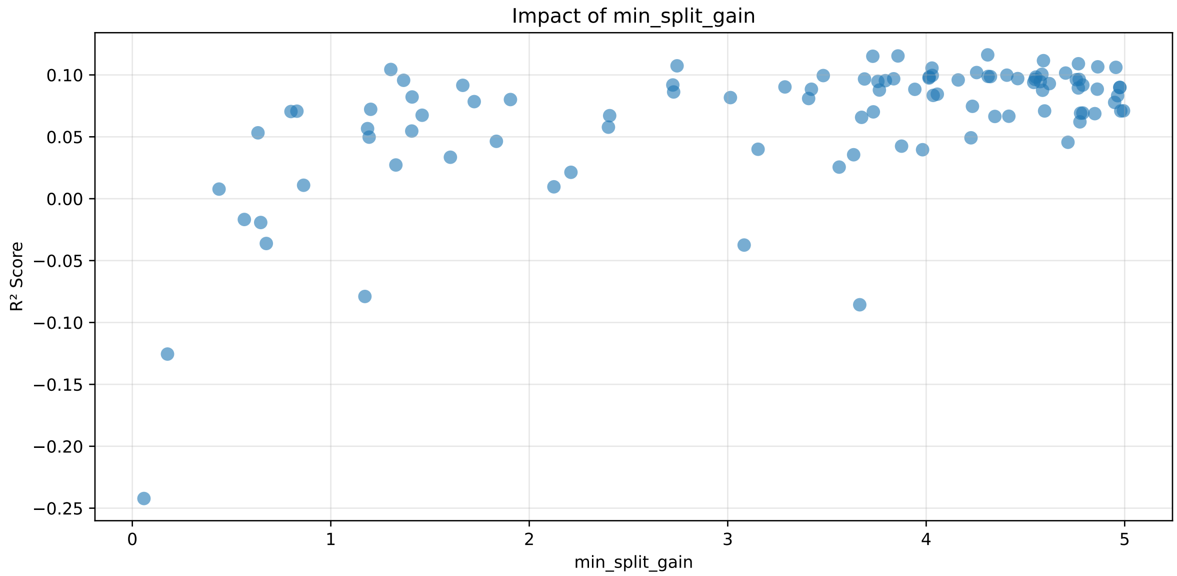
Distribution y_train vs y_test: y_train: mean=2.239, std=0.831, min=0.959, max=7.383 y_test: mean=2.202, std=0.791, min=1.062, max=7.822

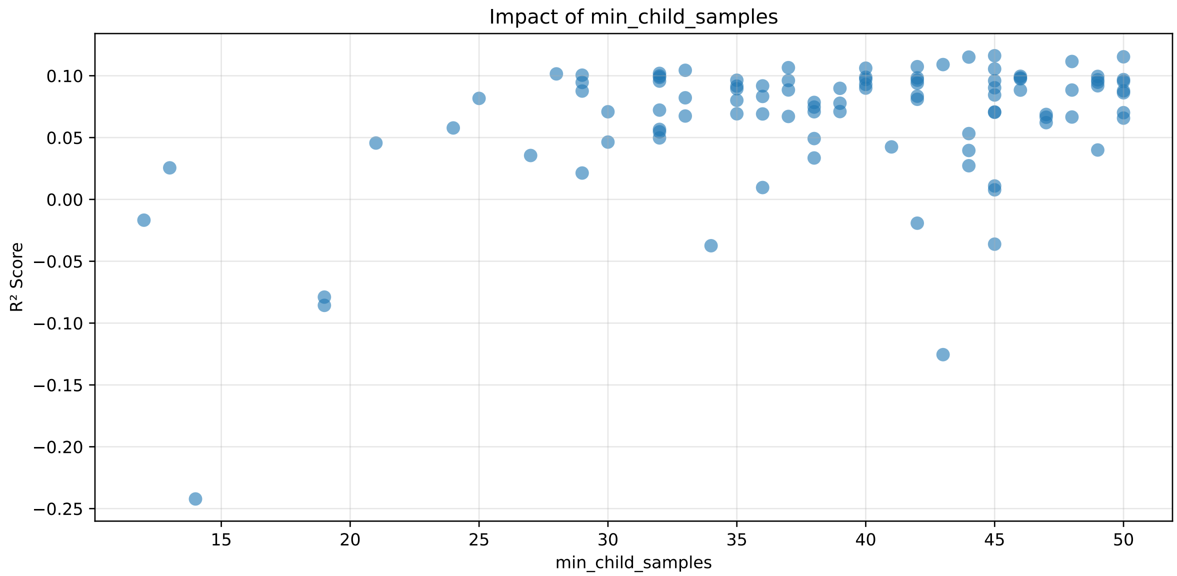
==================== TOP Importance FEATURES ==================== feature importance r_umidity 75 r_THI 45 r_CO2 36 r_temperature 21 r_NH3 14

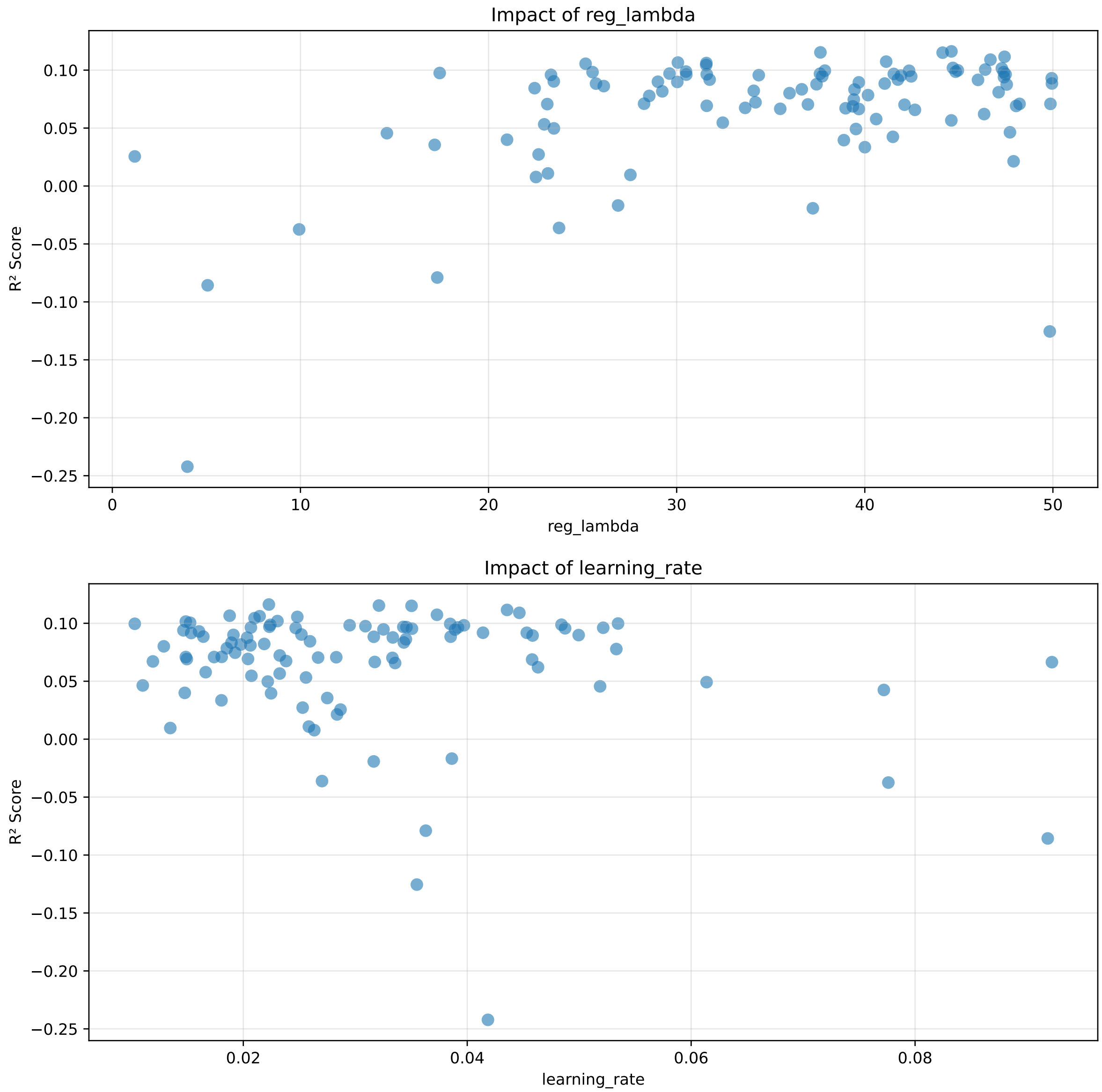




Paramètres avec >5% d'importance : ['min_split_gain', 'min_child_samples', 'reg_lambda', 'learning_rate']

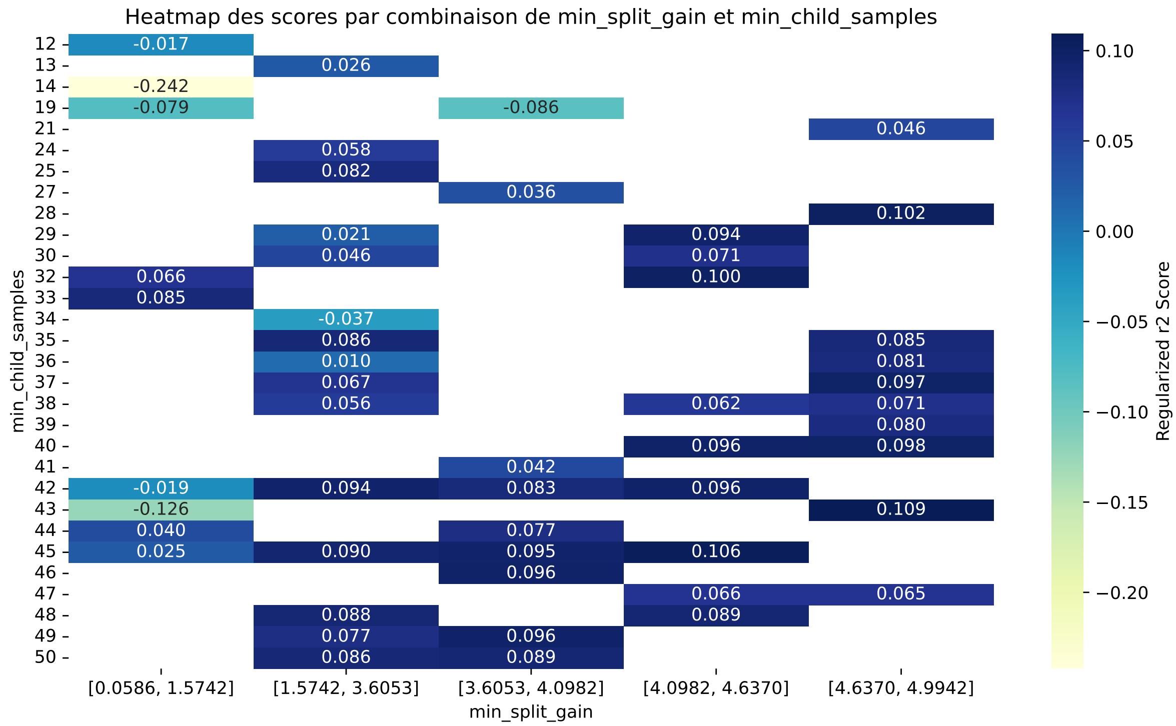

Impact of min_split_gain
R² Score
0.10
0.05
0.00
−0.05
−0.10
−0.15
−0.20
−0.25
0
1
2
3
4
5
min_split_gain


Impact of min_child_samples
R² Score
0.10
0.05
0.00
−0.05
−0.10
−0.15
−0.20
−0.25
15
20
25
30
35
40
45
50
min_child_samples




```
C:\Users\titou\Desktop\UNICT\1- Gas sensor calibration\Python\machine_learning\workflow_main

The default value of observed=False is deprecated and will change to observed=True in a futu
```



rank	mean_train_cv	mean_test_cv	test_score_cv	std_test_cv	penalized_score	test	diff	r_test	r_train
1	0.2611	0.2369	0.2384	0.0368	0.1162	0.2932	0.0563	1.2376	1.1019
2	0.3182	0.2844	0.2875	0.0509	0.1153	0.3107	0.0262	1.0923	1.1189
3	0.2871	0.2584	0.2593	0.0434	0.1151	0.3206	0.0622	1.2408	1.1109
4	0.2355	0.2149	0.2233	0.0543	0.1115	0.2891	0.0742	1.3455	1.0962
5	0.2626	0.237	0.2458	0.0575	0.109	0.3001	0.0631	1.2661	1.108
6	0.3101	0.2763	0.2875	0.0496	0.1073	0.341	0.0647	1.2343	1.1223
7	0.2452	0.2221	0.2262	0.0579	0.1065	0.2588	0.0367	1.1654	1.1041
8	0.2478	0.2242	0.2296	0.0571	0.1061	0.2473	0.0232	1.1034	1.1053
9	0.2656	0.2389	0.2439	0.0461	0.1055	0.3029	0.064	1.2679	1.1117
10	0.3697	0.3255	0.331	0.0443	0.1044	0.3745	0.049	1.1505	1.1359
---	---	---	---	---	---	---	---	---	---
100	0.6142	0.4715	0.4898	0.0371	-0.2422	0.5285	0.057	1.1209	1.3027

Hyperparameters:

Rank 1: {'reg___n_estimators': 400, 'reg___max_depth': 4, 'reg___learning_rate': 0.022284175533667207, 'reg___num_leaves': 31, 'reg___subsample': 0.6595674809484822, 'reg___colsample_bytree': 0.7781120857246108, 'reg___min_child_samples': 45, 'reg___reg_alpha': 5.005192879227375, 'reg___reg_lambda': 44.605508624261425, 'reg___min_split_gain': 4.309985679002364, 'reg___path_smooth': 2.0233573682468773, 'reg___min_child_weight':

0.0037806444438034403, 'reg___feature_fraction': 0.8918823030706903, 'reg___bagging_fraction': 0.5455819431506649, 'reg___bagging_freq': 4, 'reg___max_bin': 180, 'reg___min_data_per_group': 158}

Rank 2: {'reg___n_estimators': 200, 'reg___max_depth': 6, 'reg___learning_rate': 0.03211425966727079, 'reg___num_leaves': 51, 'reg___subsample': 0.8576158903105691, 'reg___colsample_bytree': 0.867334266790533, 'reg___min_child_samples': 50, 'reg___reg_alpha': 3.7046512730969576, 'reg___reg_lambda': 37.64128108515758, 'reg___min_split_gain': 3.857846048888227, 'reg___path_smooth': 3.38231406839908, 'reg___min_child_weight': 0.0017380938139366115, 'reg___feature_fraction': 0.541735872915493, 'reg___bagging_fraction': 0.5991470897387366, 'reg___bagging_freq': 6, 'reg___max_bin': 136, 'reg___min_data_per_group': 180}

Rank 3: {'reg___n_estimators': 200, 'reg___max_depth': 5, 'reg___learning_rate': 0.03502675543631949, 'reg___num_leaves': 27, 'reg___subsample': 0.5335041420818235, 'reg___colsample_bytree': 0.8171081574315635, 'reg___min_child_samples': 44, 'reg___reg_alpha': 5.220220095148652, 'reg___reg_lambda': 44.137595169577175, 'reg___min_split_gain': 3.7309992156251814, 'reg___path_smooth': 2.1547425036877605, 'reg___min_child_weight': 0.010288761859344764, 'reg___feature_fraction': 0.8616760320653667, 'reg___bagging_fraction': 0.5625782982980825, 'reg___bagging_freq': 5, 'reg___max_bin': 146, 'reg___min_data_per_group': 174}

Rank 4: {'reg___n_estimators': 400, 'reg___max_depth': 4, 'reg___learning_rate': 0.04355835927707821, 'reg___num_leaves': 41, 'reg___subsample': 0.591812084574854, 'reg___colsample_bytree': 0.8164959837912797, 'reg___min_child_samples': 48, 'reg___reg_alpha': 4.492338593346793, 'reg___reg_lambda': 47.429347362594484, 'reg___min_split_gain': 4.590917091301113, 'reg___path_smooth': 2.745777681140976, 'reg___min_child_weight': 0.0067054663864549785, 'reg___feature_fraction': 0.5812102011398816, 'reg___bagging_fraction': 0.5247125416451742, 'reg___bagging_freq': 5, 'reg___max_bin': 113, 'reg___min_data_per_group': 169}

Rank 5: {'reg___n_estimators': 400, 'reg___max_depth': 8, 'reg___learning_rate': 0.044656188751077165, 'reg___num_leaves': 38, 'reg___subsample': 0.7023443701815142, 'reg___colsample_bytree': 0.7582536603069336, 'reg___min_child_samples': 43, 'reg___reg_alpha': 2.566703066888876, 'reg___reg_lambda': 46.67961602419571, 'reg___min_split_gain': 4.767075208032017, 'reg___path_smooth': 2.778545383902122, 'reg___min_child_weight': 0.005430197310009218, 'reg___feature_fraction': 0.5721078587916657, 'reg___bagging_fraction': 0.5149722886666029, 'reg___bagging_freq': 5, 'reg___max_bin': 84, 'reg___min_data_per_group': 148}

Rank 6: {'reg___n_estimators': 200, 'reg___max_depth': 7, 'reg___learning_rate': 0.03729529107747443, 'reg___num_leaves': 40, 'reg___subsample': 0.8396462767351741, 'reg___colsample_bytree': 0.8201328152801979, 'reg___min_child_samples': 42, 'reg___reg_alpha': 4.619686813333731, 'reg___reg_lambda': 41.13706501997501, 'reg___min_split_gain': 2.7447603419912046, 'reg___path_smooth': 3.3949680172175043, 'reg___min_child_weight':

0.0018799125278767348, 'reg__feature_fraction': 0.5987744909678407, 'reg__bagging_fraction': 0.5234276206028512, 'reg__bagging_freq': 7, 'reg__max_bin': 115, 'reg__min_data_per_group': 127}

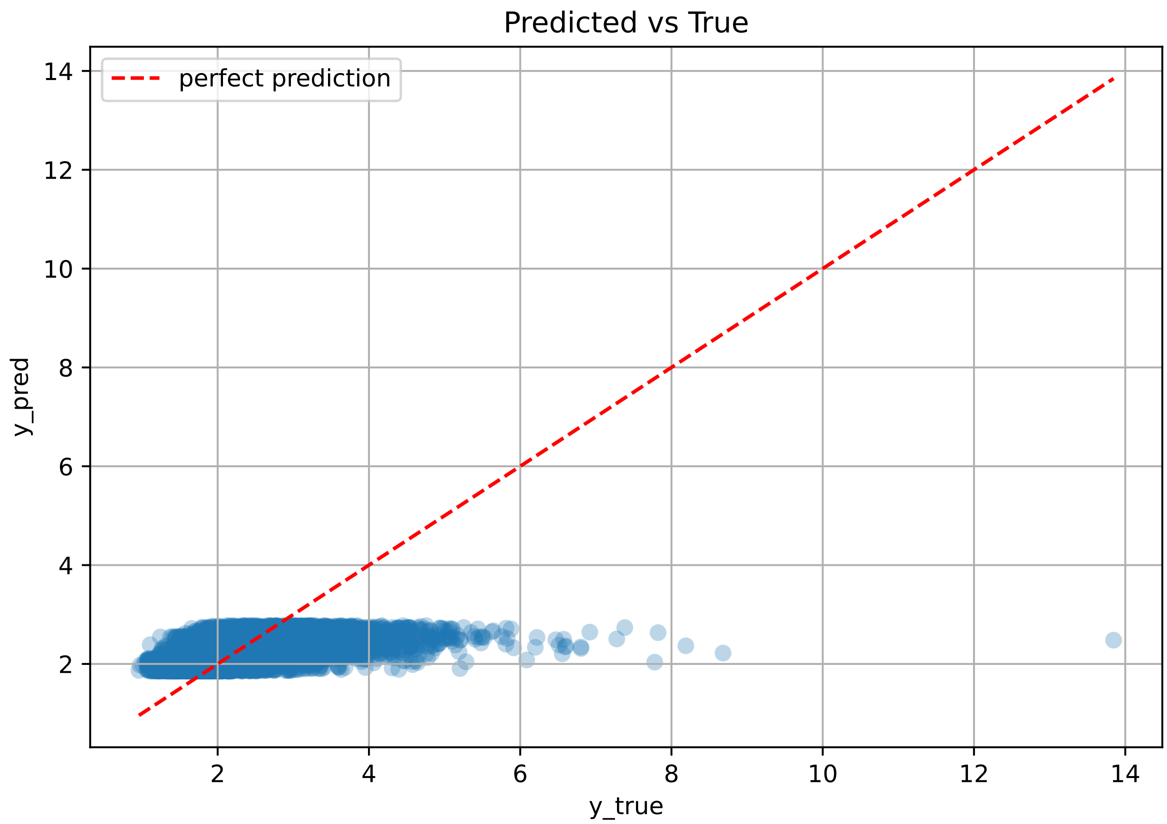
Rank 7: {'reg__n_estimators': 800, 'reg__max_depth': 4, 'reg__learning_rate': 0.018779773978831387, 'reg__num_leaves': 32, 'reg__subsample': 0.6952857316272034, 'reg__colsample_bytree': 0.7685179979170969, 'reg__min_child_samples': 37, 'reg__reg_alpha': 6.297955866759722, 'reg__reg_lambda': 30.065090045337982, 'reg__min_split_gain': 4.864568649441233, 'reg__path_smooth': 2.393298208727554, 'reg__min_child_weight': 0.006348668988602182, 'reg__feature_fraction': 0.6503142393755086, 'reg__bagging_fraction': 0.5143351464147736, 'reg__bagging_freq': 5, 'reg__max_bin': 252, 'reg__min_data_per_group': 198}

Rank 8: {'reg__n_estimators': 800, 'reg__max_depth': 4, 'reg__learning_rate': 0.021456617128157135, 'reg__num_leaves': 44, 'reg__subsample': 0.8912612952879837, 'reg__colsample_bytree': 0.7676039146761893, 'reg__min_child_samples': 40, 'reg__reg_alpha': 5.900481516685618, 'reg__reg_lambda': 31.579392873341703, 'reg__min_split_gain': 4.955830613119443, 'reg__path_smooth': 2.3936777956889843, 'reg__min_child_weight': 0.005729669897198472, 'reg__feature_fraction': 0.5564087959240578, 'reg__bagging_fraction': 0.5019344465545122, 'reg__bagging_freq': 5, 'reg__max_bin': 234, 'reg__min_data_per_group': 160}

Rank 9: {'reg__n_estimators': 700, 'reg__max_depth': 5, 'reg__learning_rate': 0.02482422911763604, 'reg__num_leaves': 50, 'reg__subsample': 0.7164415693622739, 'reg__colsample_bytree': 0.801075056984981, 'reg__min_child_samples': 45, 'reg__reg_alpha': 7.01112246565864, 'reg__reg_lambda': 25.15871876941574, 'reg__min_split_gain': 4.029116796430884, 'reg__path_smooth': 0.8442615396176403, 'reg__min_child_weight': 0.0226065633876541, 'reg__feature_fraction': 0.5465456849985977, 'reg__bagging_fraction': 0.5695947734034179, 'reg__bagging_freq': 6, 'reg__max_bin': 142, 'reg__min_data_per_group': 186}

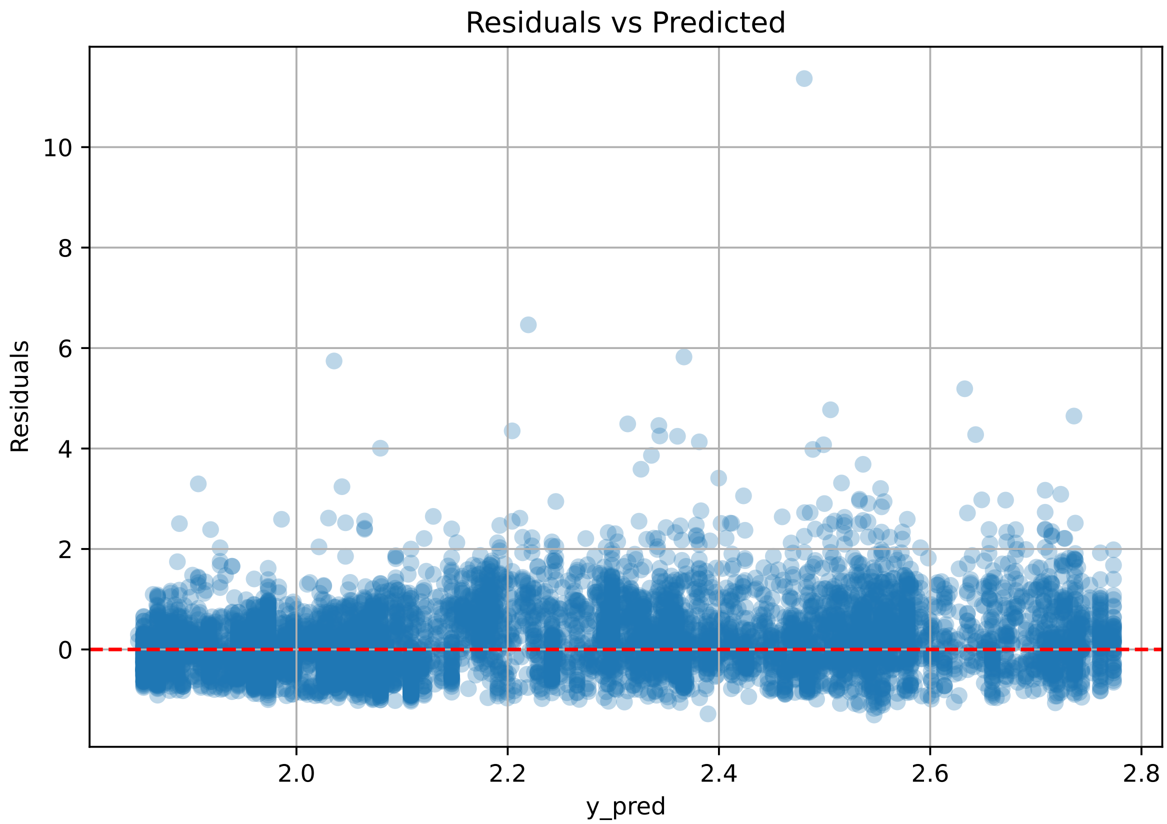
Rank 10: {'reg__n_estimators': 300, 'reg__max_depth': 4, 'reg__learning_rate': 0.021001012872946964, 'reg__num_leaves': 31, 'reg__subsample': 0.8761366865647061, 'reg__colsample_bytree': 0.585815119772607, 'reg__min_child_samples': 33, 'reg__reg_alpha': 5.768205206989291, 'reg__reg_lambda': 31.56963447340408, 'reg__min_split_gain': 1.302478288727053, 'reg__path_smooth': 2.445727547223524, 'reg__min_child_weight': 0.0010592876994020332, 'reg__feature_fraction': 0.8957161897946768, 'reg__bagging_fraction': 0.5034959048221284, 'reg__bagging_freq': 5, 'reg__max_bin': 180, 'reg__min_data_per_group': 155}

2.3.6 Scatter



Predicted vs True
perfect prediction
y_pred
y_true
2
4
6
8
10
12
14

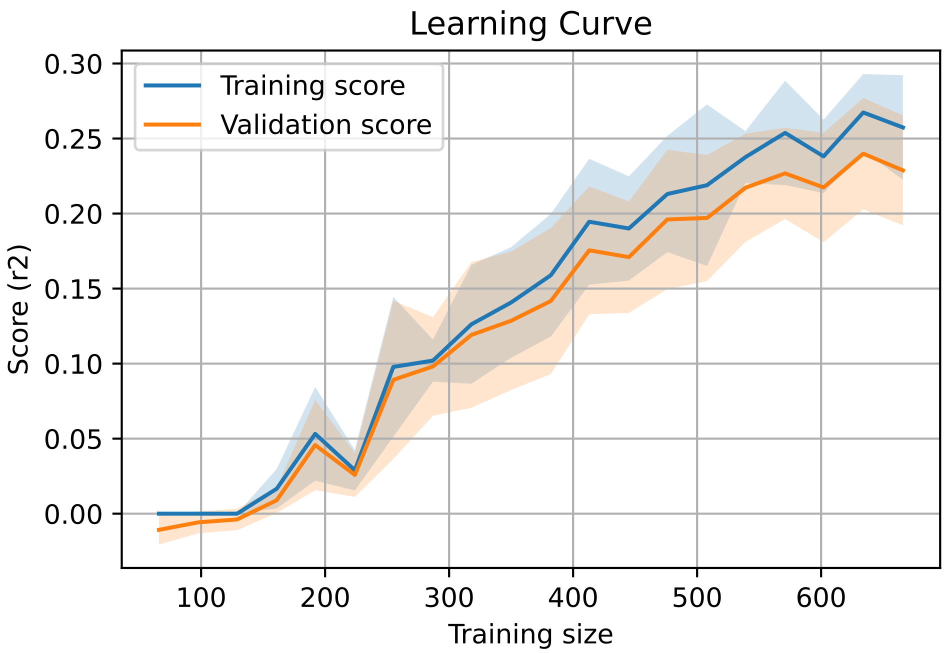
2.3.7 Scatter residuals



2.3.8 Learning curve — Group: 4

Len(group) : 1360, **Len(training)** : 952,

Len(training_real) : 667, **Len(test_of_training)** : 285, **Len(test)** : 408

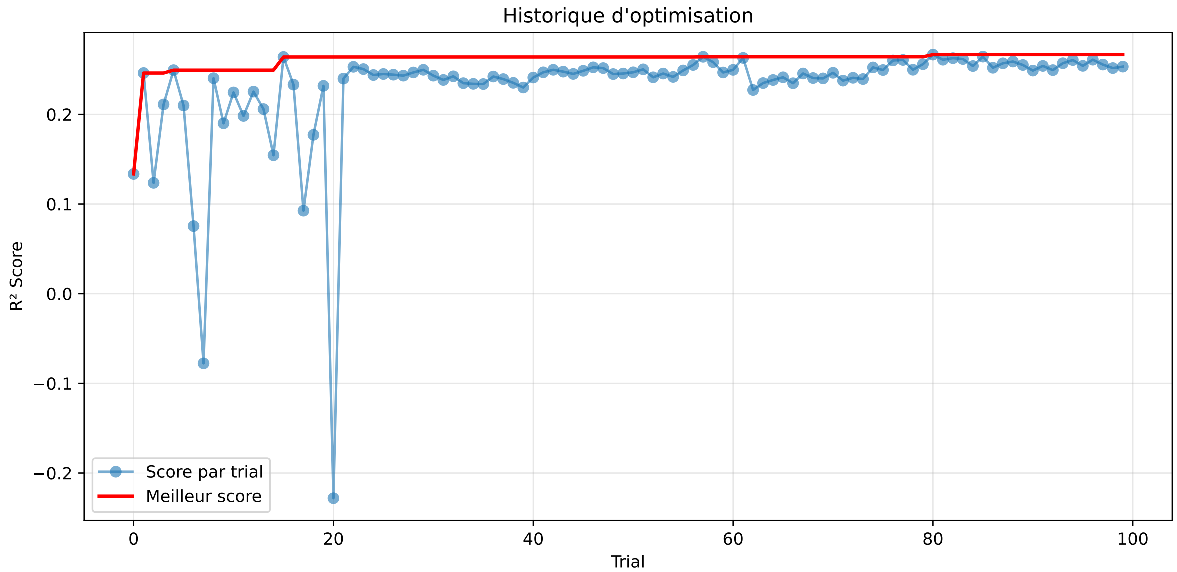


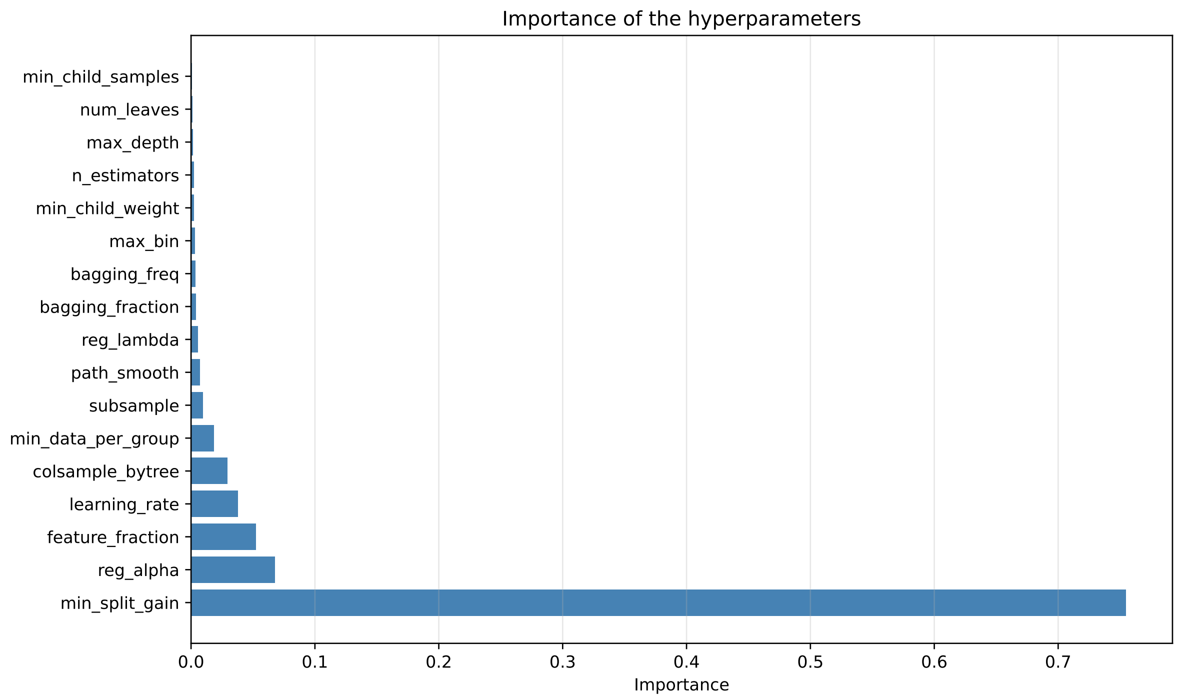
2.3.9 Gridsearch — Group: 5

Len(group) : 2726, **Len(training)** : 1909, **Len(test)** : 817

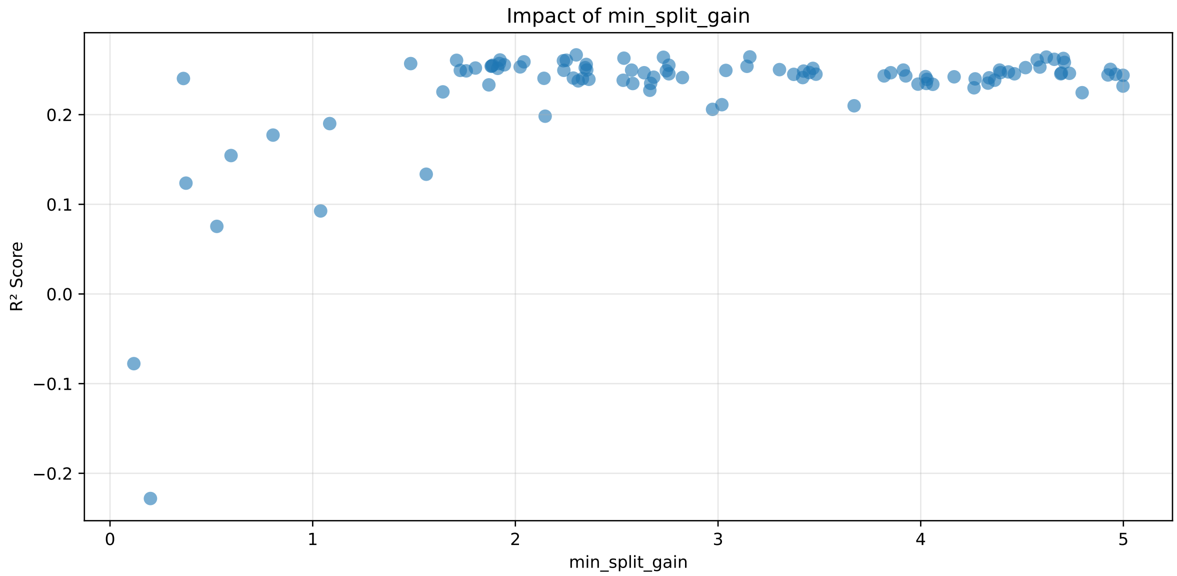
Distribution y_train vs y_test: y_train: mean=2.238, std=0.681, min=0.973, max=5.813 y_test: mean=2.212, std=0.659, min=1.071, max=4.858

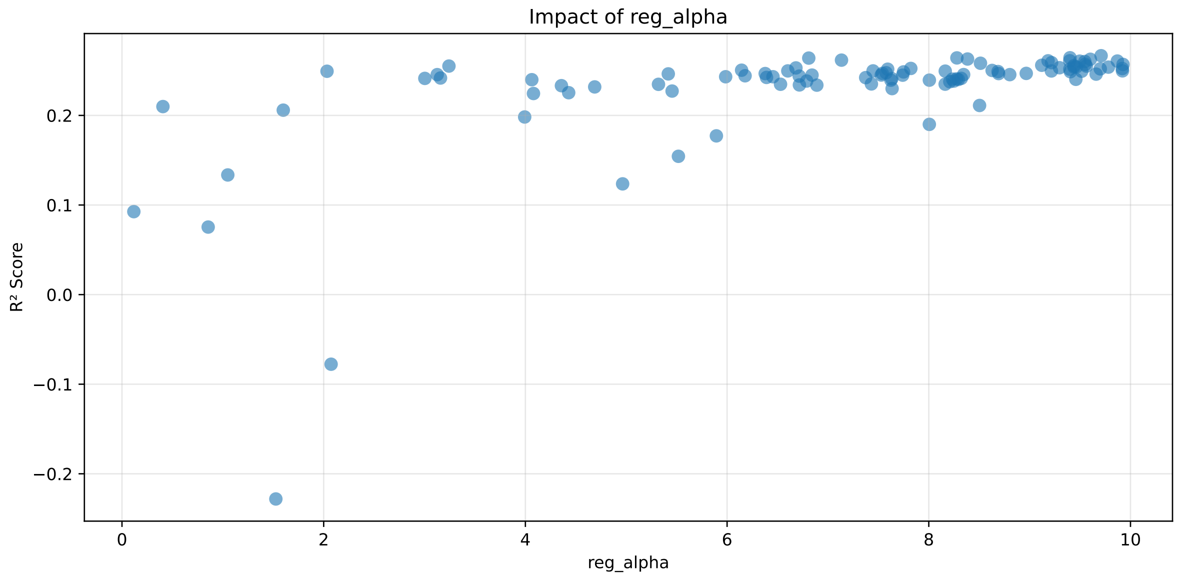
==================== TOP Importance FEATURES ==================== feature importance r_umidity 217 r_temperature 43 r_NH3 24 r_CO2 20 r_THI 3

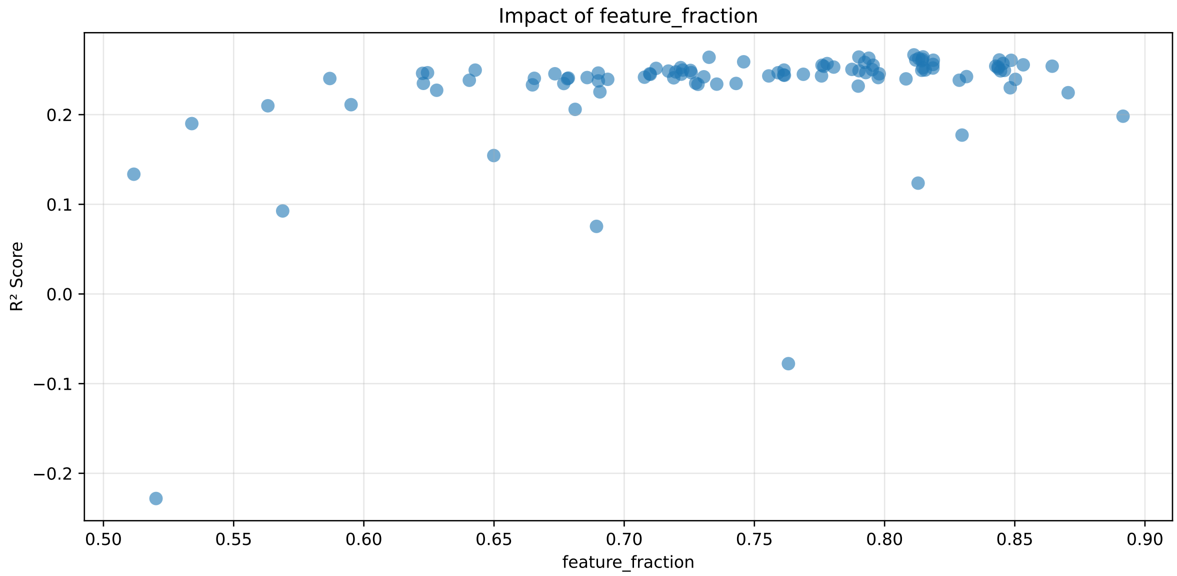




Paramètres avec >5% d'importance : ['min_split_gain', 'reg_alpha', 'feature_fraction']

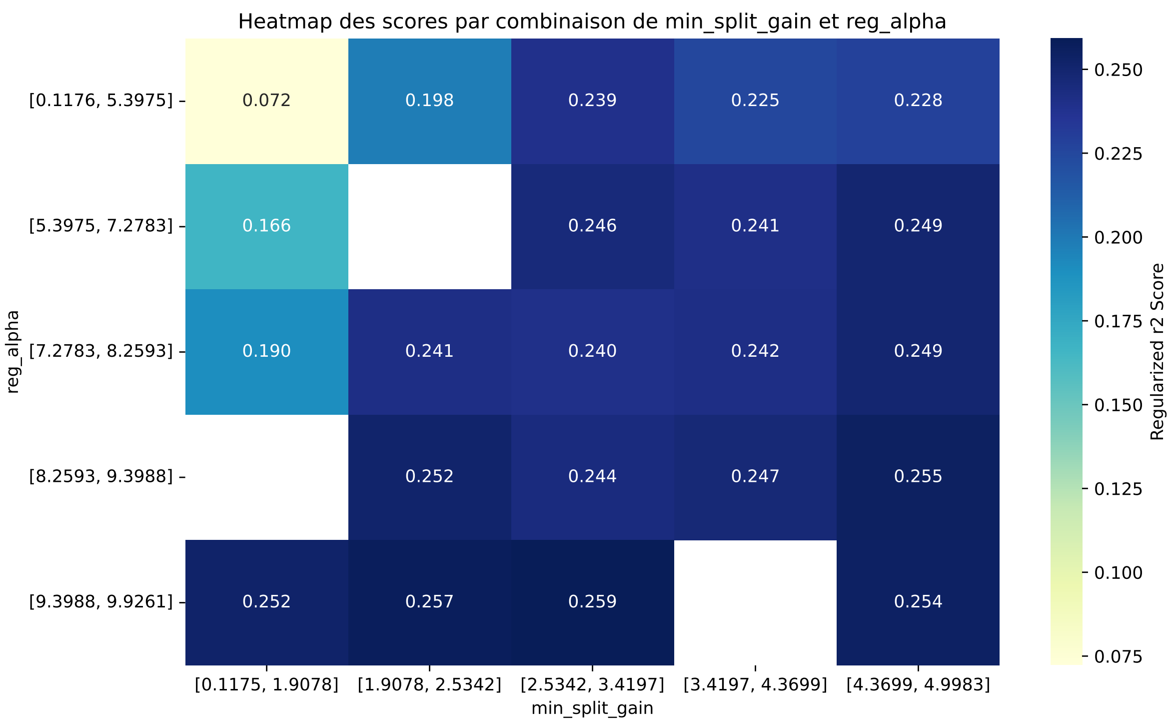

Impact of min_split_gain
R² Score
min_split_gain
0.2
0.1
0.0
−0.1
−0.2
0
1
2
3
4
5


Impact of reg_alpha
R² Score
reg_alpha
0.2
0.1
0.0
−0.1
−0.2
0
2
4
6
8
10



`C:\Users\titou\Desktop\UNICT\1- Gas sensor calibration\Python\machine_learning\workflow_main`

`The default value of observed=False is deprecated and will change to observed=True in a futu`



rank	mean_train_cv	mean_test_cv	test_score_cv	std_test_cv	penalized_score	test	diff	r_test	r_train
1	0.4053	0.3822	0.4347	0.0242	0.2666	0.4593	0.0771	1.2018	1.0605
2	0.4002	0.3776	0.43	0.021	0.2644	0.4397	0.0622	1.1646	1.0599
3	0.3969	0.3748	0.427	0.0213	0.2642	0.445	0.0702	1.1873	1.059
4	0.4113	0.3867	0.44	0.0262	0.264	0.4578	0.0711	1.1839	1.0635
5	0.4123	0.3874	0.4425	0.0259	0.263	0.4637	0.0764	1.1971	1.0642
6	0.3889	0.3679	0.4188	0.0199	0.2627	0.4258	0.0579	1.1575	1.0572
7	0.3872	0.3663	0.4163	0.0163	0.2617	0.4216	0.0553	1.151	1.0571
8	0.4142	0.3887	0.4436	0.0253	0.261	0.4419	0.0532	1.137	1.0657
9	0.3836	0.3631	0.4127	0.0155	0.261	0.4265	0.0633	1.1744	1.0563
10	0.4085	0.3839	0.4387	0.0251	0.2607	0.46	0.0761	1.1982	1.0642
100	0.6203	0.4789	0.5552	0.0274	-0.2281	0.5805	0.1016	1.2122	1.2953

Hyperparameters:

Rank 1: {'reg__n_estimators': 800, 'reg__max_depth': 4, 'reg__learning_rate': 0.017388293865671256, 'reg__num_leaves': 39, 'reg__subsample': 0.6774333943908856, 'reg__colsample_bytree': 0.5387337949903744, 'reg__min_child_samples': 48, 'reg__reg_alpha': 9.708806730381388, 'reg__reg_lambda': 38.324639415612886, 'reg__min_split_gain': 2.3003839682682385, 'reg__path_smooth': 0.20722064008148508, 'reg__min_child_weight': 1.1536238385168909, 'reg__feature_fraction': 0.8112712912239914, 'reg__bagging_fraction': 0.5099150134818184, 'reg__bagging_freq': 5, 'reg__max_bin': 150, 'reg__min_data_per_group': 179}

Rank 2: {'reg__n_estimators': 800, 'reg__max_depth': 5, 'reg__learning_rate': 0.017492134306770397, 'reg__num_leaves': 39, 'reg__subsample': 0.6786614903930008, 'reg__colsample_bytree': 0.70168393408529, 'reg__min_child_samples': 32, 'reg__reg_alpha': 9.400226651207968, 'reg__reg_lambda': 22.857939693542974, 'reg__min_split_gain': 3.1569315699361686, 'reg__path_smooth': 1.5921093800405277, 'reg__min_child_weight': 0.6000547817365537, 'reg__feature_fraction': 0.8147513474839415, 'reg__bagging_fraction': 0.5236533528219338, 'reg__bagging_freq': 6, 'reg__max_bin': 114, 'reg__min_data_per_group': 91}

Rank 3: {'reg__n_estimators': 800, 'reg__max_depth': 8, 'reg__learning_rate': 0.02010515833097892, 'reg__num_leaves': 44, 'reg__subsample': 0.6322708311710371, 'reg__colsample_bytree': 0.6029622485174009, 'reg__min_child_samples': 38, 'reg__reg_alpha': 8.278732704282337, 'reg__reg_lambda': 41.381160255190615, 'reg__min_split_gain': 4.6198490300119115, 'reg__path_smooth': 1.030227399725848, 'reg__min_child_weight': 0.4069033499196606, 'reg__feature_fraction': 0.790122052531758, 'reg__bagging_fraction': 0.598382788279117, 'reg__bagging_freq': 4, 'reg__max_bin': 106, 'reg__min_data_per_group': 98}

Rank 4: {'reg__n_estimators': 800, 'reg__max_depth': 8, 'reg__learning_rate': 0.020728686696857342, 'reg__num_leaves': 62, 'reg__subsample': 0.6875357269465758, 'reg__colsample_bytree': 0.6400166510485085, 'reg__min_child_samples': 32, 'reg__reg_alpha': 6.809915896279893, 'reg__reg_lambda': 25.153531023412157, 'reg__min_split_gain': 2.7305776622143596, 'reg__path_smooth': 0.20970913935415392, 'reg__min_child_weight': 9.620960517759366, 'reg__feature_fraction': 0.7325895008313111, 'reg__bagging_fraction': 0.5126739593752216, 'reg__bagging_freq': 4, 'reg__max_bin': 234, 'reg__min_data_per_group': 105}

Rank 5: {'reg__n_estimators': 300, 'reg__max_depth': 8, 'reg__learning_rate': 0.019587194687250554, 'reg__num_leaves': 36, 'reg__subsample': 0.628740403509535, 'reg__colsample_bytree': 0.7177954500118789, 'reg__min_child_samples': 16, 'reg__reg_alpha': 8.384453363009548, 'reg__reg_lambda': 40.99573328066971, 'reg__min_split_gain': 2.5356438864671933, 'reg__path_smooth': 1.1621686782710552, 'reg__min_child_weight': 0.10418774010448598, 'reg__feature_fraction': 0.7939425502693558, 'reg__bagging_fraction': 0.6633320363040716, 'reg__bagging_freq': 4, 'reg__max_bin': 106, 'reg__min_data_per_group': 152}

Rank 6: {'reg__n_estimators': 800, 'reg__max_depth': 6, 'reg__learning_rate': 0.016626781098319516, 'reg__num_leaves': 40, 'reg__subsample': 0.671898397175169, 'reg__colsample_bytree': 0.6612697560878283, 'reg__min_child_samples': 49, 'reg__reg_alpha': 9.602015274888327, 'reg__reg_lambda': 38.632530024422245, 'reg__min_split_gain': 4.704058596976071, 'reg__path_smooth': 0.28553026732922543, 'reg__min_child_weight': 1.1428236030951375, 'reg__feature_fraction': 0.8131261104764452, 'reg__bagging_fraction': 0.5795279339963357, 'reg__bagging_freq': 5, 'reg__max_bin': 116, 'reg__min_data_per_group': 87}

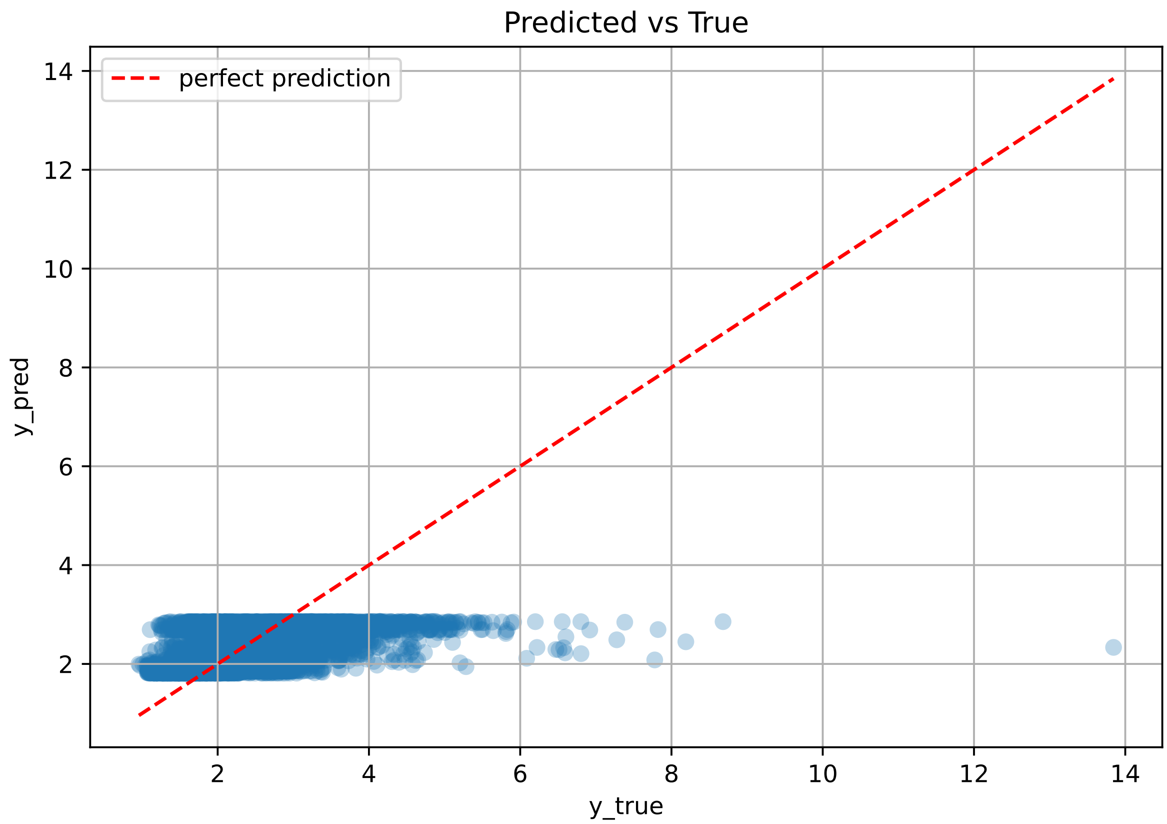
Rank 7: {'reg__n_estimators': 800, 'reg__max_depth': 5, 'reg__learning_rate': 0.017489059255515596, 'reg__num_leaves': 41, 'reg__subsample': 0.6842748306822134, 'reg__colsample_bytree': 0.7025037180338006, 'reg__min_child_samples': 16, 'reg__reg_alpha': 7.1344756679067185, 'reg__reg_lambda': 38.09592143133578, 'reg__min_split_gain': 4.658959716589575, 'reg__path_smooth': 0.18987991736166143, 'reg__min_child_weight': 1.4088568737959197, 'reg__feature_fraction': 0.8145356689966053, 'reg__bagging_fraction': 0.5151583666445374, 'reg__bagging_freq': 5, 'reg__max_bin': 115, 'reg__min_data_per_group': 91}

Rank 8: {'reg__n_estimators': 800, 'reg__max_depth': 4, 'reg__learning_rate': 0.016272578184311492, 'reg__num_leaves': 42, 'reg__subsample': 0.6454237702543515, 'reg__colsample_bytree': 0.5261460111818101, 'reg__min_child_samples': 48, 'reg__reg_alpha': 9.397858459021059, 'reg__reg_lambda': 36.57287459809363, 'reg__min_split_gain': 1.924122702939087, 'reg__path_smooth': 2.148725071084157, 'reg__min_child_weight': 4.359869374061099, 'reg__feature_fraction': 0.8440592443379342, 'reg__bagging_fraction': 0.5209828231840877, 'reg__bagging_freq': 6, 'reg__max_bin': 166, 'reg__min_data_per_group': 200}

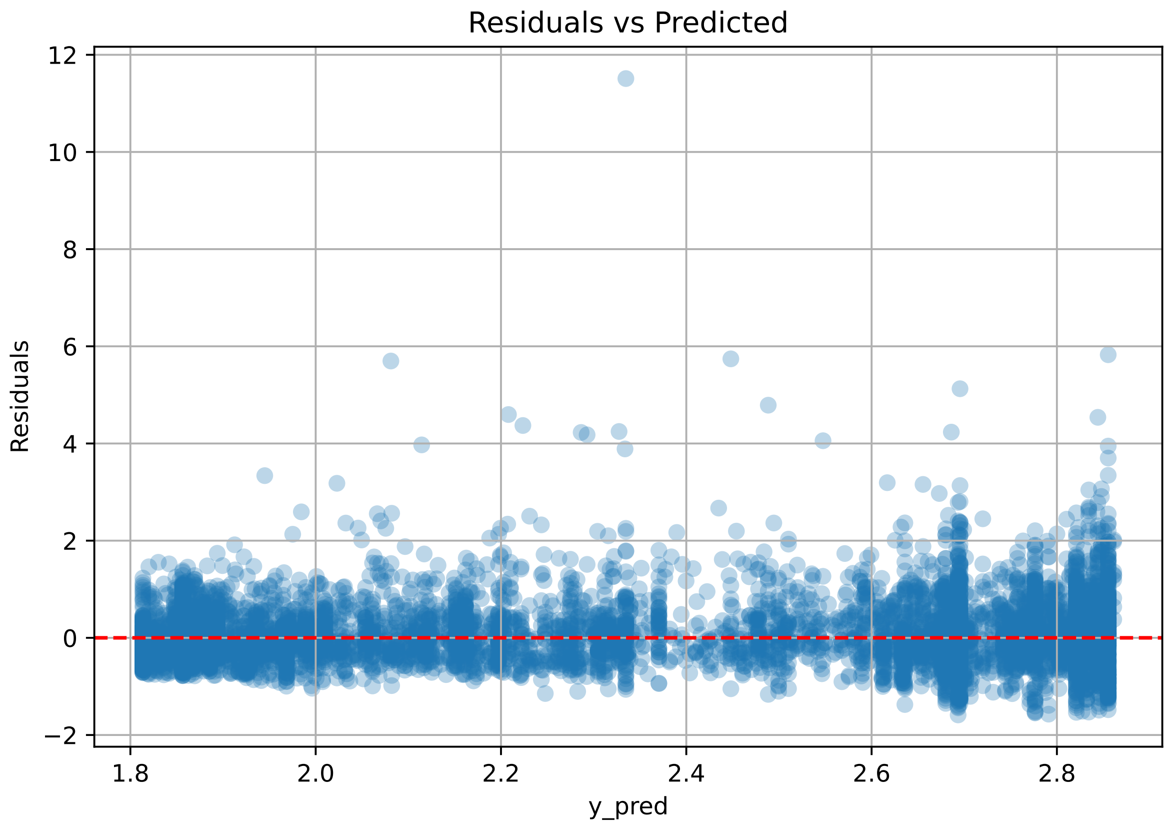
Rank 9: {'reg__n_estimators': 800, 'reg__max_depth': 5, 'reg__learning_rate': 0.01739056459079341, 'reg__num_leaves': 53, 'reg__subsample': 0.6722636979697311, 'reg__colsample_bytree': 0.661886084120904, 'reg__min_child_samples': 49, 'reg__reg_alpha': 9.183057940652162, 'reg__reg_lambda': 38.12539026611289, 'reg__min_split_gain': 4.574711557668593, 'reg__path_smooth': 1.6888004857952414, 'reg__min_child_weight': 1.179088515467469, 'reg__feature_fraction': 0.8119633971634569, 'reg__bagging_fraction': 0.5022526462774879, 'reg__bagging_freq': 5, 'reg__max_bin': 114, 'reg__min_data_per_group': 177}

Rank 10: {'reg__n_estimators': 800, 'reg__max_depth': 5, 'reg__learning_rate': 0.017629236242987574, 'reg__num_leaves': 41, 'reg__subsample': 0.6796047422689878, 'reg__colsample_bytree': 0.5338468792466566, 'reg__min_child_samples': 48, 'reg__reg_alpha': 9.870032676377757, 'reg__reg_lambda': 38.162684957027025, 'reg__min_split_gain': 2.2514894068538567, 'reg__path_smooth': 1.3014516041961333, 'reg__min_child_weight': 0.9840890235058809, 'reg__feature_fraction': 0.8187264171787894, 'reg__bagging_fraction': 0.5747617010880866, 'reg__bagging_freq': 5, 'reg__max_bin': 156, 'reg__min_data_per_group': 177}

2.3.10 Scatter



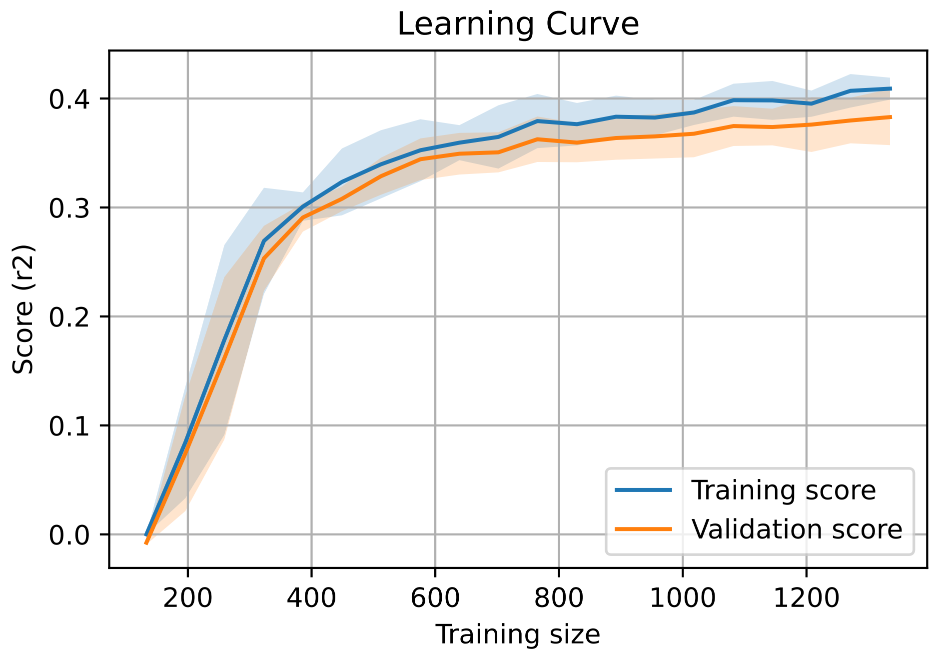
2.3.11 Scatter residuals



2.3.12 Learning curve — Group: 5

Len(group) : 2726, **Len(training)** : 1909,

Len(training_real) : 1337, **Len(test_of_training)** : 572, **Len(test)** : 817

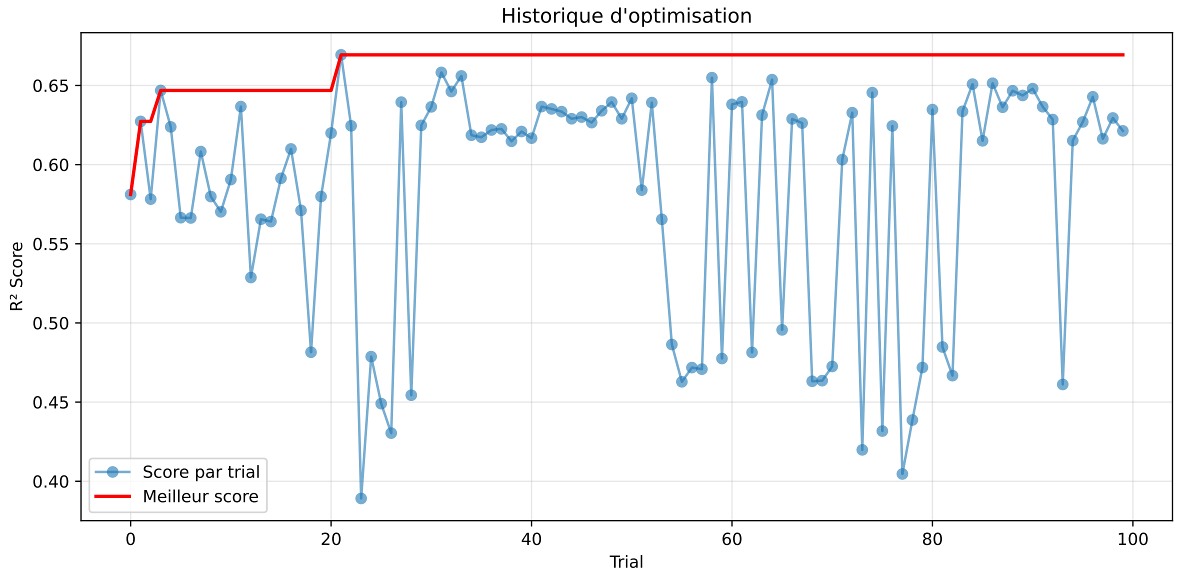


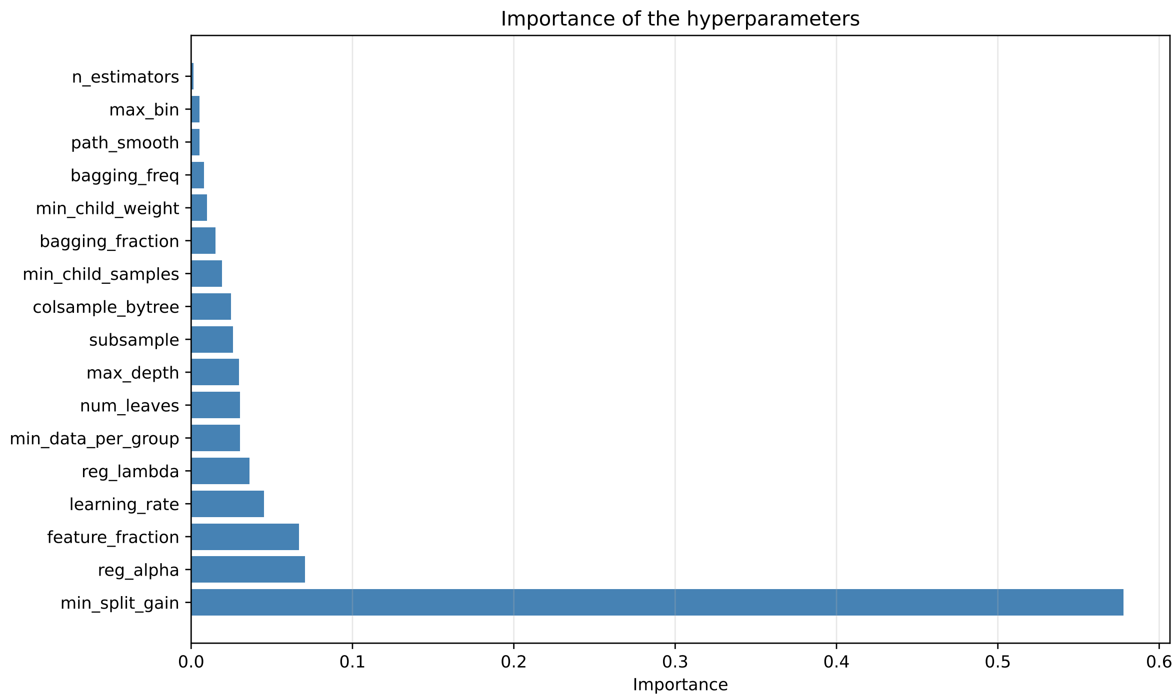
2.3.13 Gridsearch — Group: 11

Len(group) : 2726, **Len(training)** : 1909, **Len(test)** : 817

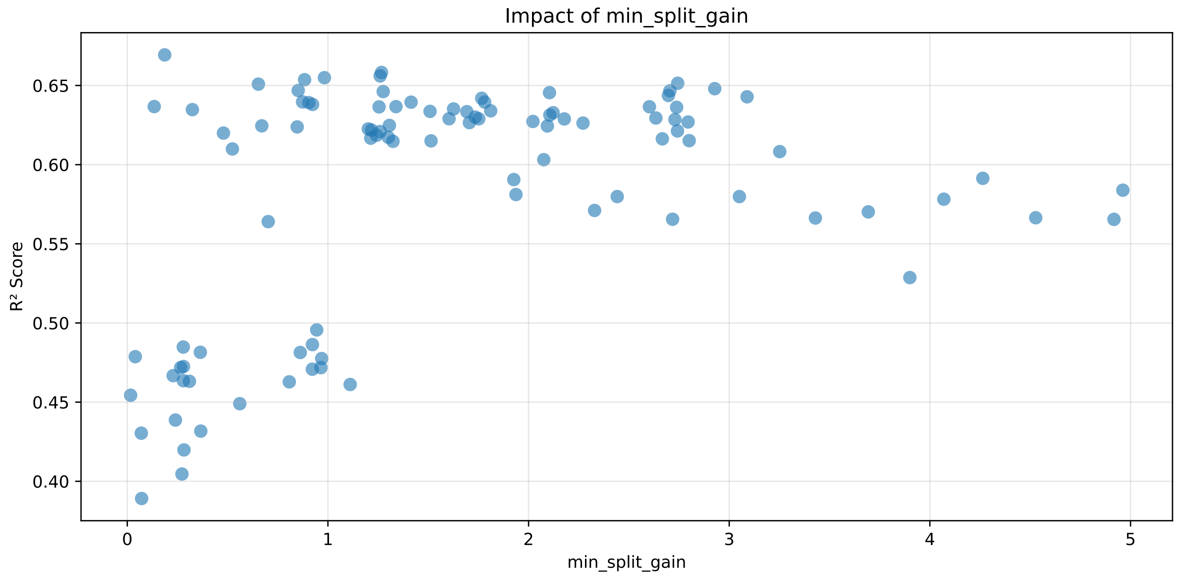
Distribution y_train vs y_test: y_train: mean=2.540, std=0.859, min=1.150, max=8.682 y_test: mean=2.561, std=0.947, min=1.120, max=13.845

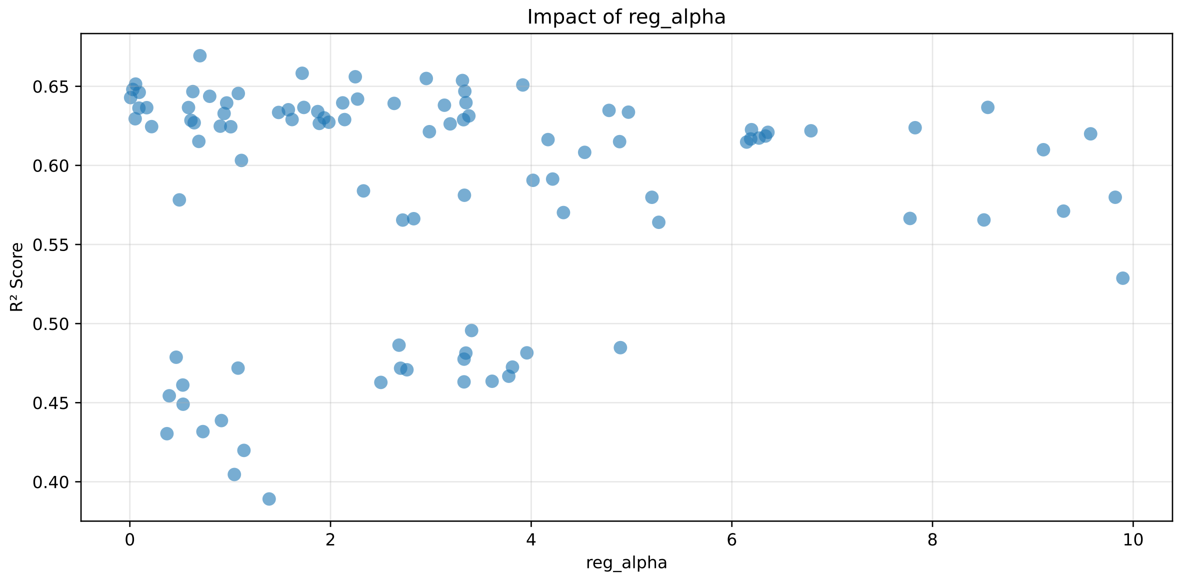
==================== TOP Importance FEATURES ==================== feature importance r_NH3 1177 r_umidity 1025 r_CO2 808 r_THI 625 r_temperature 471

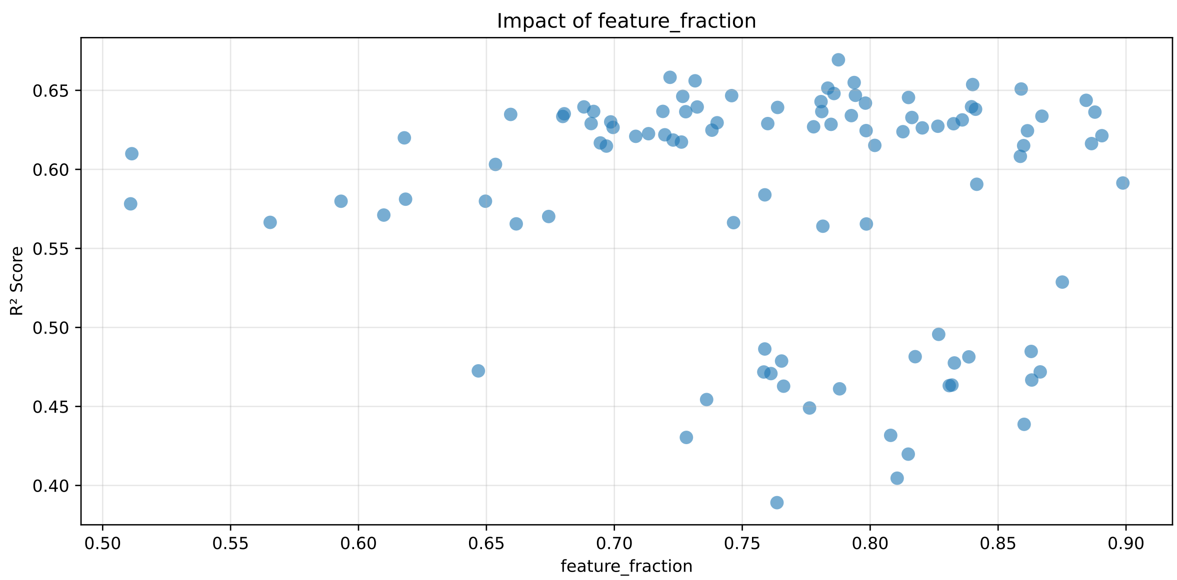




Paramètres avec >5% d'importance : ['min_split_gain', 'reg_alpha', 'feature_fraction']

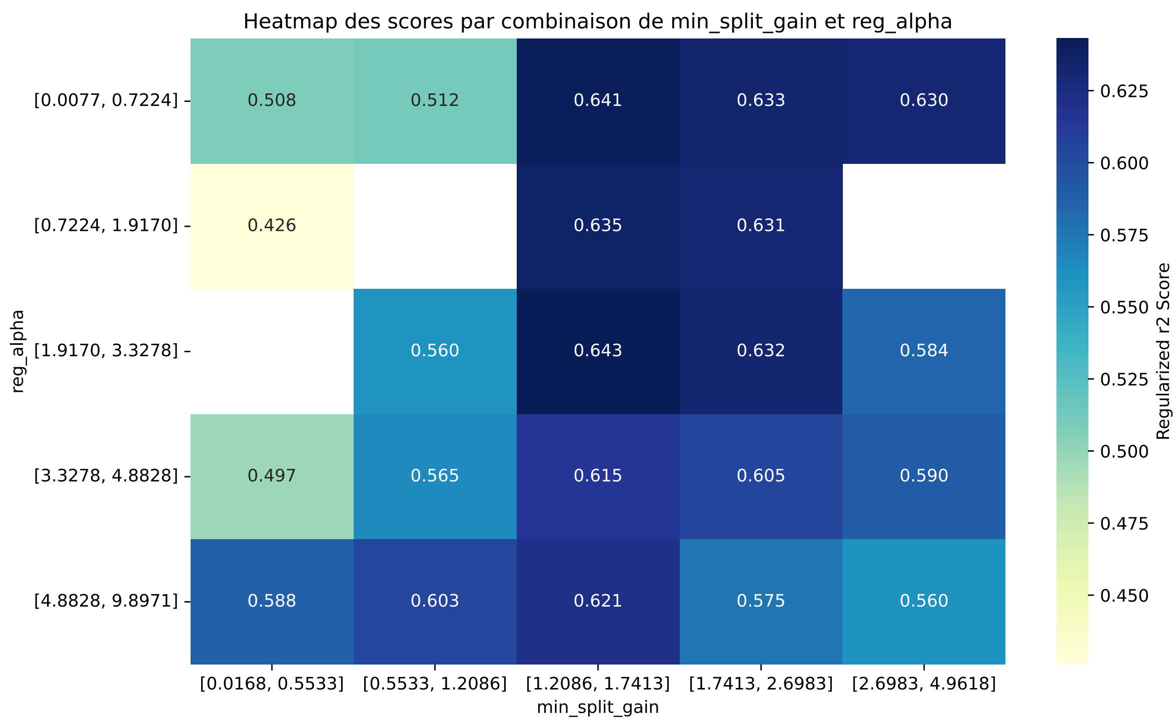

Impact of min_split_gain
R² Score
0.40
0.45
0.50
0.55
0.60
0.65
0
1
2
3
4
5
min_split_gain


Impact of reg_alpha
R² Score
0.40
0.45
0.50
0.55
0.60
0.65
0
2
4
6
8
10
reg_alpha




```
C:\Users\titou\Desktop\UNICT\1- Gas sensor calibration\Python\machine_learning\workflow_main

The default value of observed=False is deprecated and will change to observed=True in a futu
```



rank	mean_train_cv	mean_test_cv	test_score_cv	std_test_cv	penalized_score	test	diff	r_test	r_train
1	0.7013	0.6693	0.5685	0.0133	0.6693	0.5736	-0.0958	0.8569	1.0478
2	0.6904	0.6582	0.5668	0.0132	0.6582	0.5827	-0.0755	0.8853	1.0489
3	0.6883	0.656	0.5643	0.0122	0.656	0.5815	-0.0746	0.8863	1.0492
4	0.6792	0.6549	0.5575	0.0151	0.6549	0.5725	-0.0824	0.8742	1.0371
5	0.6842	0.6537	0.5572	0.0108	0.6537	0.5749	-0.0788	0.8795	1.0468
6	0.6754	0.6514	0.5567	0.0139	0.6514	0.5672	-0.0842	0.8708	1.0368
7	0.6789	0.6509	0.5526	0.0108	0.6509	0.5648	-0.0861	0.8678	1.0431
8	0.6732	0.648	0.5515	0.0123	0.648	0.566	-0.082	0.8735	1.0389
9	0.6753	0.6469	0.5544	0.0129	0.6469	0.572	-0.0748	0.8843	1.044
10	0.6697	0.6467	0.5522	0.0097	0.6467	0.5664	-0.0802	0.8759	1.0356
---	---	---	---	---	---	---	---	---	---
100	0.7412	0.6825	0.5764	0.0095	0.3891	0.5883	-0.0942	0.862	1.086

Hyperparameters:

Rank 1: {'reg__n_estimators': 800, 'reg__max_depth': 3, 'reg__learning_rate': 0.010273656034888619, 'reg__num_leaves': 63, 'reg__subsample': 0.7389061799443557, 'reg__colsample_bytree': 0.5242785013016644, 'reg__min_child_samples': 24, 'reg__reg_alpha': 0.6984789622910534, 'reg__reg_lambda': 15.551077825521336, 'reg__min_split_gain': 0.18642611478269133, 'reg__path_smooth': 0.27491424822000354, 'reg__min_child_weight': 5.621765856113254, 'reg__feature_fraction': 0.7875844404539123, 'reg__bagging_fraction': 0.5029540640880655, 'reg__bagging_freq': 5, 'reg__max_bin': 69, 'reg__min_data_per_group': 56}

Rank 2: {'reg__n_estimators': 500, 'reg__max_depth': 5, 'reg__learning_rate': 0.021562711112460452, 'reg__num_leaves': 56, 'reg__subsample': 0.7546726997783914, 'reg__colsample_bytree': 0.5854447125135174, 'reg__min_child_samples': 16, 'reg__reg_alpha': 1.7176923141143585, 'reg__reg_lambda': 21.1837593822792, 'reg__min_split_gain': 1.2669920028686505, 'reg__path_smooth': 4.276393117308949, 'reg__min_child_weight': 5.063227621824639, 'reg__feature_fraction': 0.7217748561081828, 'reg__bagging_fraction': 0.7733615586016966, 'reg__bagging_freq': 1, 'reg__max_bin': 254, 'reg__min_data_per_group': 103}

Rank 3: {'reg__n_estimators': 500, 'reg__max_depth': 7, 'reg__learning_rate': 0.021917298610023873, 'reg__num_leaves': 24, 'reg__subsample': 0.5438093648521146, 'reg__colsample_bytree': 0.59154465531964, 'reg__min_child_samples': 19, 'reg__reg_alpha': 2.2469697211230404, 'reg__reg_lambda': 20.054456851600808, 'reg__min_split_gain': 1.2602486935534376, 'reg__path_smooth': 4.2751364391201, 'reg__min_child_weight': 5.536696825715495, 'reg__feature_fraction': 0.7315656711721393, 'reg__bagging_fraction': 0.8130783276098437, 'reg__bagging_freq': 1, 'reg__max_bin': 249, 'reg__min_data_per_group': 99}

Rank 4: {’reg__n_estimators’: 800, ’reg__max_depth’: 6, ’reg__learning_rate’: 0.02680881185791052, ’reg__num_leaves’: 22, ’reg__subsample’: 0.8024811415683993, ’reg__colsample_bytree’: 0.5583475487836856, ’reg__min_child_samples’: 22, ’reg__reg_alpha’: 2.954536563894224, ’reg__reg_lambda’: 6.825981601487566, ’reg__min_split_gain’: 0.9824831736478286, ’reg__path_smooth’: 4.852105453056005, ’reg__min_child_weight’: 0.05605677746693938, ’reg__feature_fraction’: 0.7937434029425795, ’reg__bagging_fraction’: 0.6353119824124889, ’reg__bagging_freq’: 3, ’reg__max_bin’: 210, ’reg__min_data_per_group’: 86}

Rank 5: {’reg__n_estimators’: 800, ’reg__max_depth’: 6, ’reg__learning_rate’: 0.03382674654224147, ’reg__num_leaves’: 33, ’reg__subsample’: 0.7970065295088202, ’reg__colsample_bytree’: 0.8982499036832369, ’reg__min_child_samples’: 34, ’reg__reg_alpha’: 3.315247951165426, ’reg__reg_lambda’: 16.708370401498073, ’reg__min_split_gain’: 0.8835994114770434, ’reg__path_smooth’: 4.2932360753106185, ’reg__min_child_weight’: 0.24903244948284547, ’reg__feature_fraction’: 0.8400512576063024, ’reg__bagging_fraction’: 0.8707679308193494, ’reg__bagging_freq’: 3, ’reg__max_bin’: 211, ’reg__min_data_per_group’: 89}

Rank 6: {’reg__n_estimators’: 400, ’reg__max_depth’: 4, ’reg__learning_rate’: 0.010887356237860427, ’reg__num_leaves’: 38, ’reg__subsample’: 0.7727626523169744, ’reg__colsample_bytree’: 0.8558713454535879, ’reg__min_child_samples’: 14, ’reg__reg_alpha’: 0.05734987451072082, ’reg__reg_lambda’: 3.6690403560059903, ’reg__min_split_gain’: 2.743220672149606, ’reg__path_smooth’: 3.045226090441489, ’reg__min_child_weight’: 0.018252835203787506, ’reg__feature_fraction’: 0.7834350466709243, ’reg__bagging_fraction’: 0.6989095415816291, ’reg__bagging_freq’: 2, ’reg__max_bin’: 246, ’reg__min_data_per_group’: 114}

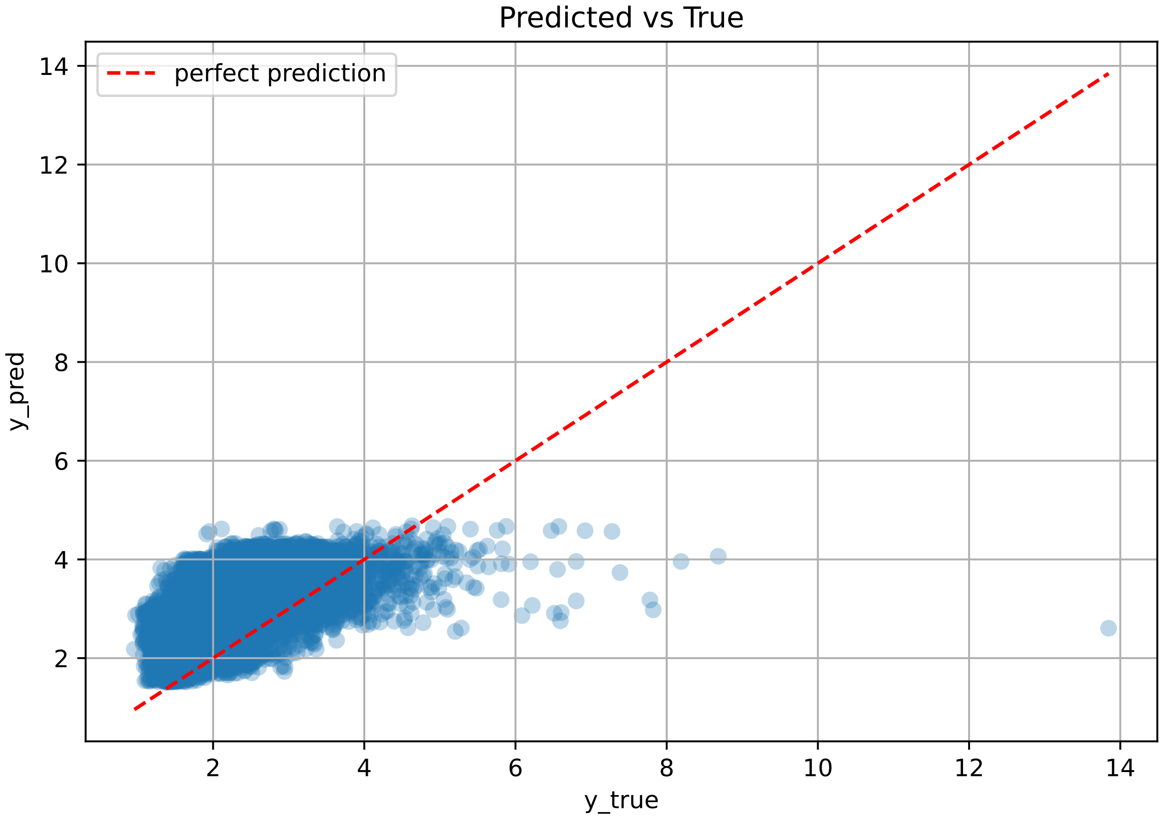
Rank 7: {’reg__n_estimators’: 400, ’reg__max_depth’: 4, ’reg__learning_rate’: 0.010880443558191684, ’reg__num_leaves’: 38, ’reg__subsample’: 0.8308878753824742, ’reg__colsample_bytree’: 0.8596834994936404, ’reg__min_child_samples’: 37, ’reg__reg_alpha’: 3.916406900554245, ’reg__reg_lambda’: 0.5261848065471195, ’reg__min_split_gain’: 0.6534173457614054, ’reg__path_smooth’: 3.4802526063295525, ’reg__min_child_weight’: 4.683971693474173, ’reg__feature_fraction’: 0.8590254030841374, ’reg__bagging_fraction’: 0.7348431808230831, ’reg__bagging_freq’: 1, ’reg__max_bin’: 236, ’reg__min_data_per_group’: 62}

Rank 8: {’reg__n_estimators’: 800, ’reg__max_depth’: 5, ’reg__learning_rate’: 0.010960699125410487, ’reg__num_leaves’: 37, ’reg__subsample’: 0.7445889613996696, ’reg__colsample_bytree’: 0.5264213818796625, ’reg__min_child_samples’: 20, ’reg__reg_alpha’: 0.02922263071687882, ’reg__reg_lambda’: 0.45221653437221043, ’reg__min_split_gain’: 2.927663837073486, ’reg__path_smooth’: 3.9884604890008104, ’reg__min_child_weight’: 4.651118070739882, ’reg__feature_fraction’: 0.7858449097272028, ’reg__bagging_fraction’: 0.6925123886941047, ’reg__bagging_freq’: 2, ’reg__max_bin’: 246, ’reg__min_data_per_group’: 77}

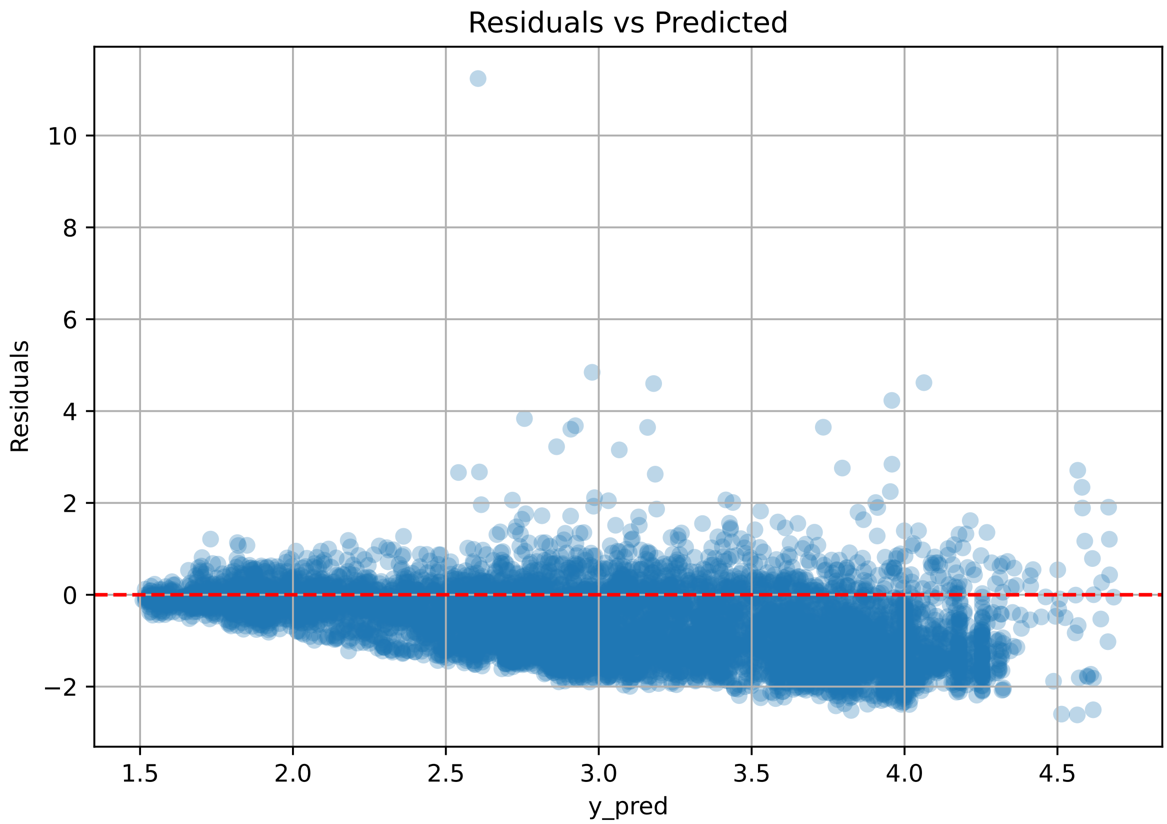
Rank 9: {'reg___n_estimators': 700, 'reg___max_depth': 7, 'reg___learning_rate': 0.010346513308696445, 'reg___num_leaves': 31, 'reg___subsample': 0.572985420552889, 'reg___colsample_bytree': 0.8462229144907608, 'reg___min_child_samples': 32, 'reg___reg_alpha': 3.3385278396463347, 'reg___reg_lambda': 32.53339346995301, 'reg___min_split_gain': 0.8518286488911259, 'reg___path_smooth': 3.1706691369875664, 'reg___min_child_weight': 1.420395848091179, 'reg___feature_fraction': 0.7942252000905301, 'reg___bagging_fraction': 0.8204615535909887, 'reg___bagging_freq': 2, 'reg___max_bin': 208, 'reg___min_data_per_group': 110}

Rank 10: {'reg___n_estimators': 800, 'reg___max_depth': 5, 'reg___learning_rate': 0.02379192681111852, 'reg___num_leaves': 38, 'reg___subsample': 0.7432181428763558, 'reg___colsample_bytree': 0.599366929257294, 'reg___min_child_samples': 15, 'reg___reg_alpha': 0.6277737677203592, 'reg___reg_lambda': 1.0675927961463856, 'reg___min_split_gain': 2.7041748683004627, 'reg___path_smooth': 2.45944689669376, 'reg___min_child_weight': 4.720214386091455, 'reg___feature_fraction': 0.7458518978350451, 'reg___bagging_fraction': 0.6870138338376882, 'reg___bagging_freq': 2, 'reg___max_bin': 64, 'reg___min_data_per_group': 93}

2.3.14 Scatter



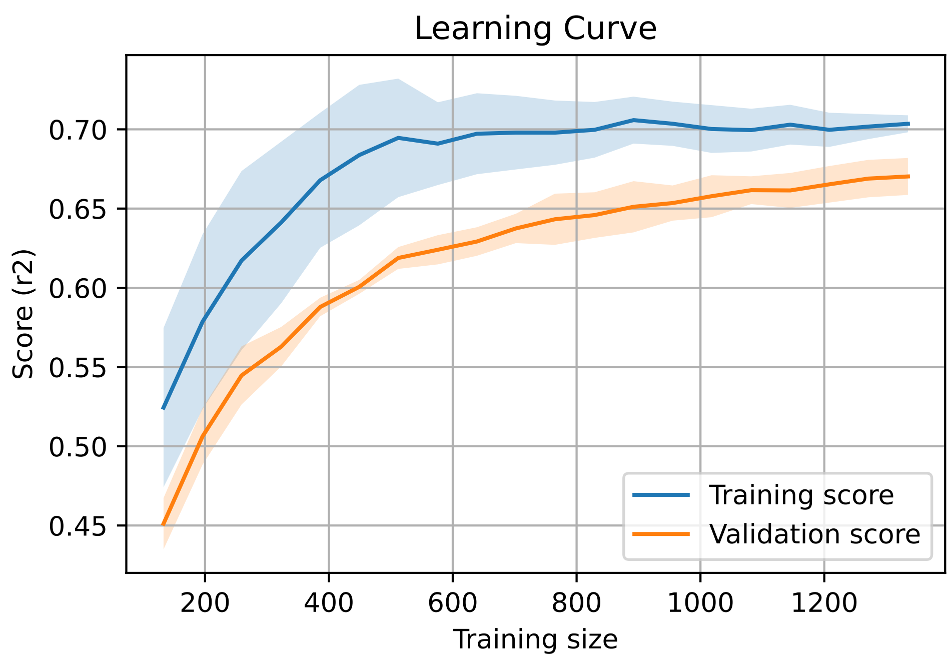
2.3.15 Scatter residuals



2.3.16 Learning curve — Group: 11

Len(group) : 2726, **Len(training)** : 1909,

Len(training_real) : 1337, **Len(test_of_training)** : 572, **Len(test)** : 817



2.4 Sumup-table

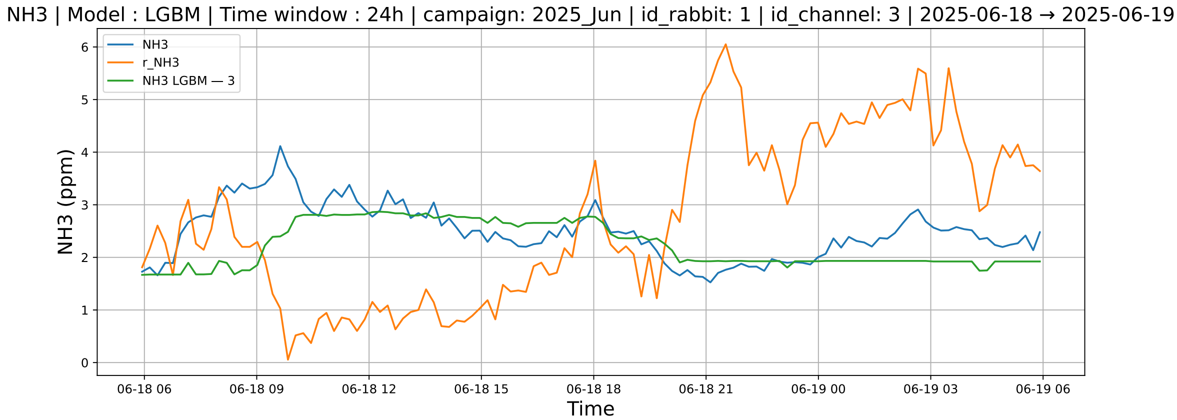
model	group	Len training	Len test	Len total	mean_train_cv	mean_test_cv	test_cv	std_test_cv	pen_score	test	diff	r_test	r_train
LGBM	3	950	406	1356	0.574	0.521	0.595	0.045	0.2564	0.64	0.118	1.227	1.102
LGBM	4	952	408	1360	0.261	0.237	0.238	0.037	0.1162	0.293	0.056	1.238	1.102
LGBM	5	1909	817	2726	0.405	0.382	0.435	0.024	0.2666	0.459	0.077	1.202	1.06
LGBM	11	1909	817	2726	0.701	0.669	0.568	0.013	0.6693	0.574	-0.096	0.857	1.048
LGBM	MEAN	5720	2448	8168	0.485	0.452	0.459	0.03	0.327	0.491	0.039	1.131	1.078

3 Plot

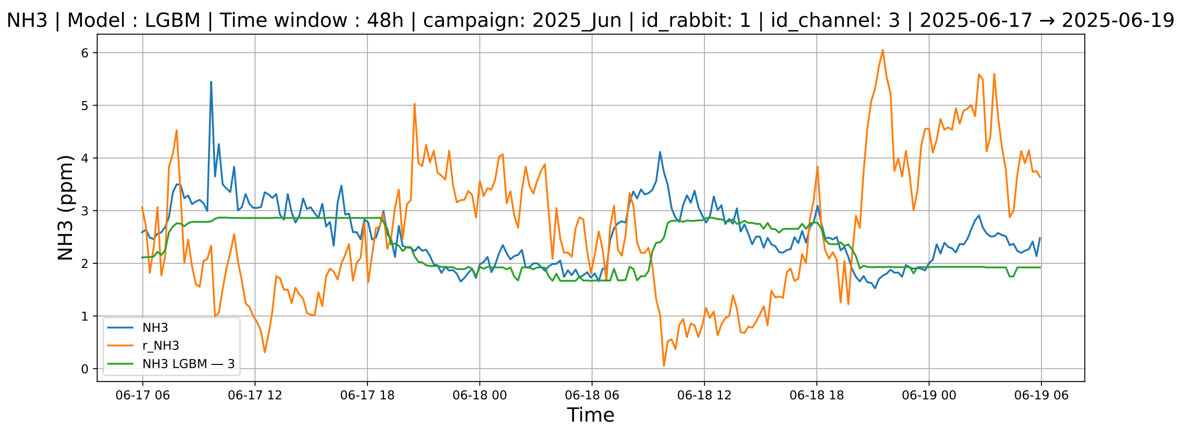
3.1 Standalone plots

3.1.1 Group: campaign: 2025_Jun,id_rabbit: 1,id_channel: 3

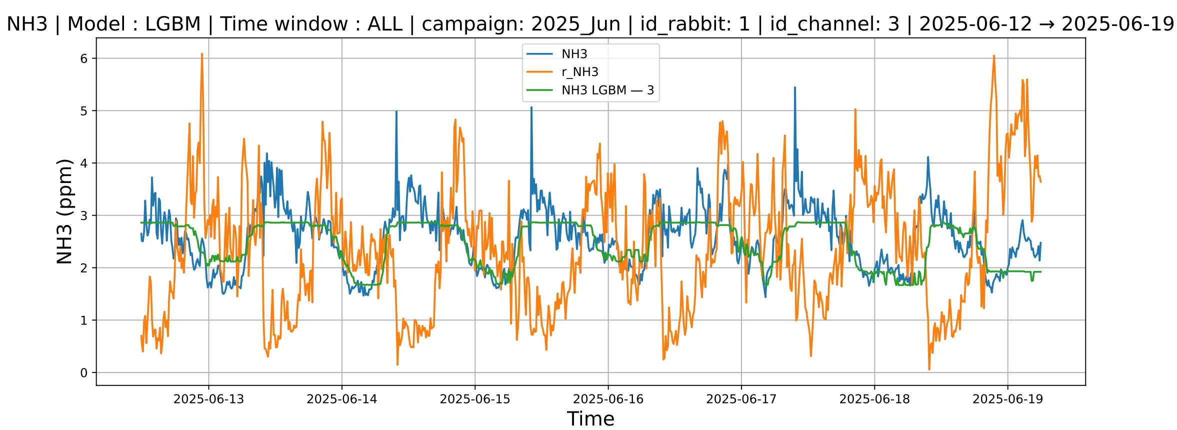
3.1.1.1 Time window : 24



3.1.1.2 Time window : 48

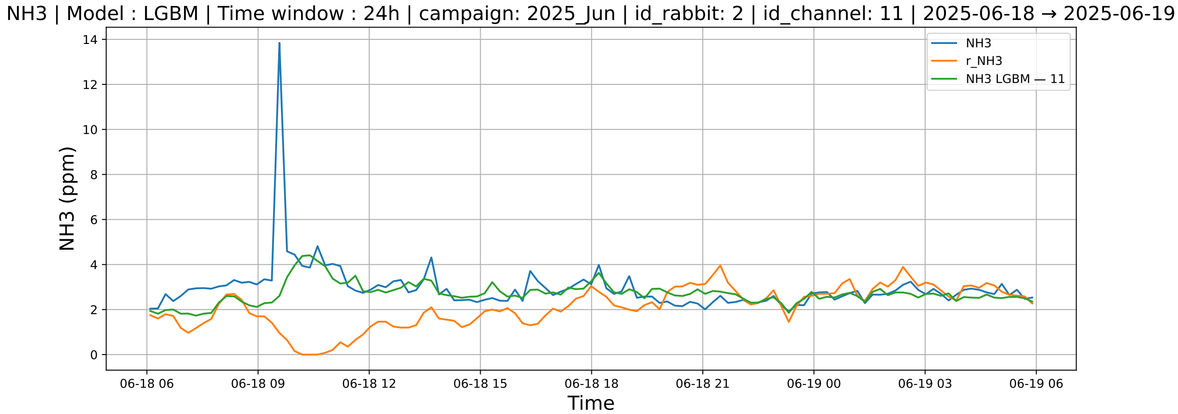


3.1.1.3 Time window : ALL

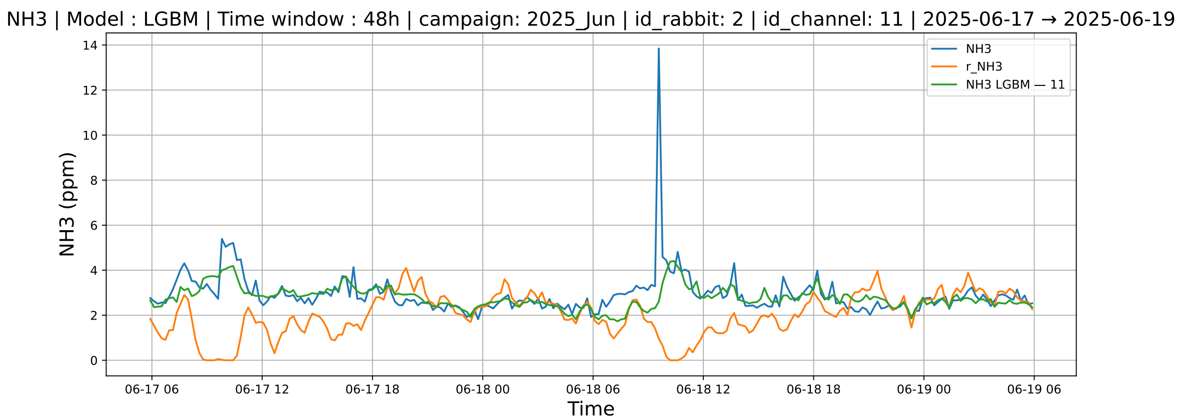


3.1.2 Group: campaign: 2025_Jun,id_rabbit: 2,id_channel: 11

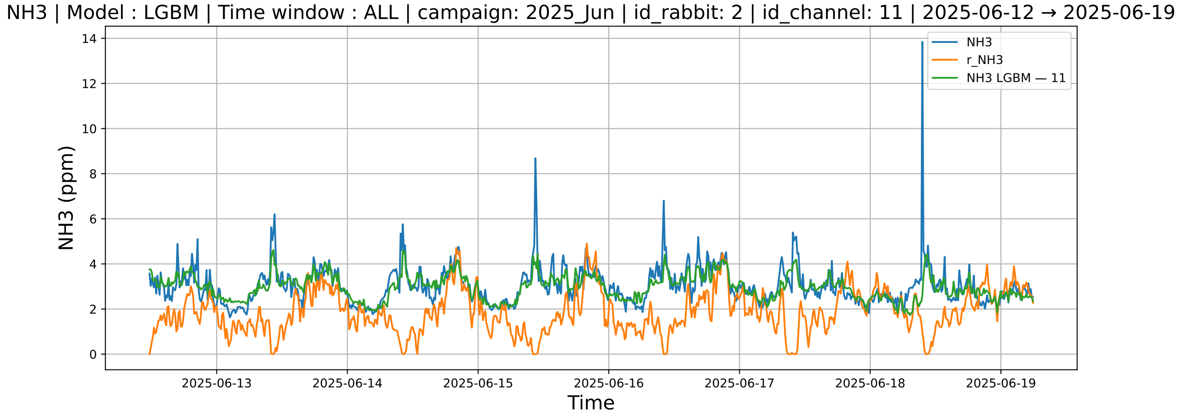
3.1.2.1 Time window : 24



3.1.2.2 Time window : 48

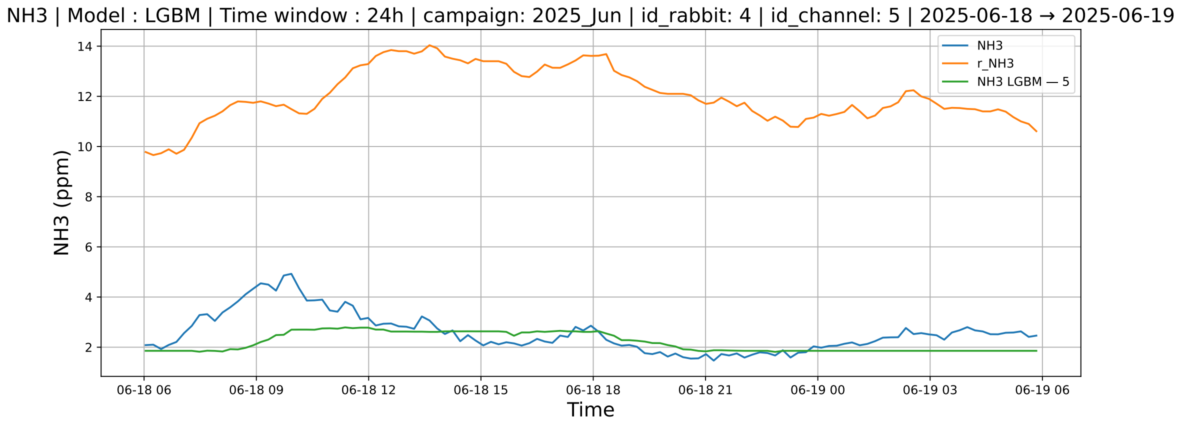


3.1.2.3 Time window : ALL

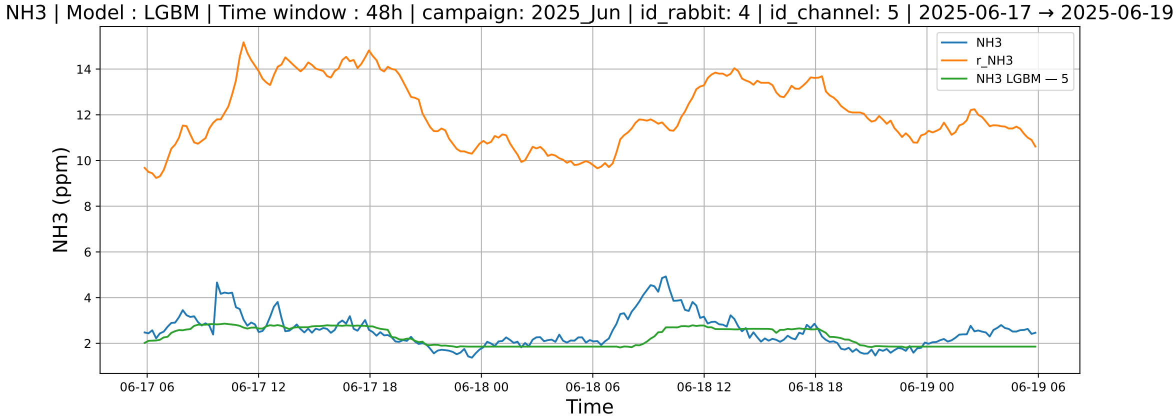


3.1.3 Group: campaign: 2025_Jun,id_rabbit: 4,id_channel: 5

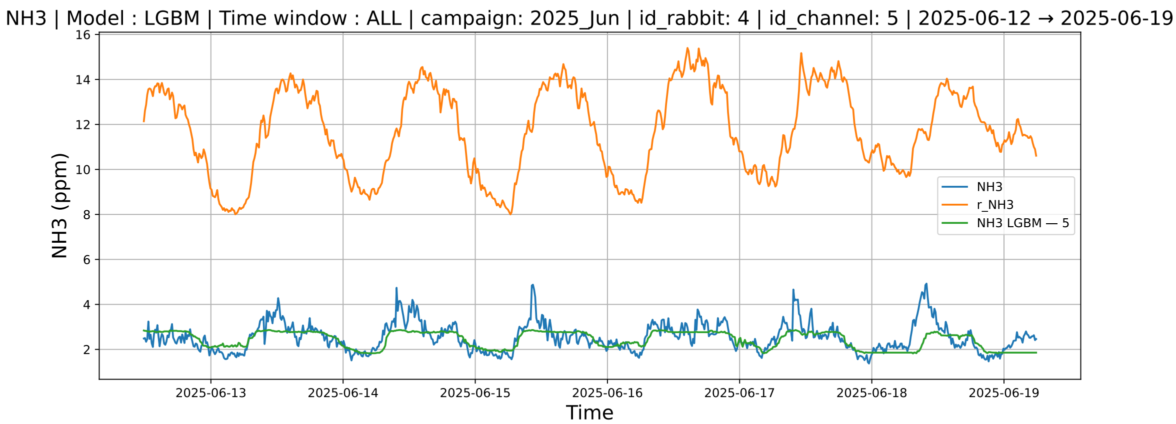
3.1.3.1 Time window : 24



3.1.3.2 Time window : 48

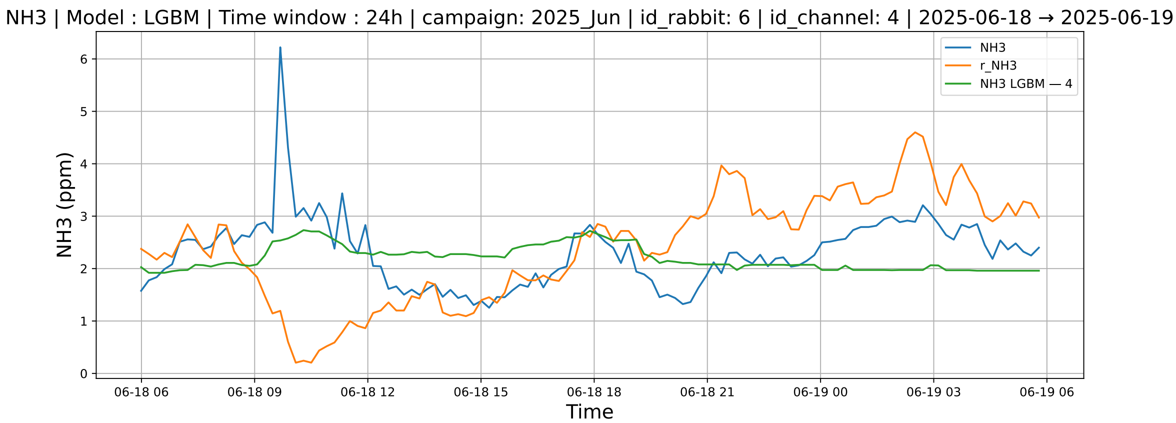


3.1.3.3 Time window : ALL

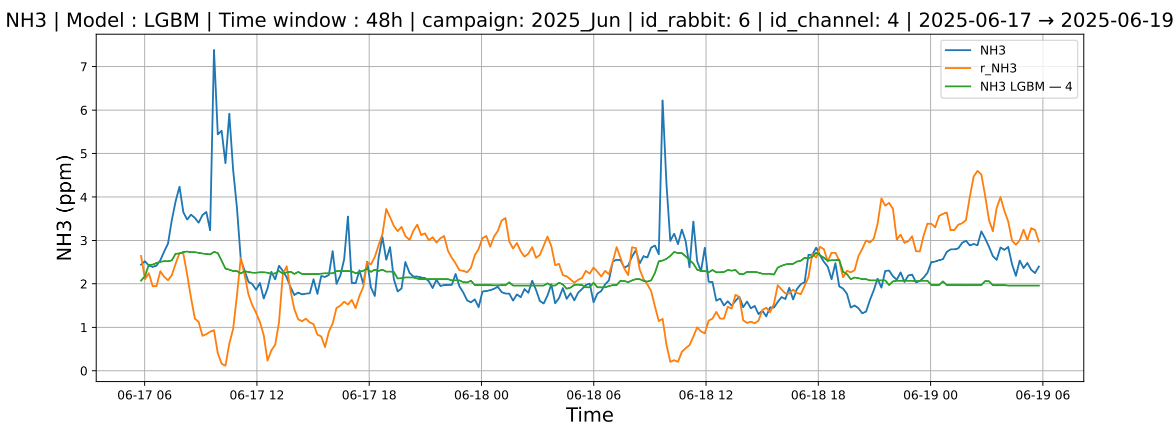


3.1.4 Group: campaign: 2025_Jun,id_rabbit: 6,id_channel: 4

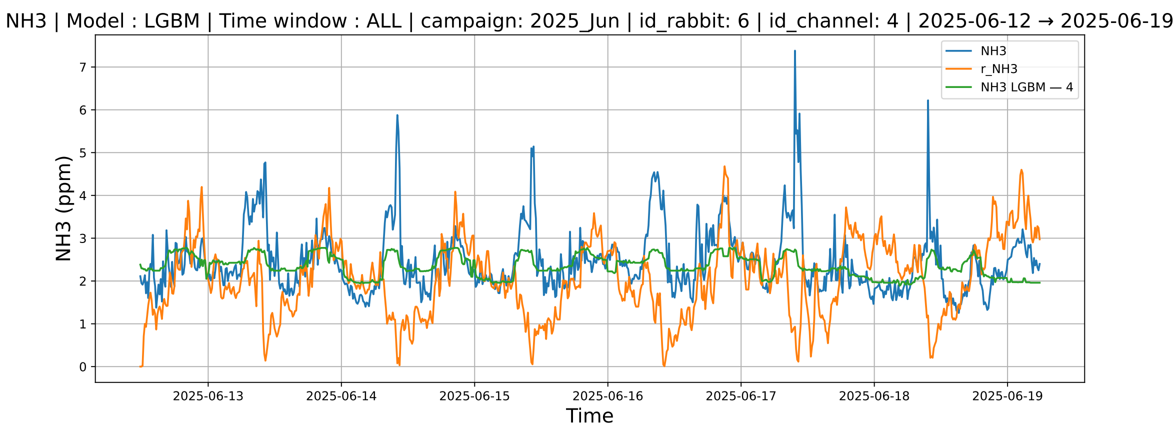
3.1.4.1 Time window : 24



3.1.4.2 Time window : 48



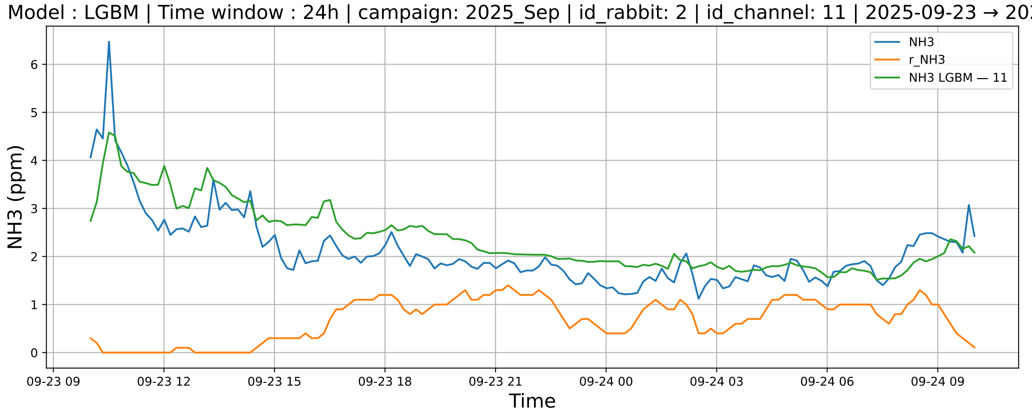
3.1.4.3 Time window : ALL



3.1.5 Group: campaign: 2025_Sep,id_rabbit: 2,id_channel: 11

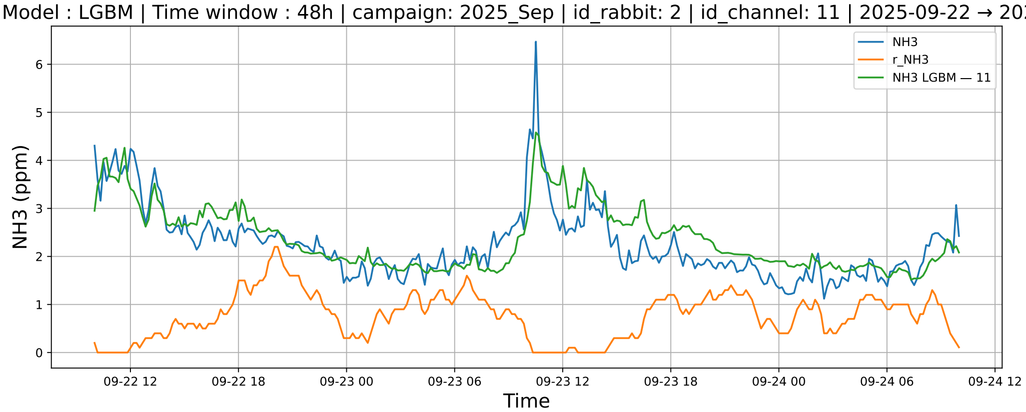
3.1.5.1 Time window : 24

NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24

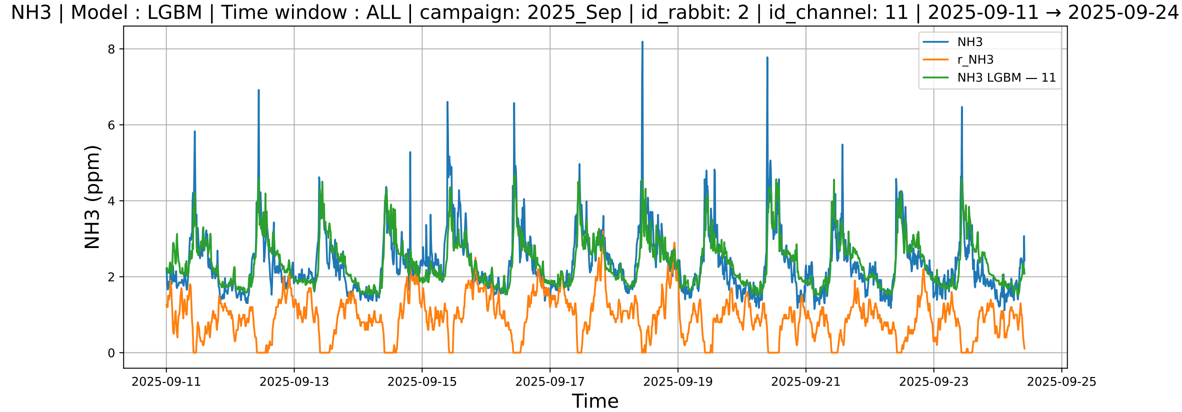


3.1.5.2 Time window : 48

NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24

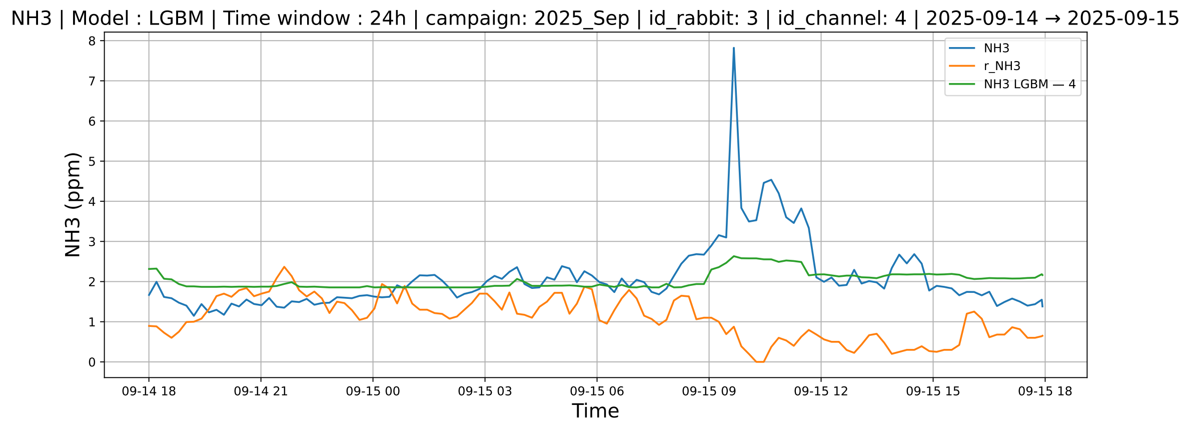


3.1.5.3 Time window : ALL

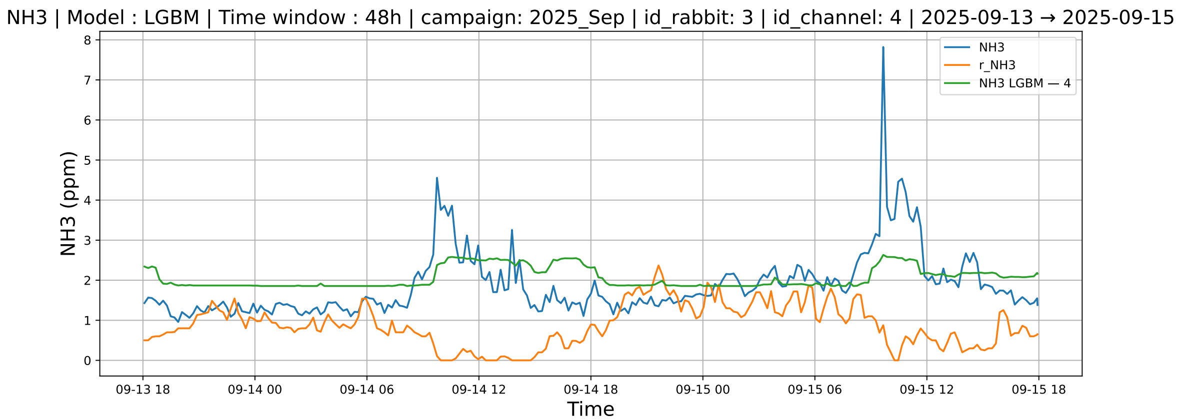


3.1.6 Group: campaign: 2025_Sep,id_rabbit: 3,id_channel: 4

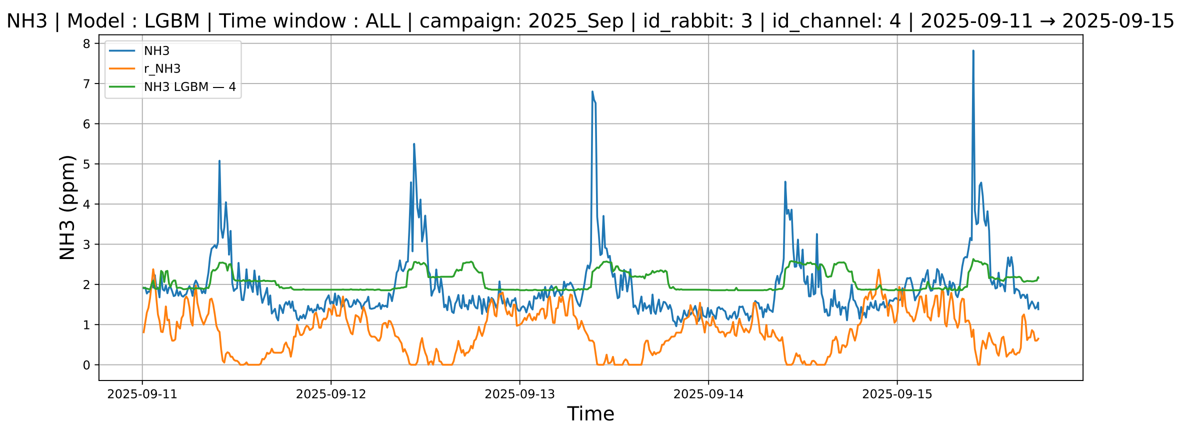
3.1.6.1 Time window : 24



3.1.6.2 Time window : 48



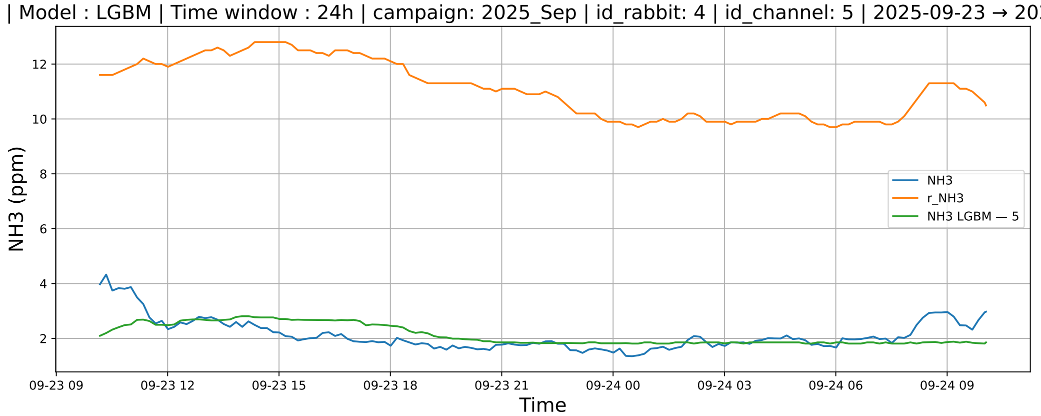
3.1.6.3 Time window : ALL



3.1.7 Group: campaign: 2025_Sep,id_rabbit: 4,id_channel: 5

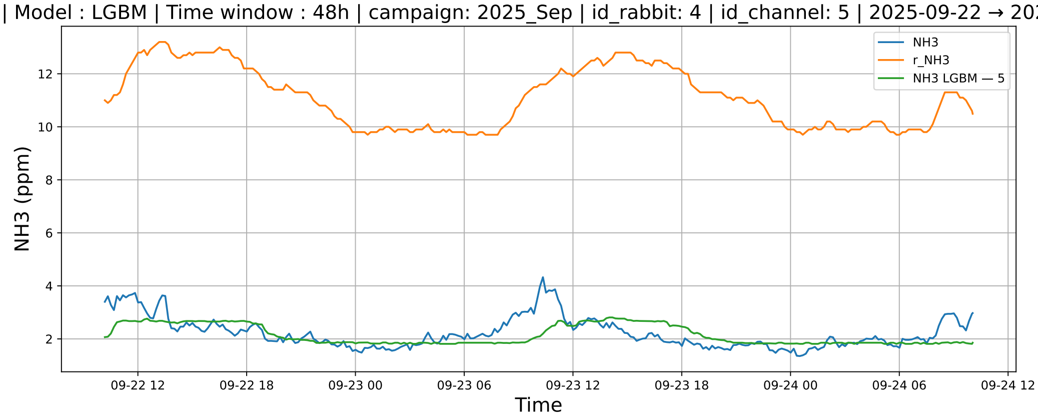
3.1.7.1 Time window : 24

NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-23 → 2025-09-24



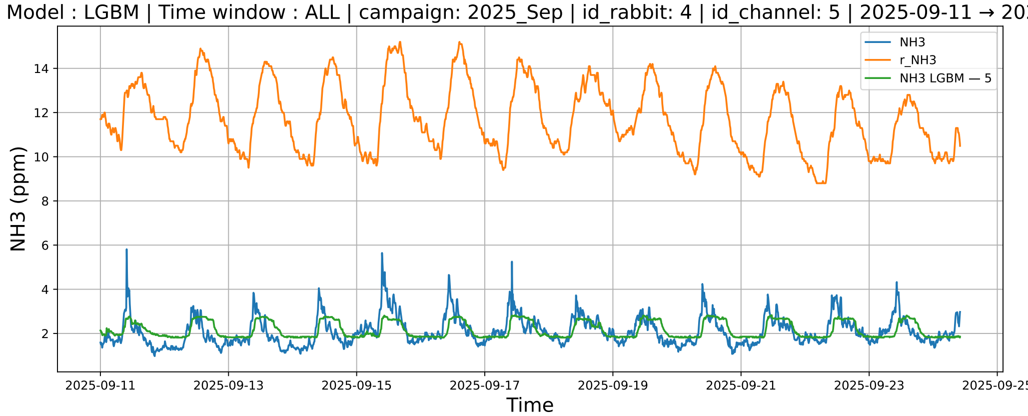
3.1.7.2 Time window : 48

NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-22 → 2025-09-24



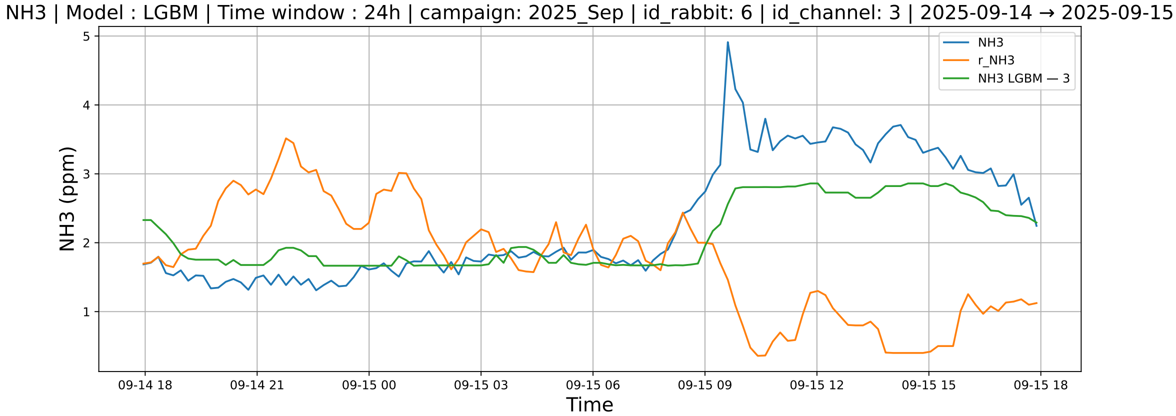
3.1.7.3 Time window : ALL

NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-11 → 2025-09-24

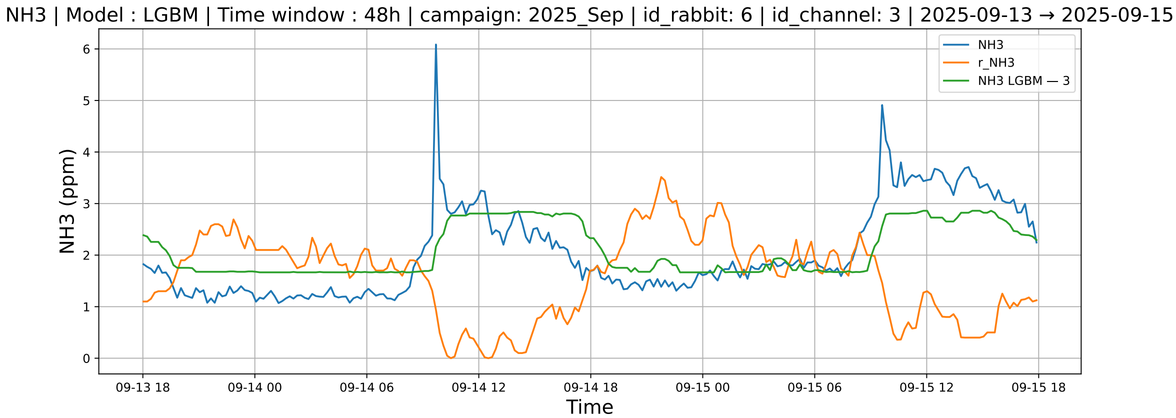


3.1.8 Group: campaign: 2025_Sep,id_rabbit: 6,id_channel: 3

3.1.8.1 Time window : 24

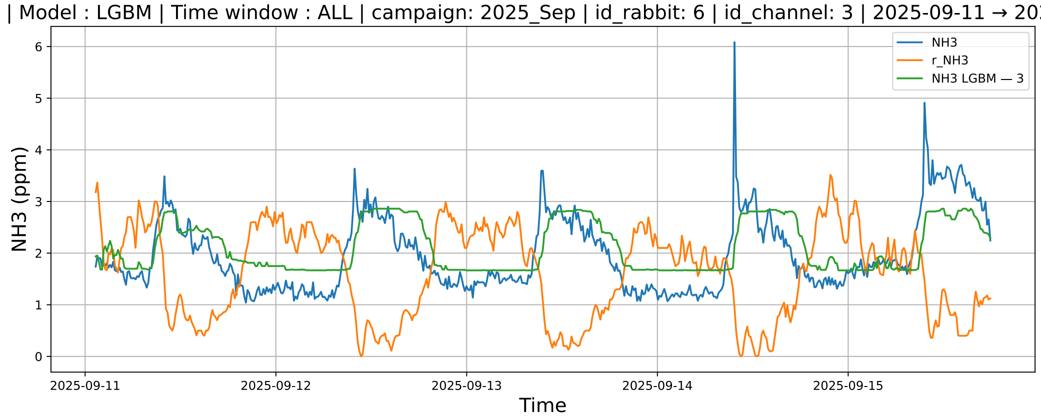


3.1.8.2 Time window : 48



3.1.8.3 Time window : ALL

NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15

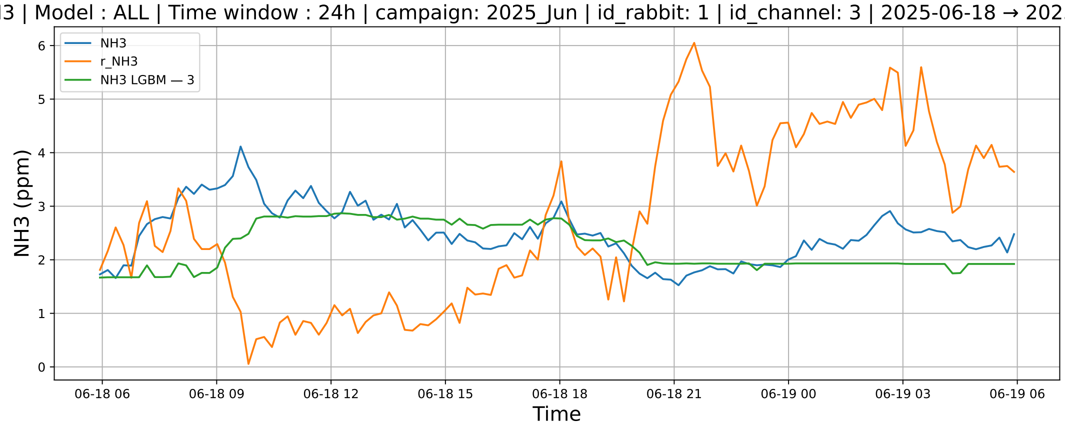


3.2 Compare plots

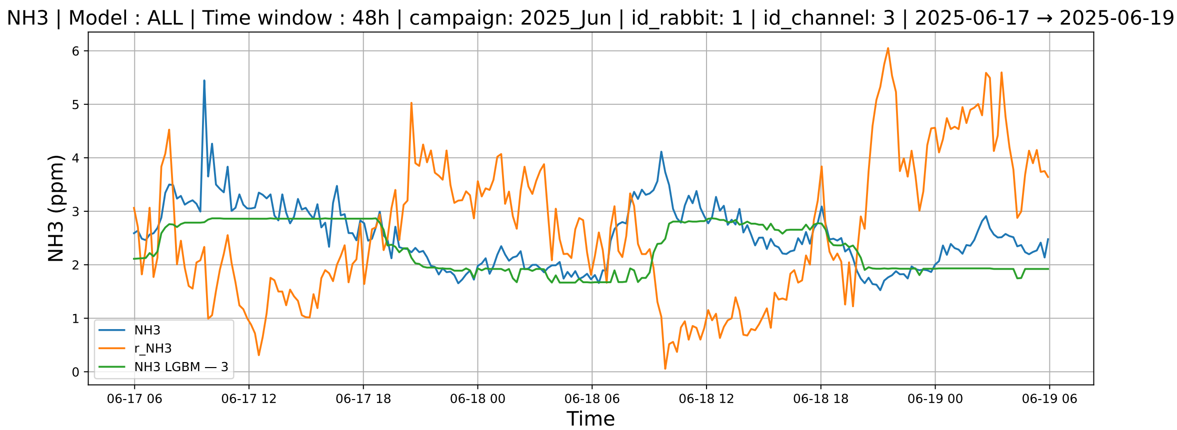
3.2.1 Group: campaign: 2025_Jun,id_rabbit: 1,id_channel: 3

3.2.1.1 Time window : 24

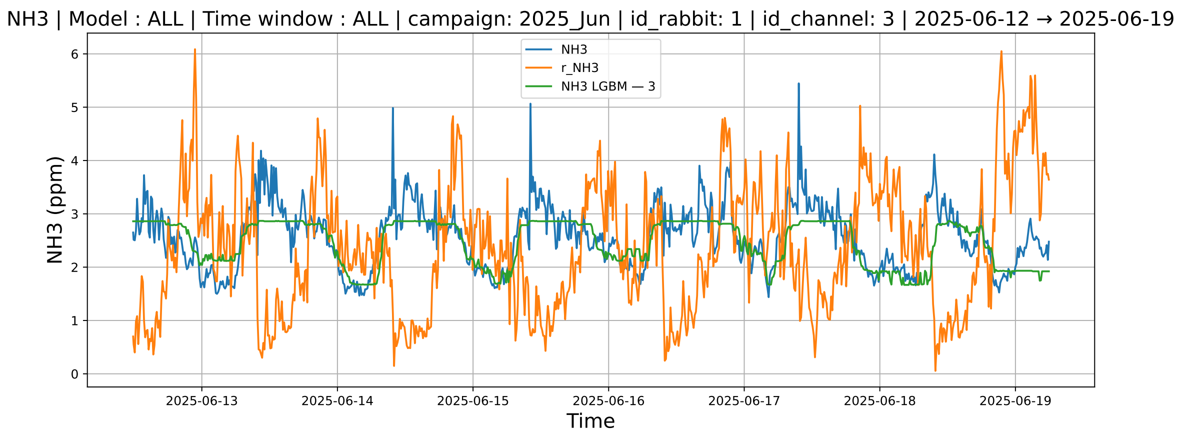
NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-18 → 2025-06-19



3.2.1.2 Time window : 48



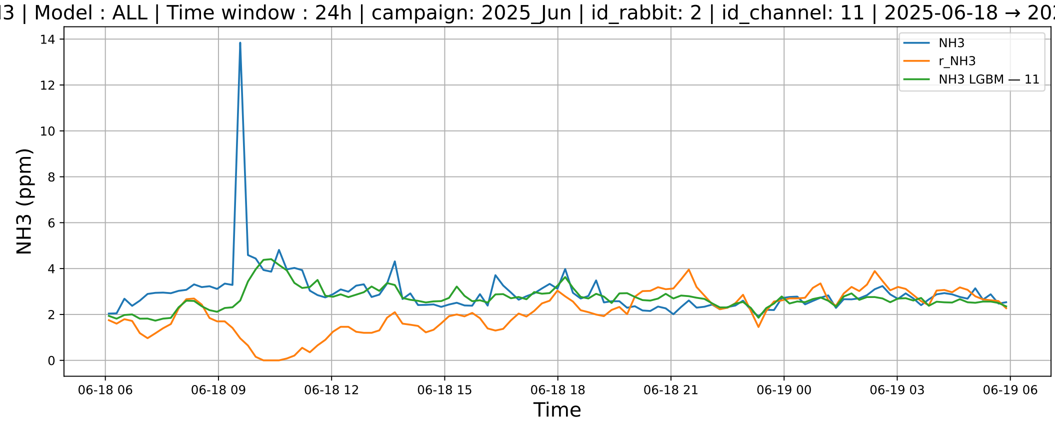
3.2.1.3 Time window : ALL



3.2.2 Group: campaign: 2025_Jun,id_rabbit: 2,id_channel: 11

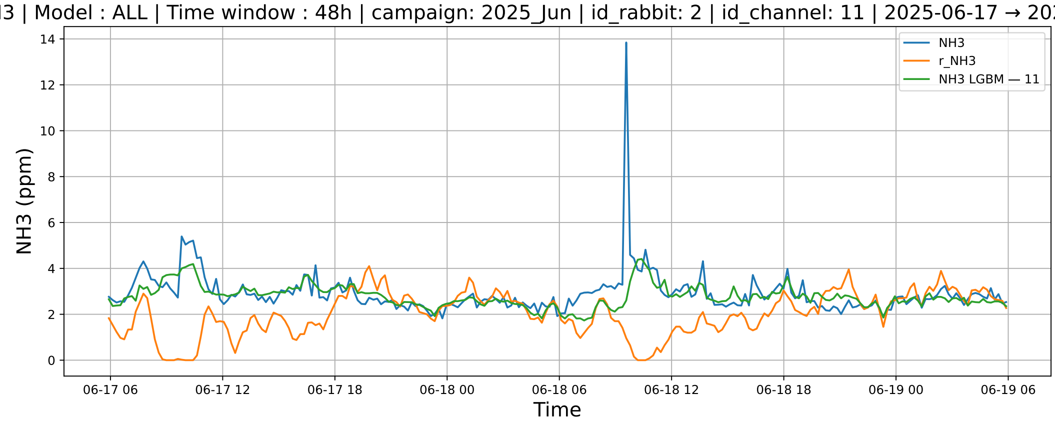
3.2.2.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19



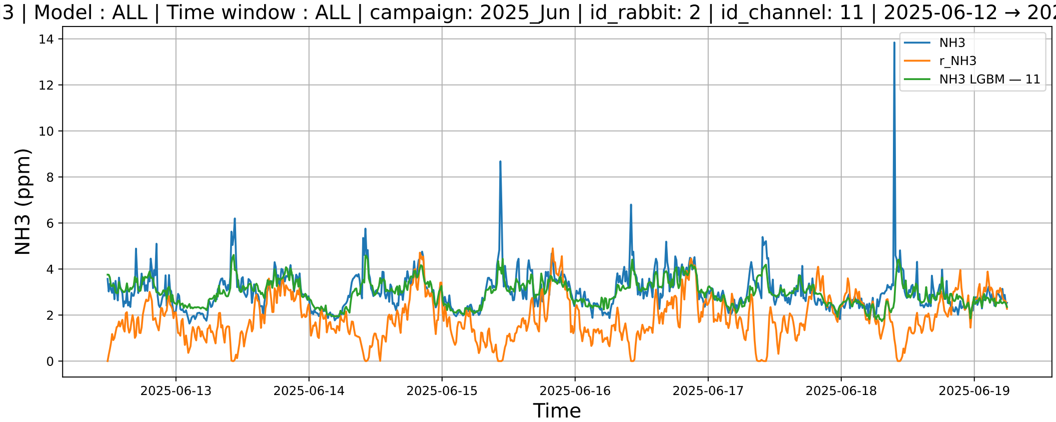
3.2.2.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19



3.2.2.3 Time window : ALL

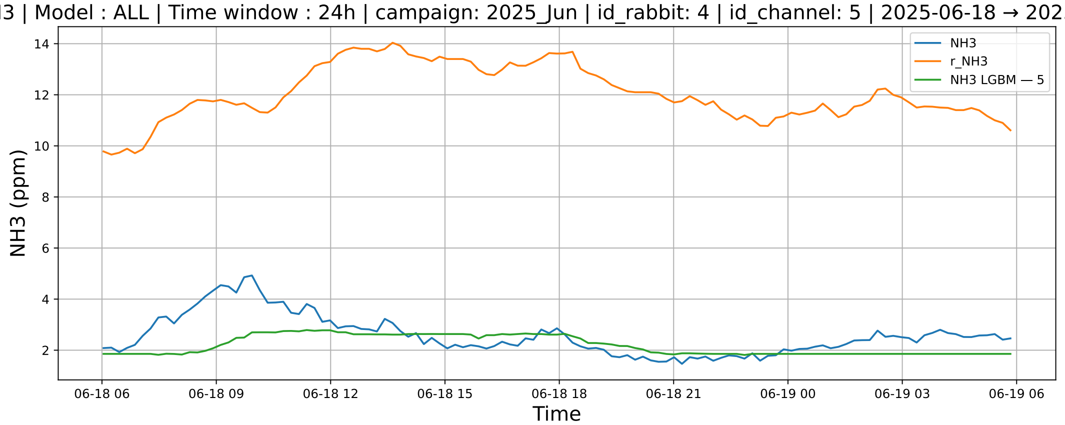
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-12 → 2025-06-19



3.2.3 Group: campaign: 2025_Jun,id_rabbit: 4,id_channel: 5

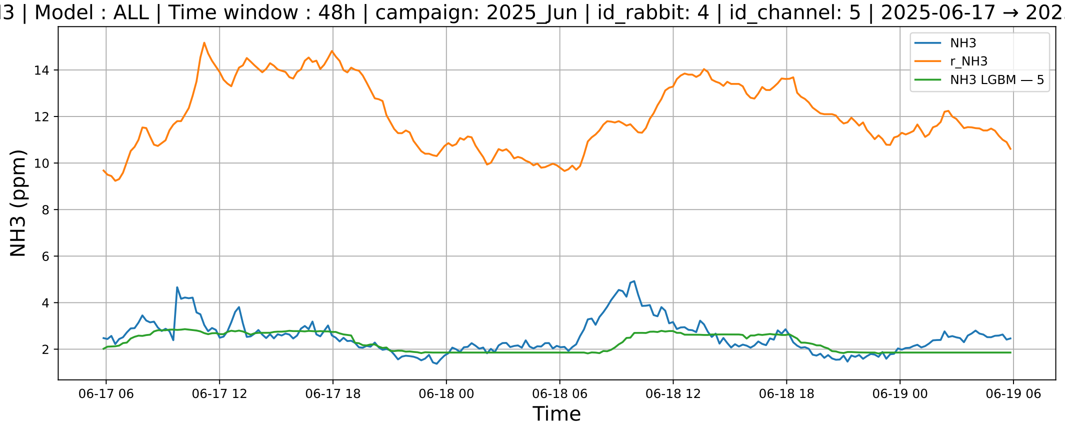
3.2.3.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-18 → 2025-06-19



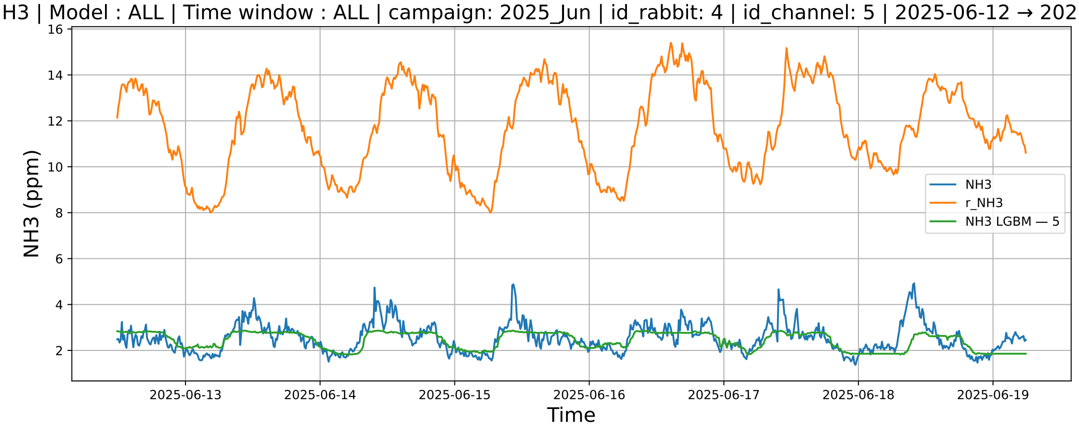
3.2.3.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-17 → 2025-06-19



3.2.3.3 Time window : ALL

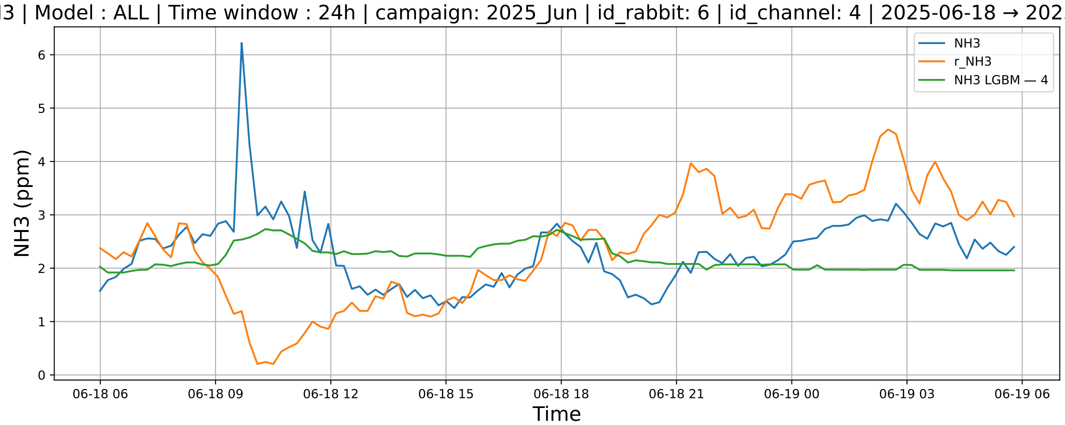
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-12 → 2025-06-19



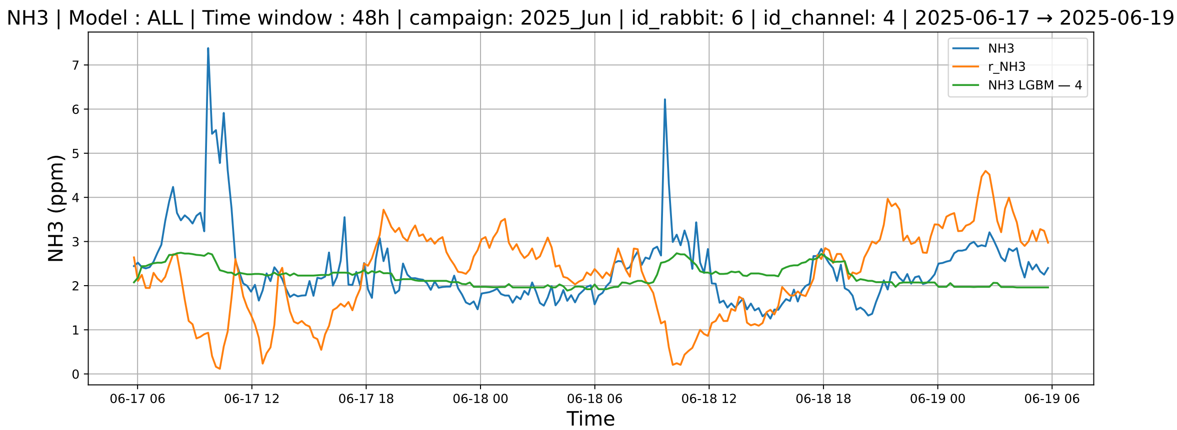
3.2.4 Group: campaign: 2025_Jun,id_rabbit: 6,id_channel: 4

3.2.4.1 Time window : 24

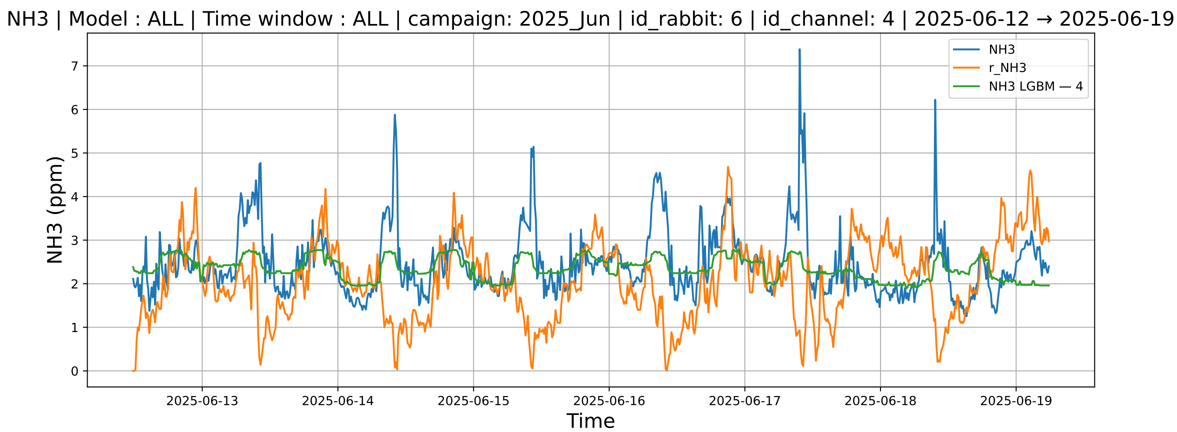
NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-18 → 2025-06-19



3.2.4.2 Time window : 48

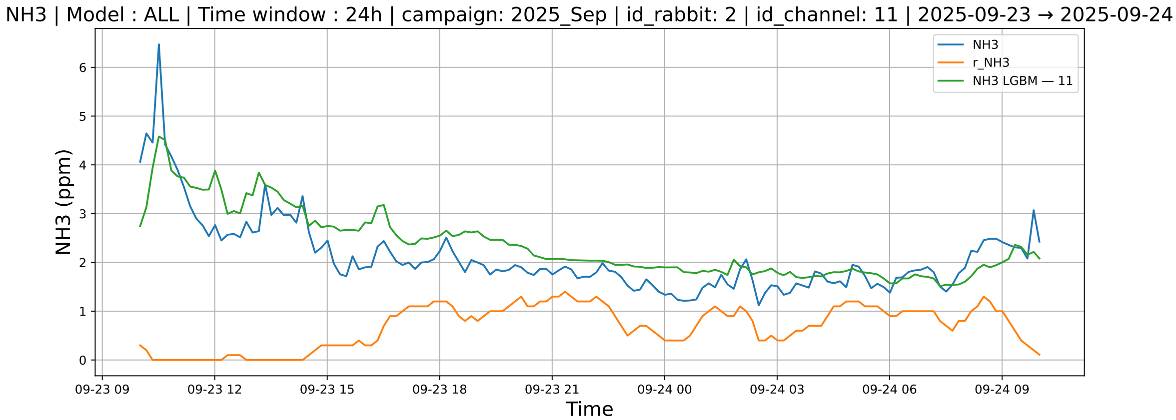


3.2.4.3 Time window : ALL

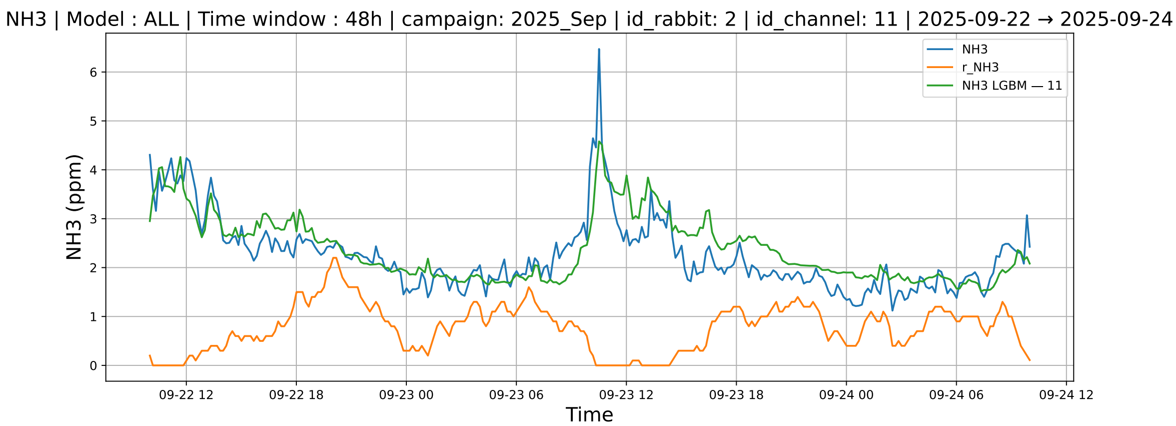


3.2.5 Group: campaign: 2025_Sep,id_rabbit: 2,id_channel: 11

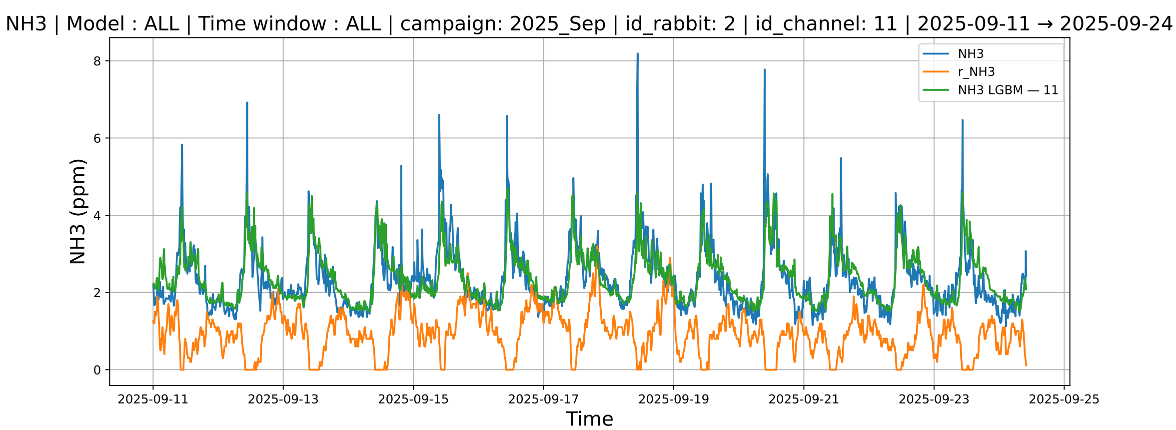
3.2.5.1 Time window : 24



3.2.5.2 Time window : 48

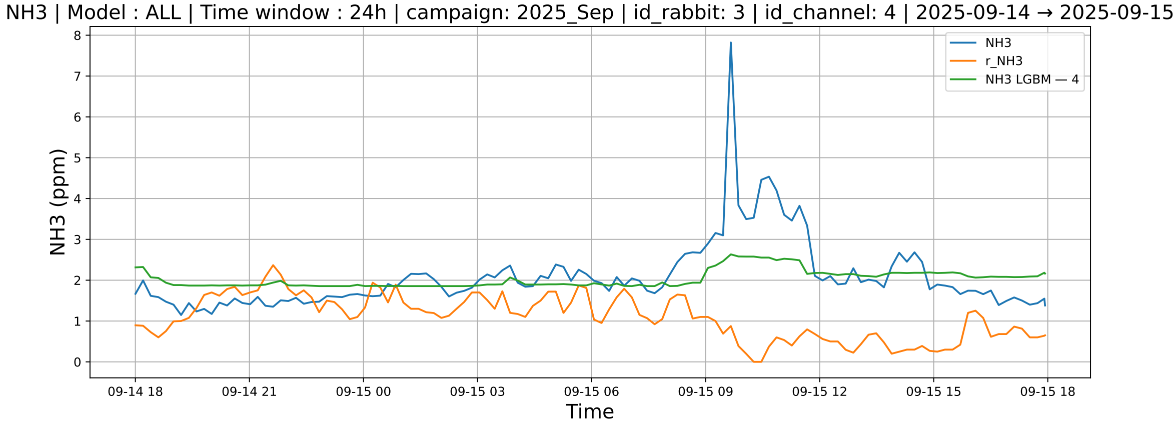


3.2.5.3 Time window : ALL

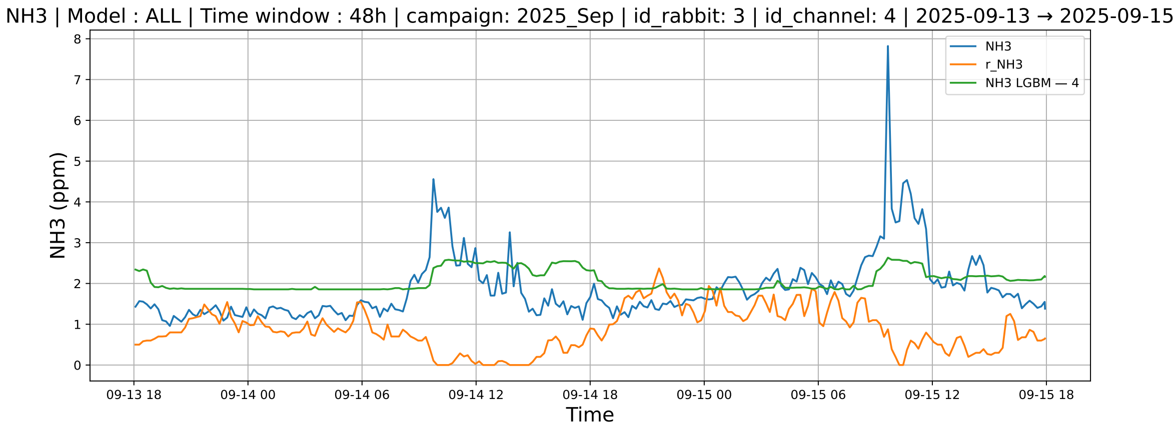


3.2.6 Group: campaign: 2025_Sep,id_rabbit: 3,id_channel: 4

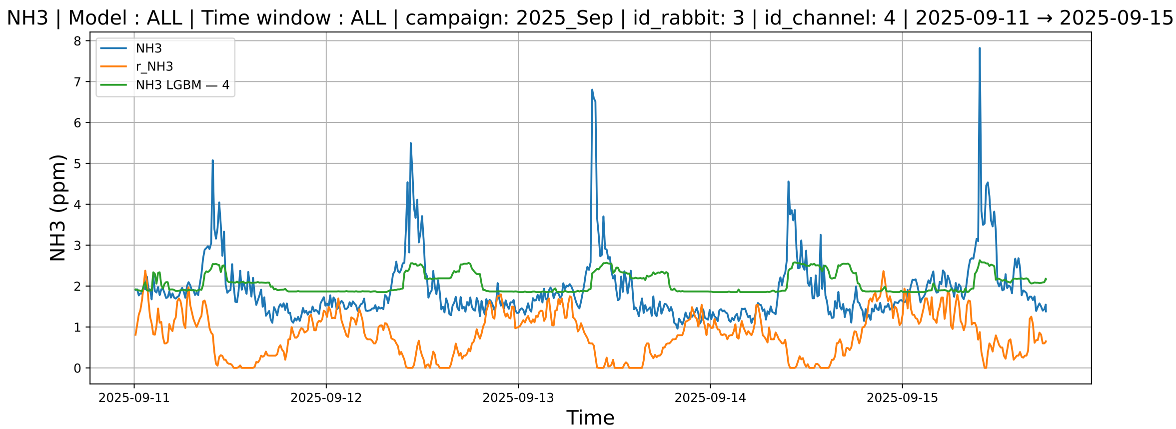
3.2.6.1 Time window : 24



3.2.6.2 Time window : 48

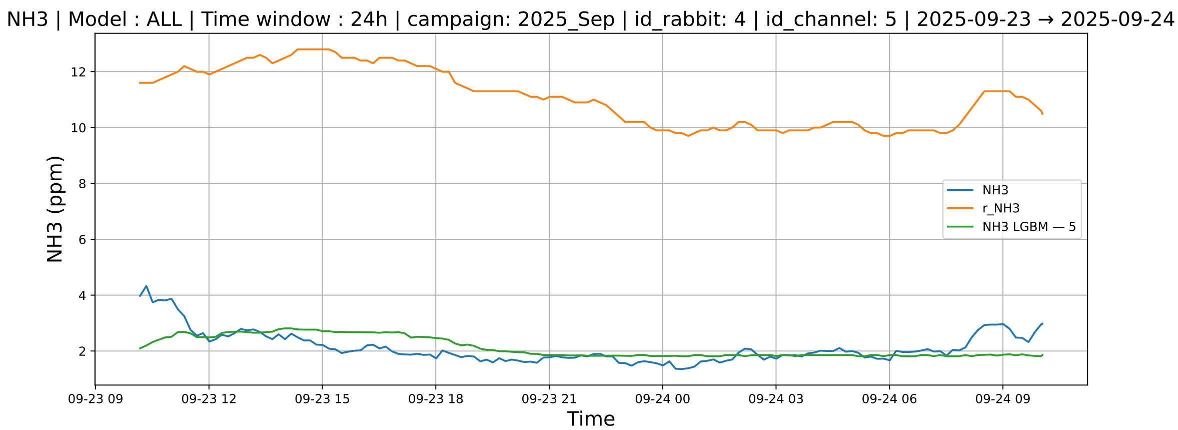


3.2.6.3 Time window : ALL

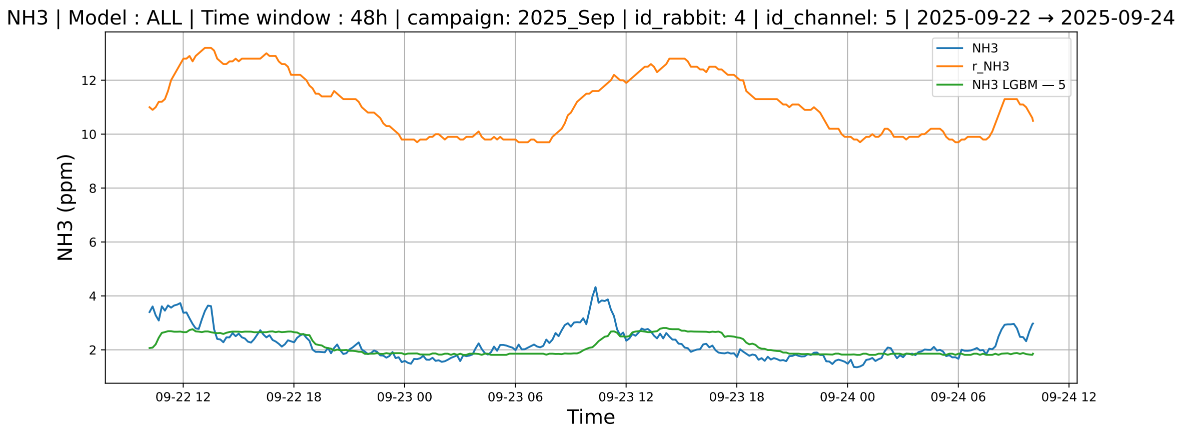


3.2.7 Group: campaign: 2025_Sep,id_rabbit: 4,id_channel: 5

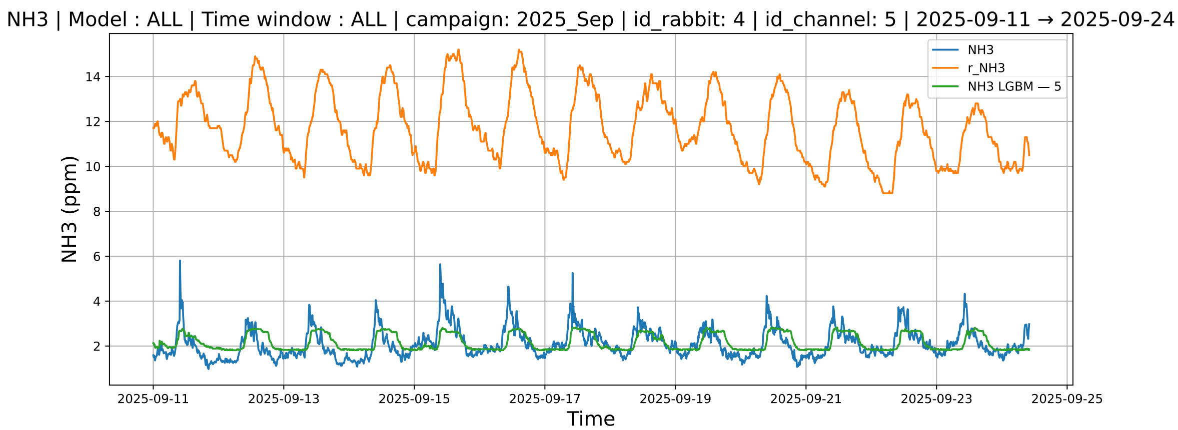
3.2.7.1 Time window : 24



3.2.7.2 Time window : 48



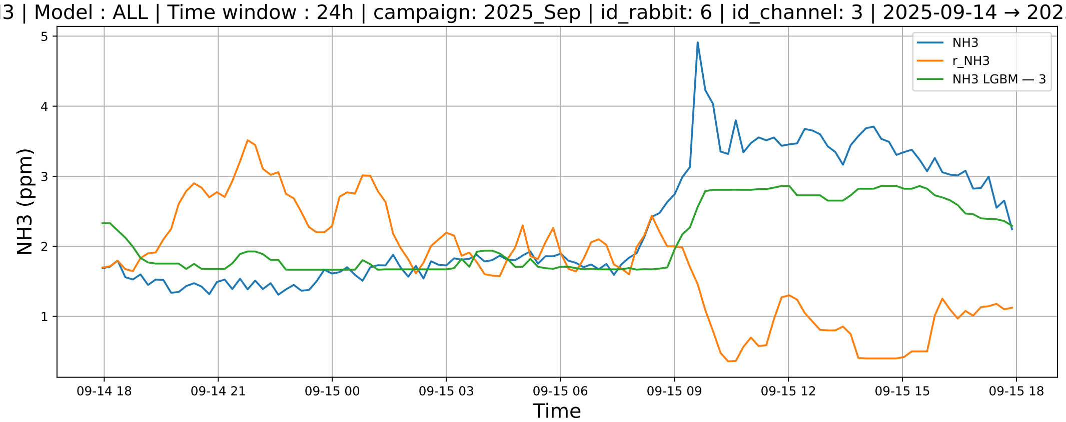
3.2.7.3 Time window : ALL



3.2.8 Group: campaign: 2025_Sep,id_rabbit: 6,id_channel: 3

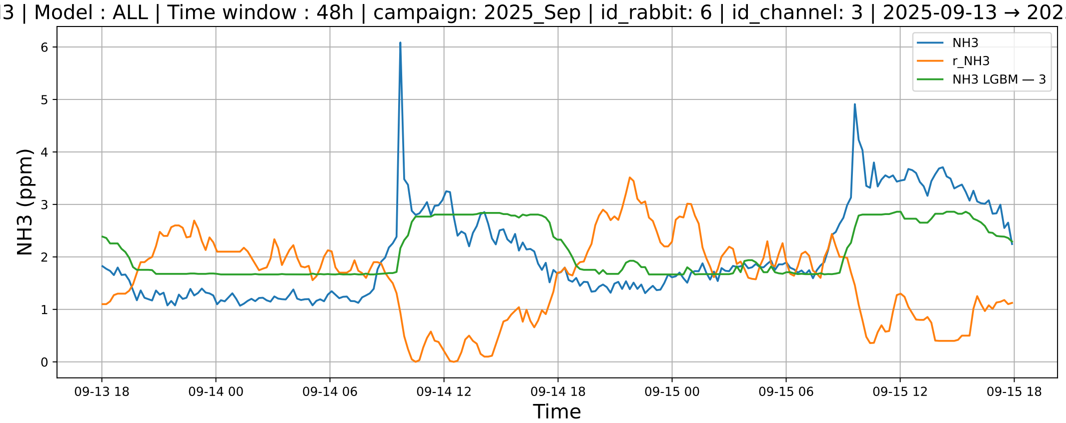
3.2.8.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-14 → 2025-09-15



3.2.8.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15



3.2.8.3 Time window : ALL

NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15